Design Package for the 2026 Cube Design

Team PoliSat - Mission Hermes

Daniel Levi de Souza Ferreira, Davi de Oliveira Mendonça, David Riboldi Junior, Diego Fontes de Avila, Emanuel Silvestre do Nascimento, Flávio Yuji Hashimoto, Gabriel Lopes Prodossimo, Gustavo Carvalho de Mesquita, Gustavo Lucas Teixeira Brito, Henrique Ettinger Cores, Henrique Grolla Ferreira, Igor Araújo Nunes, Julia Freitas Lima, Júlia Teixeira Borges dos Santos, Luca Pieroni Pimenta, Mateus Trindade Pimenta, Misael Jesuíno da Silva Carvalho, Natália Sakane Stucki, Nathan Cunha Oliveira do Prado Amaral, Nicolas Leal Baffi, Sofia Souvarni Favaro Jayanthi, Sophia Soares Mariano, Vitor Layme Izzo.
Escola Politécnica da Universidade de São Paulo, São Paulo, SP, 05508-010, Brazil

Igor Rocha Santos, Davi de Oliveira Mendonça, João Pedro Rufino Sandamil
Instituto de Matemática e Estatística da Universidade de São Paulo, São Paulo, SP, 05508-090, Brazil

Patrick Avelar Santos Silva
Instituto de Física da Universidade de São Paulo, São Paulo, SP, 05508-010, Brazil

I.Nomenclature

a = semi-major axis
Isat = moment of inertia of the satellite
b = viscous damping constant
Tz = resulting mechanical torque
θ = angular position of the satellite
Ra = armature resistance of the motor
ia = armature current of the motor
La = armature inductance of the motor
ea = armature voltage of the motor
ke = electrical constant of the motor
ωm = angular velocity of the motor
Tm = mechanical torque of the motor
Te = electrical torque of the motor
Irr = moment of inertia of the reaction wheel
ω_z *= satellite angular velocity*
τ_m *= motor torque*
J_{wheel} *= reaction wheel moment of inertia*
LoRa = Long Range
CSS = Chirp Spread Spectrum
UART = Universal Asynchronous Receiver/Transmitter
VSWR = Voltage Standing Wave Ratio
l = antenna length
λ = Wavelength
c = light speed on the vacuum
v0 = light speed on the material
f0 = communication frequency
LCD = Liquid Crystal Display
FSPL = Free Space Path Loss
P_t *= Power Transmitted by the transmitter*
P_r *= Power Received by the receiver*
G_{go} *= Directional gain of the transmitter's antenna*
G_{gi} *= Directional gain of the receiver's antenna*
d = Distance between CubeSat and ground station
L = losses
Ap = projected area
S = solar constant

Fsat-earth = view factor of the satellite in relation to the Earth
fa = albedo factor
θ = angle between the satellite's position and the zenith
ε = emissivity of the external surfaces
G = Earth's radiation flux
BATT = battery charge percentage
Psun = solar panels
MPPT = maximum power point tracking
BHeater = battery heater
PSD = power spectral density
f□ = natural frequency
S_γ = material yield strength
σ₃σ = estimated three-sigma equivalent stress
FS = factor of safety
gRMS = root-mean-square acceleration expressed in multiples of standard gravity

II. Introduction

II. Introduction

PoliSat is the first student-led extension group at the Polytechnic School of the University of São Paulo (USP) dedicated to the design, development, and construction of nanosatellites. Founded in 2021 during the COVID-19 pandemic, the group began with fewer than ten members and an initial grant of R$10,000 from the Amigos da Poli endowment fund.

As in-person activities resumed and the group expanded its outreach, PoliSat grew in both membership and project scope. At the time of this report, the group had six sponsors, with total funding exceeding R$90,000 since its foundation.

Initially composed entirely of mechanical engineering students, PoliSat has since expanded to include students from mechanical, mechatronics, electrical, computer, materials, and naval engineering, as well as physics, computer science, and applied mathematics, totaling 27 members. This range of backgrounds allows members to work across the different disciplines involved in satellite development.

The group is organized around two leadership roles. The administrative captain coordinates fundraising, sponsor relations, recruitment, documentation, external communications, and competition logistics. The project captain oversees technical development, coordinates work across subsystems, tracks progress, and helps resolve integration issues.

Each subsystem has its own leader, who coordinates development and holds regular meetings with its members. These meetings complement the group's general assemblies and provide a structure for discussing technical decisions, reviewing progress, and coordinating work between teams. Subsystem teams typically consist of four to seven members, including the leader.

PoliSat organizes its projects around small, multidisciplinary teams. This structure supports close collaboration while giving members responsibility for specific parts of the satellite and requiring them to consider how their work fits into the complete system.

The group has achieved results in national and regional satellite competitions, including first place in the Southeast region of the 2022 Brazilian Satellite Olympiad, second place at LASC 2022, and first place at LASC 2023.

PoliSat is based at USP's Interdisciplinary Center for Interactive Technologies (CITI), where the team holds meetings and stores its satellite hardware, tools, and other project materials.

Under the guidance of faculty advisor Professor Renato Machado Monaro, PoliSat continues to develop its nanosatellite projects and participate in engineering competitions. Its long-term goal is to launch a satellite built by the team into orbit, contributing to the development of satellite engineering at USP.

III. System Architecture

General overview

The Hermes Mission was developed to demonstrate the feasibility of a low-cost and compact ADS-B payload capable of receiving and processing ADS-B signals transmitted by aircraft. The payload is designed to receive 1090 MHz ADS-B signals, decode the transmitted messages, and extract relevant information such as aircraft identification, position, altitude, and velocity. The implementation of this system in a CubeSat presents significant challenges, particularly regarding signal reception, data processing, antenna performance, and the

integration of the payload with the satellite's communication and onboard computing systems. Since several of the technologies and integration methods adopted by the mission are new to the group, extensive testing, validation, and subsystem integration were required throughout the development process. Overall, the mission represents a multidisciplinary challenge for the PoliSat team, requiring the development and validation of new solutions while promoting practical experience in satellite systems engineering, RF design, embedded systems, software, and mission operations.

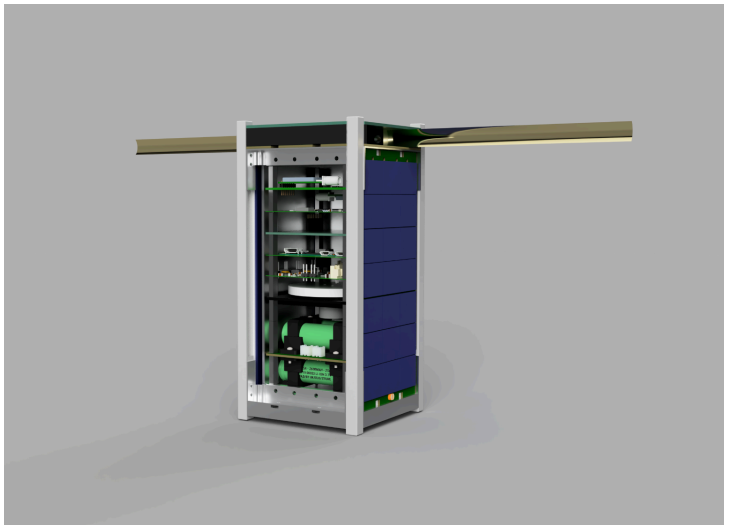


Figure 1: CAD design for the satellite composition

A. Attitude Determination and Control System (ADCS):

1) Introduction

The Attitude Determination and Control System (ADCS) is responsible for stabilizing the angular rate of the platform, estimating or determining its current orientation, and controlling that orientation according to a commanded reference. The ADCS uses specific sensors and actuators to accomplish this objective, such as gyroscopes, magnetometers, star trackers, reaction wheels, and light sensors.

For the Hermes mission, the ADCS development was initially validated through a one-axis prototype mounted on a rotating platform. This experimental setup was designed to reproduce the relevant rotational dynamics of the satellite while allowing the team to develop and validate the attitude determination and control algorithms in a controlled environment. The prototype implements two main operational functions: inertial rate damping and pointing toward a light source. The first function reduces the initial angular velocity of the platform, while the second uses the relative intensities measured by the light sensors to determine the direction of the light source and control the platform accordingly. The prototype therefore serves as a validation platform for the ADCS algorithms and hardware, rather than representing the complete mechanical configuration of the flight satellite.

The architecture of the subsystem is shown in the diagram below.

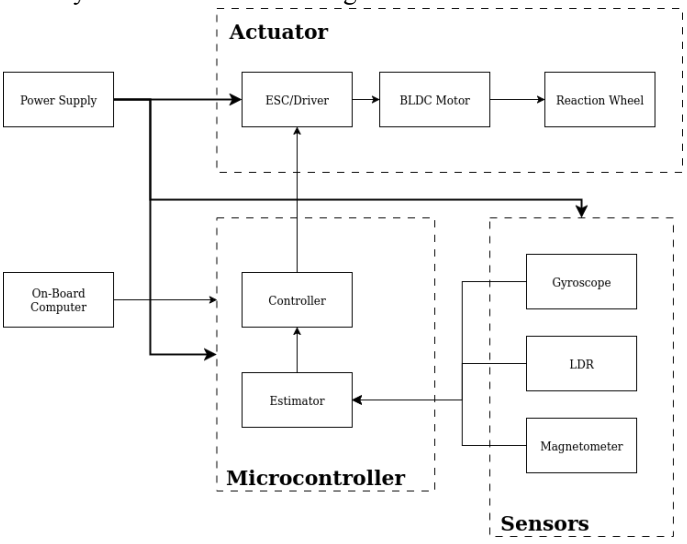


Figure 2: ADCS Architecture

Basic operation is described below.

The primary actuator is a reaction wheel driven by a BLDC motor. The actuator is controlled by the microcontroller. For attitude estimation, the sensors are used to obtain an estimate of the current orientation. More than one position sensor is used (LDRs and the magnetometer) to perform sensor fusion. Data from the gyroscope is also used. Sensor data fusion is performed on the microcontroller, which uses the final estimate in the control law.

The ADCS interfaces with the rest of the satellite's systems, in particular with the Electrical Power System (EPS) and the On-Board Computer (OBC).

The main quantitative requirements of the ADCS are listed below:

- Light-source direction determination accuracy: < 5°
- Ability to stabilize the satellite (about the z-axis) from 60 RPM to 5 RPM in 10 s
- Gain margin: 6 dB
- Phase margin: 45°

These requirements arise both from the needs of the prototype under development and from good control-system design practices.

In addition to the quantitative requirements, we add the following qualitative requirements:

- The control system must be stable.
- The control system must be robust to parametric variations in the plant.

2) Modelling

The system is modeled as a free body rotating on a surface with friction about one axis, coupled to a rotating system (the reaction wheel). The kinematic model is given:

$$\dot{\theta} = \omega \tag{1}$$

The dynamic equations governing the satellite are:

$$I_{sat}\dot{\omega} + b\omega = -\tau_{motor} \tag{2}$$

While the dynamic equation governing the wheel is:

$$I_{rw}\dot{\omega}_{rw} = \tau_{motor} \tag{3}$$

Where we neglect the internal friction of the wheel. In the frequency domain, the dynamics of the body are given:

$$I_{sat}\dot{\omega} + b\omega = -\tau_{motor} \tag{4}$$

$$I_{sat}s\Omega(s) + b\Omega = -u(s) \tag{5}$$

Leading to the following transfer function:

$$\frac{\Theta(s)}{u(s)} = -\frac{1}{s(I_{sat}s + b)} \tag{6}$$

Finally, the actuator assembly is modeled as a first-order lag:

$$H(s) = \frac{1}{T_{act}s + 1} \tag{7}$$

System identification tests showed that T_{act} = 0.05 s is a valid choice. Below, we show the root locus plot of the open-loop plant:

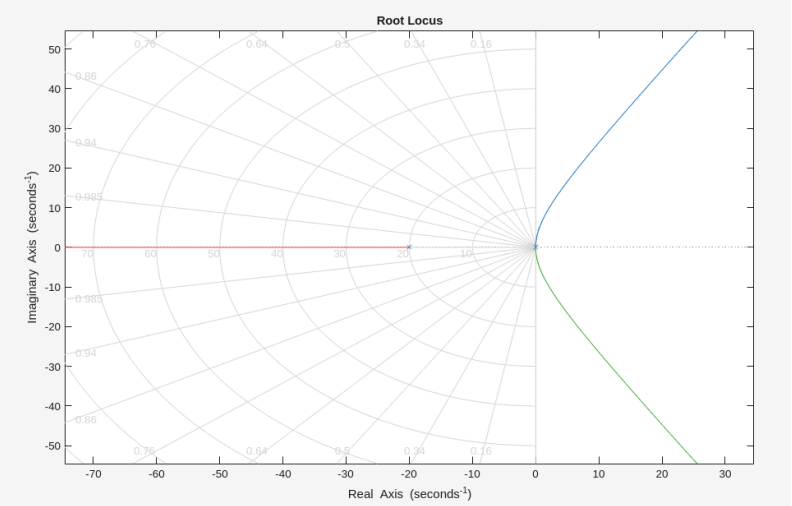


Figure 3: Root Locus Plot Open Loop Plant

As can be seen from the plot, the plant is marginally stable and, due to the influence of actuator lag, can't be controlled with proportional gain only.

Another thing to note is that the actuator pole is much faster than the other two poles of the system.

3) Control

a) Control Law - Rate Damping

Before beginning any type of operation, whether communications or science, the satellite must first be stabilized to a reasonable angular rate. In the Hermes mission, this is simulated by stabilizing the satellite from a high initial angular rate. For this control mode, we propose the following control law:

$$u = -K\omega \tag{8}$$

Essentially, a negative feedback proportional to the current angular rate measured by the gyroscope. This law has the advantage of being simple and of directly using data from the most reliable sensor. The gain is selected so as to satisfy the design requirement:

- Ability to stabilize the satellite (about the z-axis) from 60 RPM to 5 RPM in 10 s

The problem is easily solved through an analysis of the governing dynamical equation on angular rate:

$$-K\omega = I_{sat}\dot{\omega} \tag{9}$$

Where we have a pole at $-\frac{K}{I_{sat}}$. In this analysis we are neglecting the effect of the delay caused by the actuator*. For a settling time of 15 s, we need a time constant τ equal to 2 s, where we use the approximation of 5τ for the settling time of the system given an initial condition. Given the satellite's inertia, we obtain a time constant τ = J/K = 4,47e−3 / 1,0e−3 = 4,47 s.

*The actuator dynamics are much faster than those of the plant; as a result, the response is dominated by the dynamics imposed by the controller, which justifies neglecting the actuator's lag.

Below we show the control system simulation, starting from an initial rotation of 60 RPM ($\omega_0 = 6.28$):

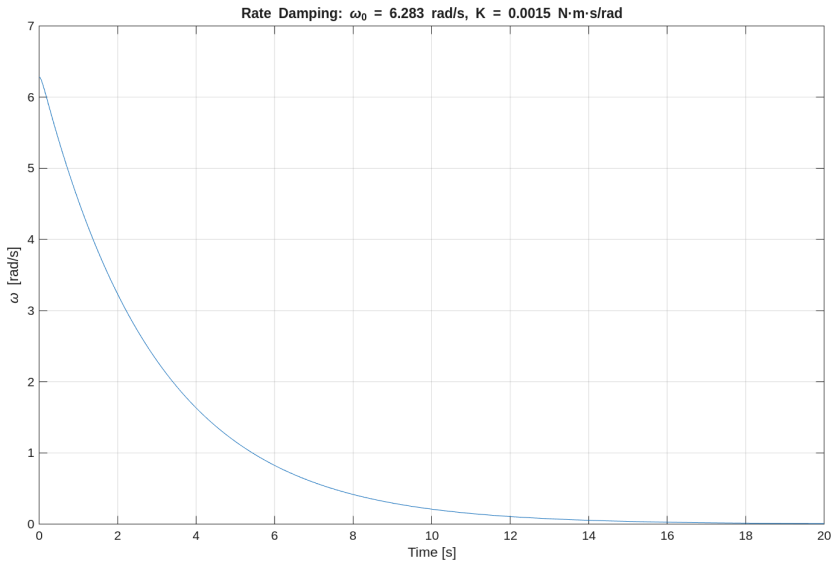


Figure 4: Closed Loop Response

b) Estimation

In both modes of operation, the subsystem uses two sensors—one for orientation (a light sensor or a magnetometer) and one for angular rate (a gyroscope). The sensor data is fused using a Complementary Filter, which makes it possible to combine the slower signals measured by the orientation sensors with the faster signals from the gyroscope.

The estimation loop is shown below:

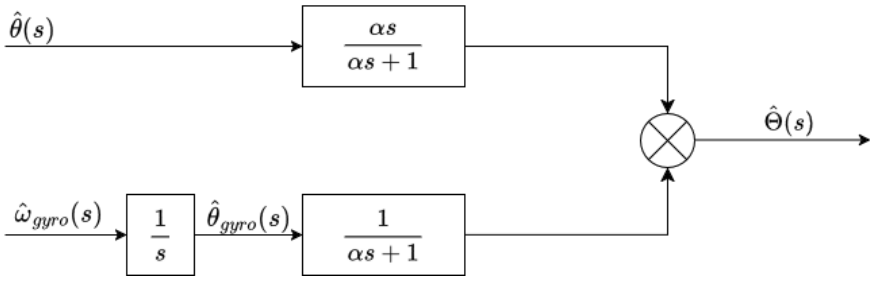


Figure 5: Complementary Filter

c) Discretization

To be implemented on the microcontroller, the controller must be discretized. A convenient sampling frequency is 10–20× the system bandwidth. For a frequency of 0.706 rad/s, a reasonable range of sampling rates is 1.12 to 2.25 Hz. For safety, we choose a slightly higher rate of 10 Hz.

The controller is discretized using the bilinear transform (3). The transfer function in the z-domain is:

$$C(z) = \frac{0.04935\,z - 0.04645}{z+1} \tag{13}$$

The response is simulated below:

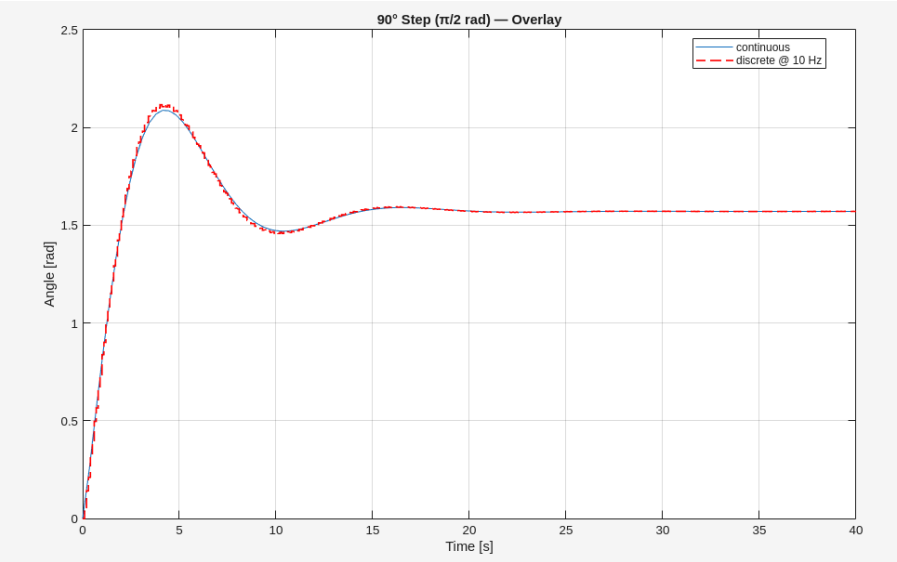


Figure 6: Discrete Closed Loop Step Response

4) Hardware

a) Motor

A brushless DC motor was selected as the primary actuator, due to its better performance compared with conventional DC motors. The chosen model was a **Maxon EC-20 FLAT**, shown below:



Figure 7: Maxon EC 20 Flat BLDC

The motor was selected so as to ensure the design requirements, namely:

- Ability to stabilize the satellite (about the z-axis) from 60 RPM to 5 RPM in 10 s

The settling time of 10 s implies an angular acceleration of $6.28 / 15 = 0.576$ rad/s². The torque to be applied is, therefore:

$$\tau = I_{sat}\dot{\omega} = 4.47 \times 10^{-3} \cdot 0.419 \approx 2.6\ mNm$$

The EC-20 FLAT can deliver 3.2 mN·m of continuous torque, providing a margin of approximately 1.2× and satisfying the requirement.

The speed control of the motor is performed by the *Electronic Speed Controller* (the motor driver). We chose the **ESCON 24/2**, also by Maxon as per the manufacturer's recommendations. In our design it operates in Speed Control Mode.

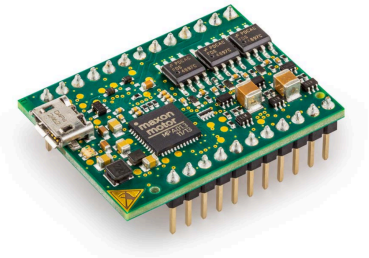


Figure 8: Maxon ESCON 24/2 Electronic Speed Controller

b) Inertial Measurement Unit

The ADCS uses a 9-degree-of-freedom inertial measurement unit; the selected IC is the **MPU-9250**, chosen for availability and modularity instead of implementing separate inertial sensors.

c) Light Dependent Resistor (LDR)

The ADCS uses four light sensors to measure light intensity and determine the direction of the light source from these measurements. The resistors are mounted on the faces of the satellite.

d) Microcontroller

The microcontroller used is an Arduino Nano. The Arduino Nano is a development board based on the ATmega328 microcontroller. It is a low-cost board and easy to set up.

e) Printed Circuit Board

The printed circuit board is the hardware component that supports the ADCS sensors and electronics. The PCB was made using the KiCAD software suite, a free and open source EDA tool. Our design is shown below:

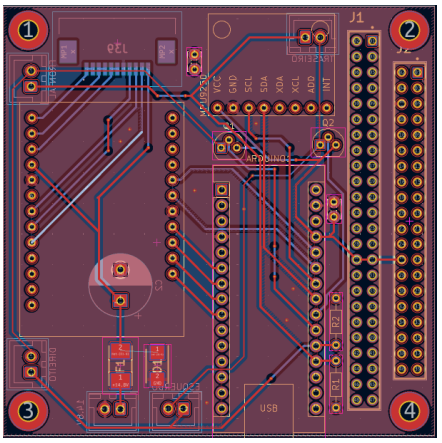


Figure 9: ADCS Printed Circuit Board

f) Reaction Wheel

The reaction wheel is the element that exchanges angular momentum with the satellite. By conservation of angular momentum (and assuming no external torques) spinning the wheel up or down with the motor torque changes the wheel's angular momentum; the satellite's body takes the equal-and-opposite change so that the total remains conserved.

With that in mind, sizing the reaction wheel means deciding how much angular momentum the wheel must be able to store. For a satellite with inertia I_{sat} = 4.47e-3 kg m², the maximum angular momentum is under our assumptions (60 RPM):

$$H = I_{sat}\omega = 0.0288\ N\,m\,s$$

Assuming the wheel is rotating at 12000 RPM:

$$J_{wheel,\,min} = \frac{H}{\omega_{rw}} = 2.29 \times 10^{-5}\ kg\,m^2$$

B. On-board Computer (OBC):

1. Introduction

The On-Board Computer (OBC) subsystem is responsible for controlling the satellite's operation and managing the communication between subsystems. It uses onboard sensors to evaluate the satellite's condition and change states accordingly, enabling or disabling tasks to maintain the satellite's integrity. Besides executing its autonomous algorithm, the OBC can receive telecommands from the TT&C subsystem and execute or forward them to other subsystems. The chosen microcontroller for this subsystem is the STM32, which supports various communication protocols, of which will be used I²C and UART, for reading the sensors and communicating with other microcontrollers, respectively. The state machine implements prioritized tasks to handle sensor data collecting and interpreting, as well as transmitting this information to the TT&C subsystem for eventual downlink to the ground station.

2. Microcontroller

This subsystem will make use of the STM32L476RG, with ultra-low power consumption and rich peripherals for communication. The board is also really good for testing and implementing new codes, as it uses C language, as well as has visual representation on its own application, STM32CubeMX.

3. Buses

For implementing the sensors functioning and the communication between subsystems, the OBC will use different methods.

a) I²C: The sensors will be connected using this method, such as two INA3221 for measuring current, DS3221 to keep track of the time, two BME280s, one for measuring the batteries temperature and the other for the mechanical structure, and two MAX17048, for the battery percentage and voltage. Additionally, the two battery chargers, MP2733, will also be connected on the I²C bus. In order to be able to communicate with multiple sensors with the same I²C address, a Multiplexer was implemented.
b) UART: For the mission, UART communication was chosen between subsystems because it supports various data types and is easy to use with different microcontrollers. The OBC is connected to the TT&C via UART to receive telecommands from the ground station and transmit them to other subsystems when necessary, as well as to send the sensor data collected by the OBC. It is also connected to the ADCS via UART, allowing it to enable the detumbling and receive position information in return. Finally, the PAYLOAD has a UART connection for their RPi module.
c) GPIO: The OBC is responsible for deciding, based on telecommand by Earth and current mode, when the deploy of the antennas will happen. To achieve this, specific GPIO pins on the STM32 bus act as switches: when set to high, they activate the electrical circuit with a MOSFET transistor which starts the deploy, and other GPIO pins read if the circuit is still intact or, in case the deploy successfully happened, is open.

Subsystem	Device	Type	Name	Quantity	Communication
EPS	Current/ Power Monitor	Sensor	INA3221	2	I2C1
	Voltage/ Battery Percentage Monitor	Sensor	MAX17048	2	I2C1
	Battery Charger	Other	MP2733	2	I2C1
OBC	Time Monitor	Sensor	DS3231	1	I2C1
TCS	Temperature Monitor	Sensor	BME280	2	I2C1
PAYLOAD	ADS-B Raspberry Pi	Other	RPi Zero W	1	USART3

Table 1: Communication with the Components Interface

4. Sensors

All the data collected from the sensors is transmitted on the downlink telemetry package, made by the Transmit task when the Housekeeping task is done.

a) BME280: Temperature sensor for the satellite's thermal control. Two of these sensors will be connected to the board, one for the satellite's external mechanical structure and another for the battery temperature

measurement. Different I²C addresses were assigned to each sensor by soldering the SDO port. These sensors are used in the decision-making process for state transitions, deciding if the thermal conditions are safe.

b) MAX17048: A library was implemented to use the MAX voltage sensor for voltage measurement and calculation of the battery charge percentage (State of Charge, SoC). Two main functions were created: MAX17048_GetVoltage and MAX17048_GetSOC. Through the Housekeeping, this sensor constantly provides information about the satellite's current battery status, which is essential for its safety.

c) INA3221: Both INA sensors will be fixed on the EPS PCB for reading current values and shunt voltage from 5V and 13.8V lines, and current from each solar panel. They will be connected to the OBC board using the shared buses between subsystems. The sensors will be read simultaneously, thus two different addresses are used by soldering the A0 port to GND and VCC.

d) DS3231: This sensor will have the simple task of keeping track of time so it will be easy for the base and subsystems to catch the time to send their data, or even to identify which was the time some error was committed. The team is studying the possibility of setting the time as the actual time of the mission, or starting by resetting the sensor.

e) MP2733: Despite not being a sensor itself, the MP2733 battery charger is used in a very similar way to them. A library was created to implement the necessary functions for setting the desired charge parameters and read the current status. This status reading occurs with the sensors readings, and the initialization is also made alongside them.

5. Operation Control

For Operation Control, FreeRTOS was used to manage tasks and functions. A State Machine concept was developed to better organize which tasks should be active during each stage of the mission.

a) Tasks: There are eight tasks in total: Manager, Housekeeping, Receive, Transmit, Deploy, Mission, Detumbling and Error Handling. Table 3 details them further.

Involved Subsystems	Task Name	States in which the Task is executed	Priority	Description
GENERAL	Manager	ALL	Normal	Task which handles the State Machine and activates or deactivates other tasks.
GENERAL	Housekeeping	ALL	Low	Reads sensor values and gathers the telemetry data.
OBC, EPS, TT&C	Receiver	ALL	Low	Receives telecommand from the base via TT&C using UART
OBC and TT&C	Transmit	TRANSMIT	Low	Send data from the sensors to TT&C via UART
OBC e ADCS	Detumbling	DETUMBLING	Low	Uses UART to turn the ADCS on for detumbling.
OBC, TT&C, EPS, Payload	Mission	MISSION	Low	Uses UART to activate the ADS-B Module.
OBC, EPS and TT&C	Deploy	DEPLOY	Low	Sends the deploy signal to the electric circuit and

				waits for the successful signal.
OBC, EPS and TT&C (only receiving)	Error	ALL	Low	Tries to reinitialize sensors and turns off other tasks in order to try and correct problems found.

Table 2: Detailing of the Tasks used for the State Machine

b) States: IDLE, TRANSMIT, DEPLOY, DETUMBLING and MISSION. IDLE is the base state, from which any other can be accessed. During it, only the tasks Manager, Housekeeping and Receive are active, working at regular intervals according to a timer, while the other tasks are inactive. When the telemetry from Housekeeping is collected, the state changes to TRANSMIT for the downlink before returning to IDLE. According to the telecommand obtained by Receive, the state can change into MISSION, DEPLOY or DETUMBLING, activating the respective tasks until they are finished and then returning to IDLE.

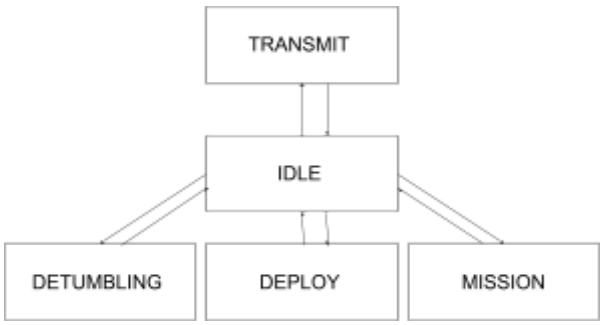


Figure 10: State Machine Diagram

c) Events: FreeRTOS Events work like warnings to the tasks so they can start their operation. Figures 11 a-f shows diagrams of these events and how they activate tasks. It's interesting to note that the BIT_MECHANICS is a way to avoid Deploy and Detumbling at the same time, two energy consuming tasks that cannot be allowed to be active simultaneously.

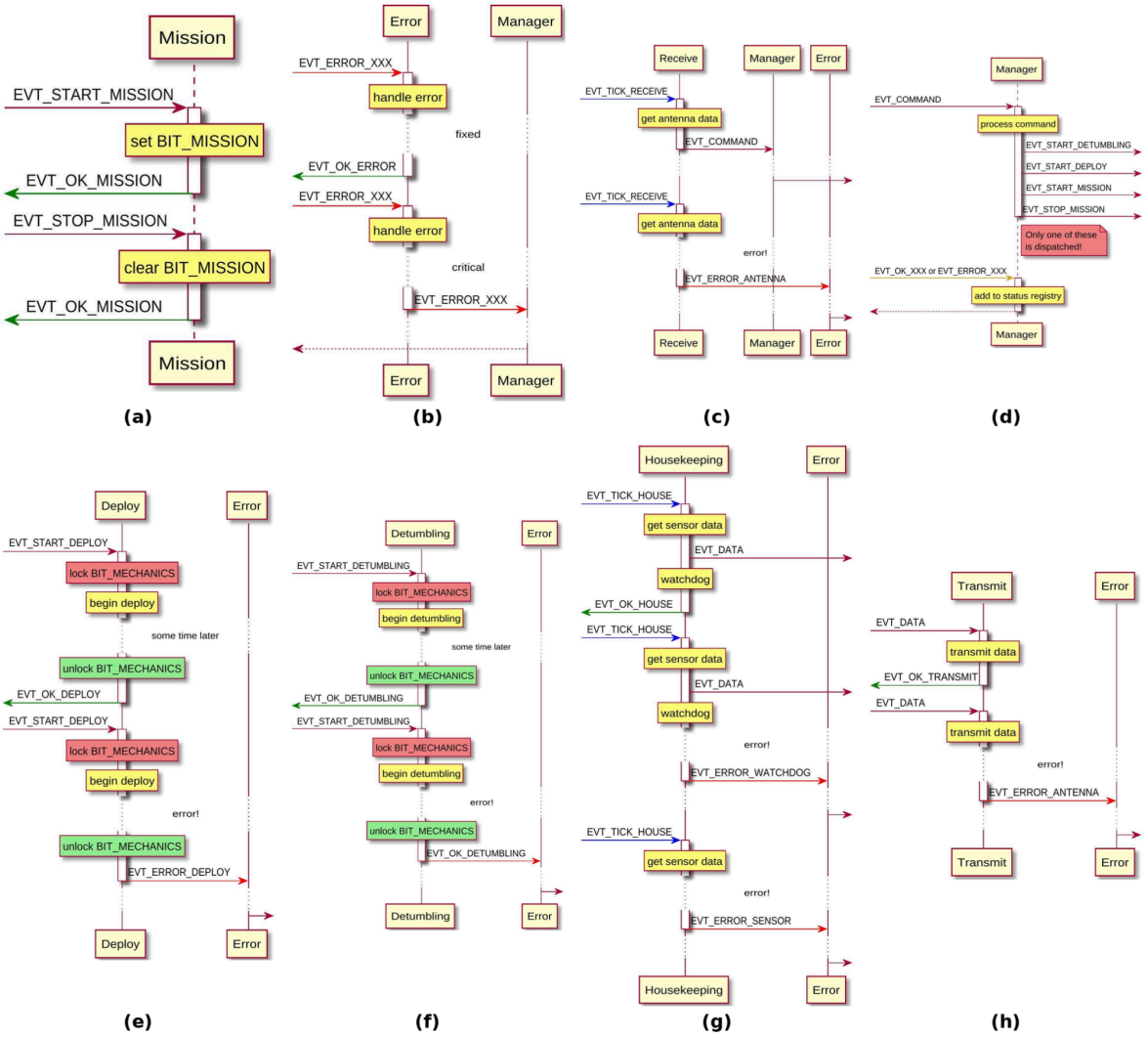


Figure 11: Tasks and Events Diagrams: (a) Mission, (b) Error, (c) Receive, (d) Manager, (e) Deploy, (f) Detumbling, (g) Housekeeping, (h) Transmit.

6. PCB

Regarding the OBC PCB, it was developed fully using the Altium application. Only two of the sensors will be fixed to the PCB, the DS3231 and one BME280 while the others will be on the EPS PCB. Besides, there are the microcontroller, the buses and some complementary elements, necessary for the PCB complete functioning.

C. Telemetry Tracking and Command (TT&C):

The main goal of the Telemetry Tracking and Command (TT&C) is to establish a fast and reliable two-way communication system between the CubeSat and ground station. It connects all CubeSat subsystems to a TT&C's microcontroller, in which data is temporarily stored and processed, then connected to the ground station. The entirety of the operation of TT&C is going to be divided in segments and explained separately. The integration of all segments is going to be explored after explaining each segment.

1. Decision and design of the antenna system

1.1. Communication with ground station

The communication was decided to be half-duplex operating in 433 MHz, meaning that data can only be either received or transmitted at a time. All communication is encoded and sent via UART to a LoRa module, using a LoRa E32-433T20D for both uplink and downlink. To ensure good and reliable communication between

cubesat and groundstation, a Reliable Data Transmission (RDT 3.0) algorithm is going to be used by the team. The implementation of the transmitter and receiver is shown below:

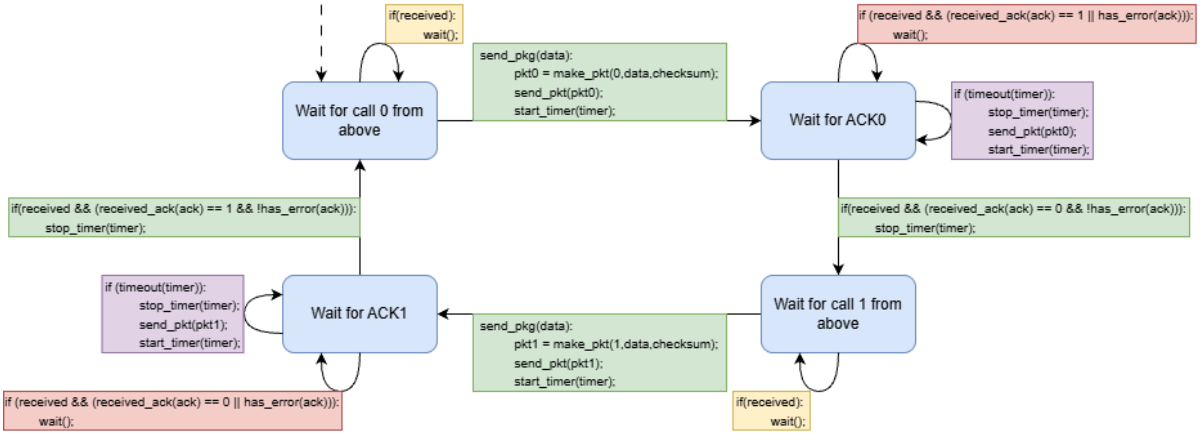


Figure 12: Transmitter's finite state machine design

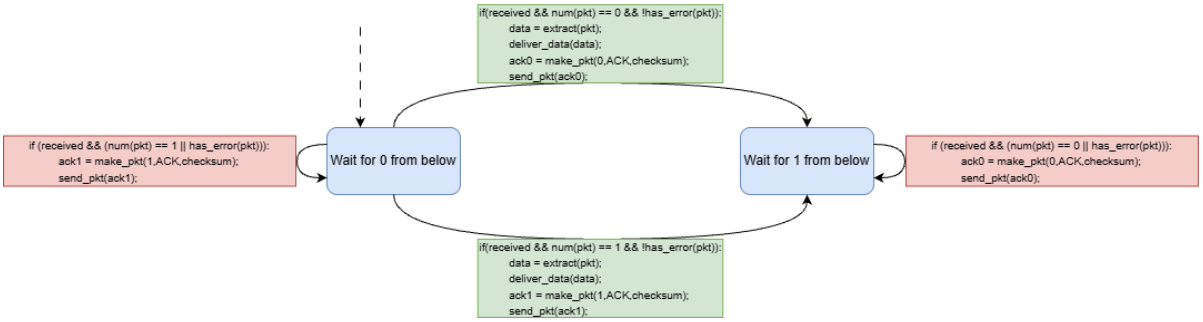


Figure 13: Receiver's finite state machine design

1.2. Communication link design

The selected antenna is a phased array consisting of two λ/4 monopoles, forming a dipole, operating at 433 MHz. The array has a total length of approximately 17.32 cm, with the antennas electrically phased 180° apart (reference Fig. 1).

The resulting radiation pattern is toroidal, with a wide half-power beamwidth, low directivity, and linear polarization. Therefore, precise axial pointing from the satellite's ADCS is not required for reliable communication. However, the ground station antenna should use circular polarization, allowing it to receive the signal effectively regardless of its orientation relative to the satellite.

For a preemptive check if transmission link is possible, Friis transmission equation (in decibel form) can be used:

$$P_R^{[dB]} = P_t^{[dB]} + G_{do}^{[dBi]} + G_{di}^{[dBi]} - FSPL^{[dB]} - A_{Pol}^{[dB]} - \sum_{i=0}^{n} A_i^{[dB]}$$

Where, $P_R^{[dBm]}$ is the power received in the receiver's antenna, $P_t^{[dBm]}$ is the power transmitted from the transmitter's antenna in decibels, $G_{do}^{[dBi]}$ is the directivity gain from the direction of the transmitter's antenna in decibels relative to an isotropic antenna, $G_{di}^{[dBi]}$ is the directivity gain from the direction of the receiver's antenna in decibels relative to an isotropic antenna, $FSPL^{[dB]}$ is the power loss due to Free Space Power Loss effect, $A_{pol}^{[dB]}$ is the power loss due to polarization mismatch and $\sum_{i=0}^{n} A_i^{[dB]}$ is the sum of all the attenuation effects from the atmosphere to medium impedance mismatch.

The biggest attenuation effect is due to the Free Space Power Loss effect and can be calculated as:

$$FSPL^{[dB]} = 20\,log(\frac{4\pi d}{\lambda}) = 140.74\,dB$$

Where $d = 600\,km$.

For the multiple attenuations the team considered the following values:

- **Impedance mismatch (Mismatch loss):** 0.18 dB (VSWR = 1.5)
- **Loss in the connectors (Connector loss):** 0.5 dB
- **Attenuation atmospheric absorption*:** 0.7 dB
- **Attenuation due to rain*:** 0.3 dB
- **Attenuation due to clouds and fog*:** 0.3 dB
- **Attenuation due to atmospheric reflection*:** 0.1 dB
- **Attenuation effect due to scintillations in the troposphere and ionosphere:** 2.7 dB

***Worst-case estimation. Likely much lower, usually negligible.**

$$\sum_{i=0}^{n} A_i^{[dB]} = 4.78\,dB$$

Considering that the LoRa433T20D has a typical sensibility of -146 dB and a transmitting power of 20 dBm, and the power loss due to polarization mismatch is approximately 3 dB, a directional gain of the ground station antenna of 2.15 dB, and a gain of 0 dB of the CubeSat antenna.

The following table shows that the CubeSat to ground station links are well designed and have a decent margin between the power received in both uplink and downlink and respective transceiver sensibilities.

Links of CubeSat to ground station		
Power Factor	Downlink	Uplink
$-A_{pol}^{[dB]}$	-3	-3
$-FSPL^{[dB]}$	-140.74	-140.74
$G_{di}^{[dBi]}$	5.27	2.15
$G_{do}^{[dBi]}$	2.15	5.27
$-\sum A_i^{[dB]}$	-4.78	-4.78
$P_t^{[dBm]}$	20	20
$P_R^{[dBm]}$	**-121.1**	**-121.1**
$S/N^{[dBm]}$	**-146**	**-146**
Margin(dBm)	**24.9**	**24.9**

Table 3: Links of CubeSat to ground station

1.3. Phase shifter circuit

Since the design for the antenna array in the CubeSat requires a phase of 180º between each monopole antenna from the antenna array, it is necessary for a transmission line that connects each antenna and a circuit that can phase 180º each antenna.

To allow for future reuse of the circuit with different configurations (such as 4 monopoles positioned at 90º from each other) the circuit was designed with 4 90-degree-shifted ports. In this scenario, the circuit uses an ADT2-1T+ to phase the outputs by -90º and 90º into 2 different branches, each of which are separated in another 2 branches, one being phased by 90º with the use of a QCN-3+.

The following circuit was created in the simulator NI Multisim and represents the described configuration:

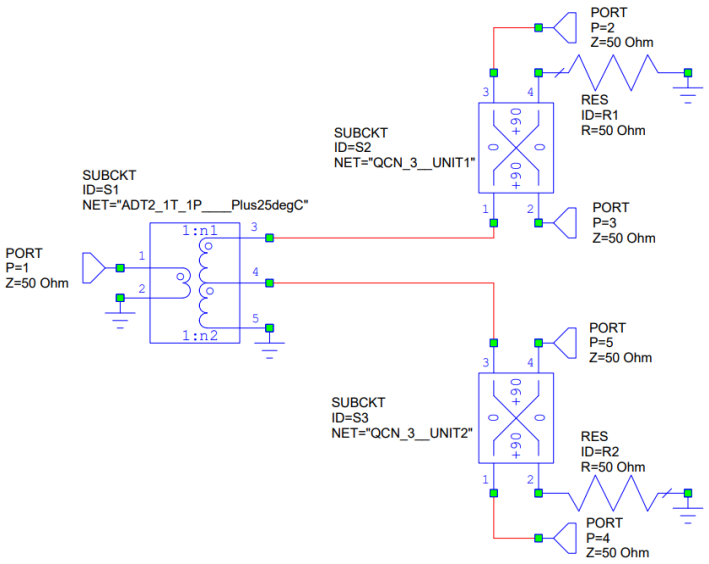


Figure 14: Representation of the circuit phase shifter circuit

In the circuit each port is represented using P=i for the i-th port. Therefore, port 1 is the LoRa433T signal input, which is then divided into 2 other branches using the ADT2-1T+, and divided into 4 ports using 2 QCN-3+. This results each port being phased 90º from each other, which is shown in the following simulation:

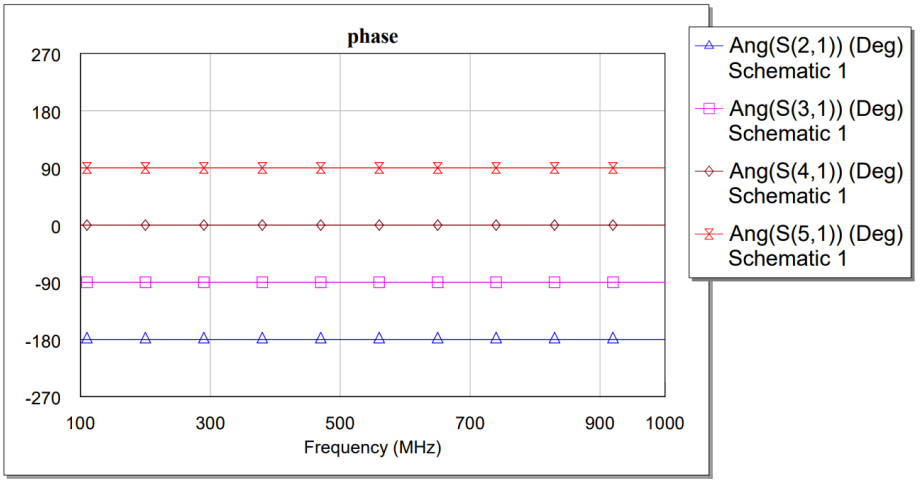


Figure 15: Simulation of the phase shifter circuit

From top to bottom, each line of the plot displays the phase for the port 5, port 4, port 3 and port 2 in relation to port 1, which demonstrates that the circuit is properly working.

***For the purposes of the current implementation, the phase shifters will not be added to the circuit. Instead, the transmission lines of 2 connected ports of opposite branches (such as 2 and 4, or 3 and 5) will be bridged across the pads for the shifters, creating a direct connection to the ADT2-1T+ Balun. The solder bridges will be made according to the correct trace width, as to keep the correct 50 ohm impedance.**

1.4. Transmission Line

In order to connect the entire circuit, the team has been working in developing its own microstrip transmission line, the line is being simulated in both Advanced Design System (ADS) and in CST Studio Suite. In order to create the full transmission line, the team simulated the lines of 50 Ω in ADS. For the dielectric the team used FR4 with relative electrical permittivity of 4.14 and height of 1.524 mm.

For the transmission line of 50 Ω, the following results were obtained:

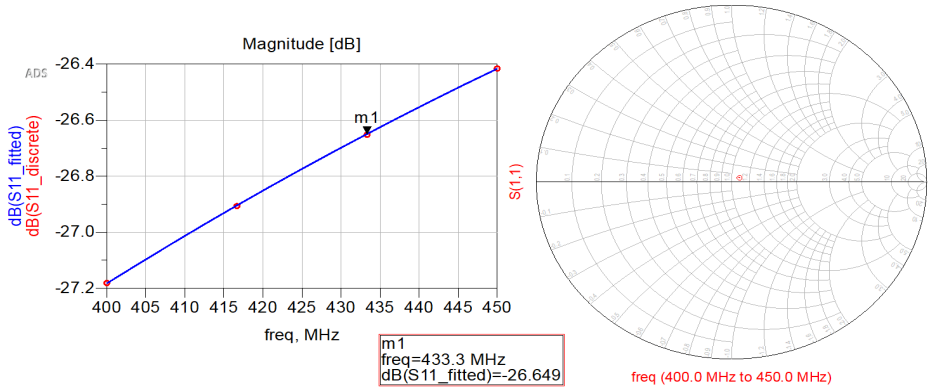


Figure 16: Reflection coefficient of 50Ω transmission line

These results permitted the team to develop the full transmission line in CST Studio to represent the phase shifter circuit, even though the team was unable to simulate the full transmission line due to licensing issues with CST studio. The following image is a layout of the transmission line, similar to the one used in the simulations.

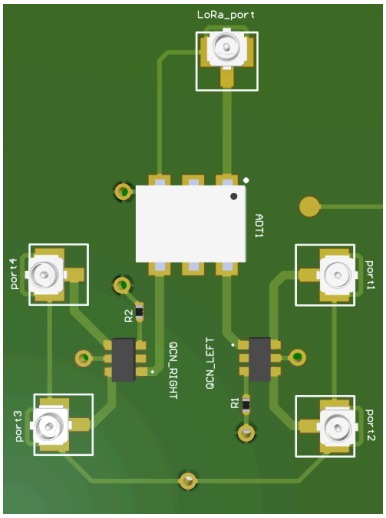


Figure 17: Full transmission line

2. *Ground Station RF Reception and Hardware Architecture:*

To close the TT&C downlink and ingest the processed ADS-B telemetry stream transmitted from the CubeSat, the ground station utilizes a dedicated physical receiver and hardware decoding chain:

- QFH 435 MHz Antenna:The ground station employs a Quadrifilar Helix (QFH) antenna tuned to 435 MHz. Featuring a wide beamwidth and omnidirectional radiation pattern with circular polarization, the QFH antenna ensures reliable reception of signals across varying satellite elevation passes without requiring complex mechanical tracking rotators. It is made from regular copper wire, wound into a 3D-printed guide, soldered onto a PCB with X-shaped traces, and connected at the center to a regular Rg58 cable, with a quarter-wave choke balun sleeve.
- LoRa Receiver Module:The QFH antenna feeds directly into a LoRa transceiver module (operating at 435 MHz), which demodulates the incoming radio frequency packets sent via the satellite's TT&C subsystem. The LoRa physical layer provides robust link margin and noise immunity suitable for receiving data packets under high-interference terrestrial conditions.
- UART-to-USB Bridge and Laptop Ingestion:The demodulated data bytes from the LoRa module are passed through a UART-to-USB bridge converter module. This interface translates serial logic levels into standard USB protocol, allowing the telemetry stream to interface directly with a laptop computer running the ground station processing pipeline.

Parameter	Specification
Standard Operating Frequency	435MHz (can shift in construction, close to the expected 433MHz link)
Antenna Type	Quadrifilar Helix (QFH)
Polarization	Circular (RHCP / LHCP compatible)
Beamwidth	Wide omnidirectional coverage (hemispherical)
Impedance	50Ω
Gain @ 435MHz	~5dB
Interface Chain	QFH Antenna → LoRa Receiver → UART-to-USB Bridge → Ground Station Laptop

Table 4: Ground Station Antenna Parameters

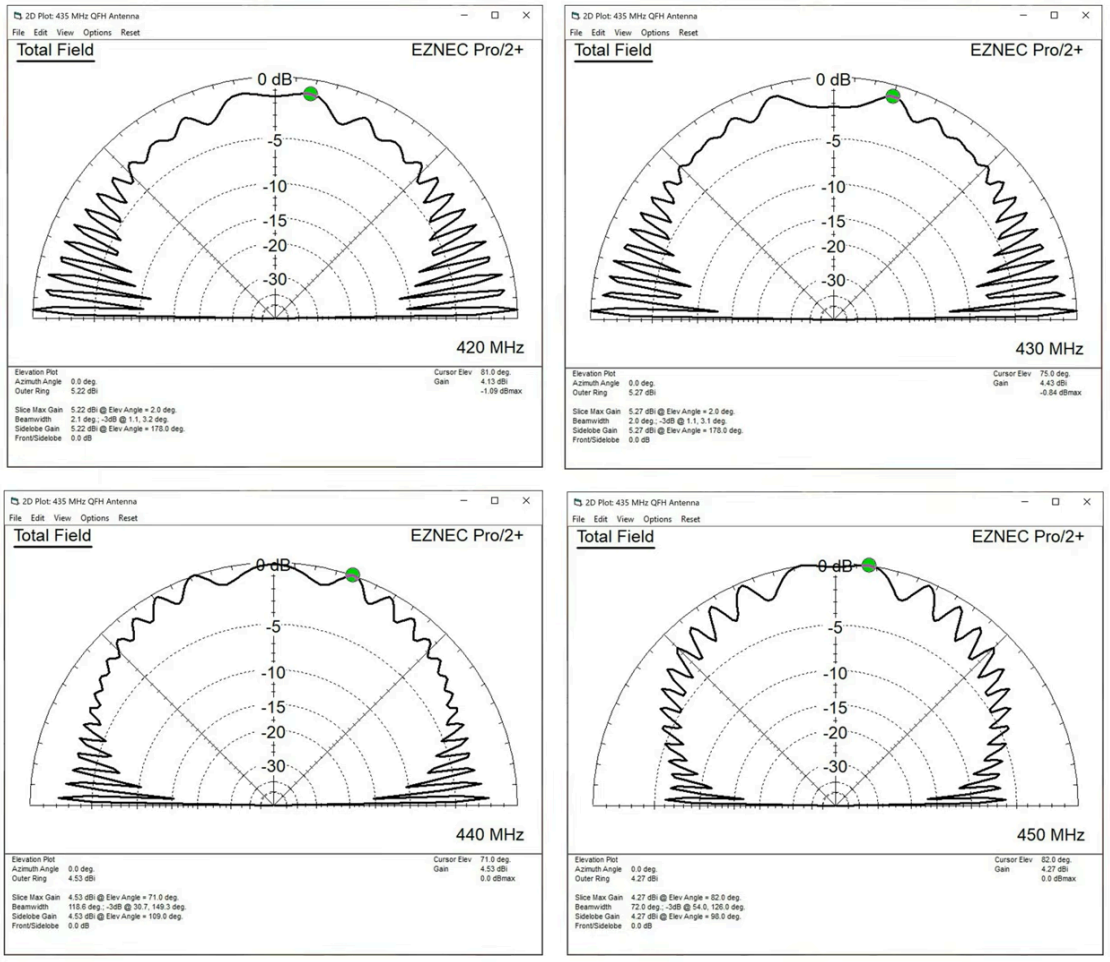


Figure 18: Simulated slices of radiation pattern and gain directivity of the 435 MHz QFH ground station antenna.

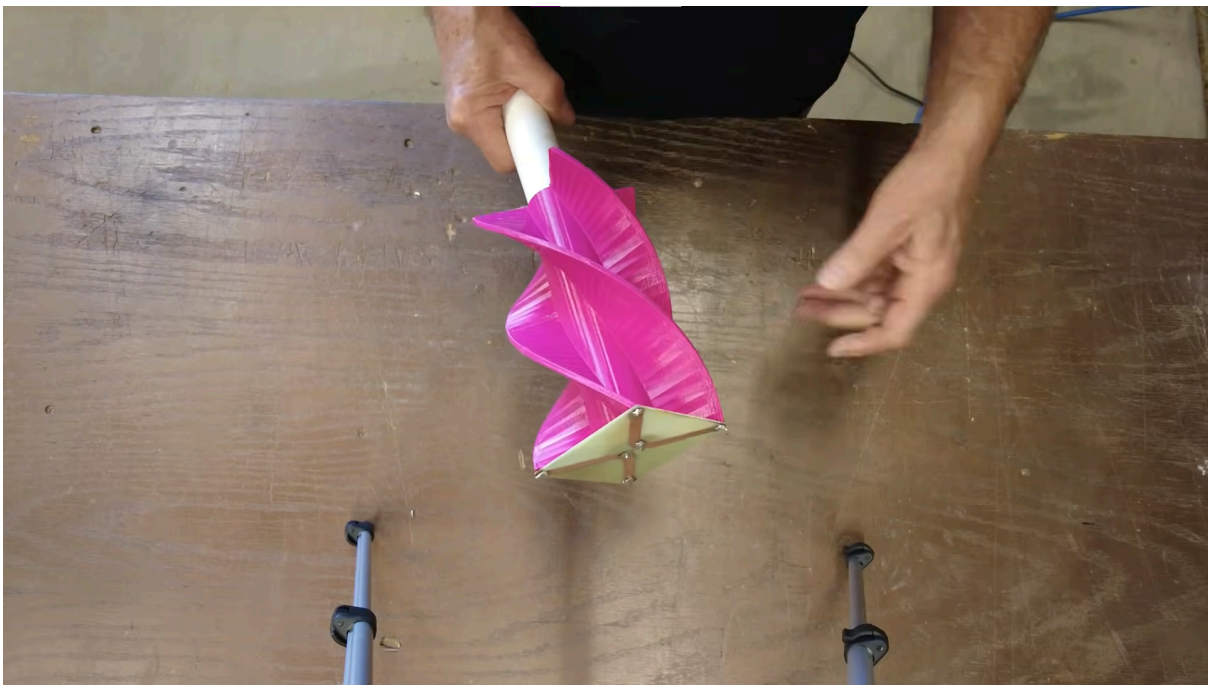


Figure 19: Assembled QFH ground station antenna prototype.

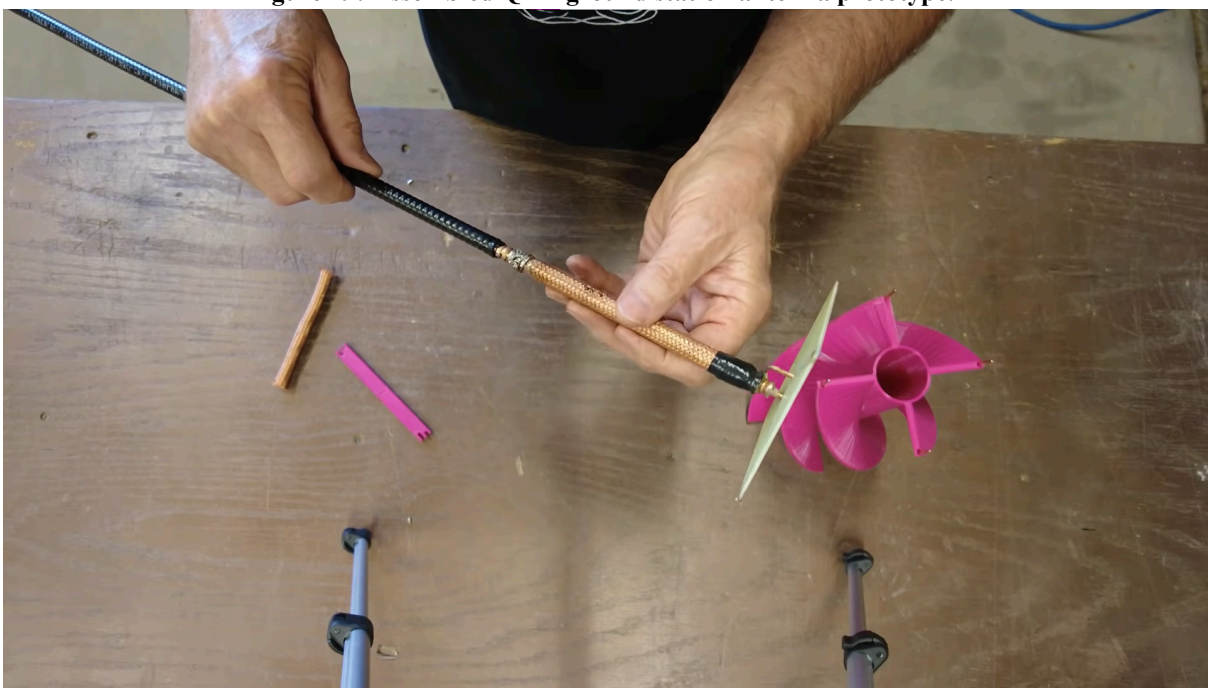


Figure 20: Quarter-wave copper sleeve choke balun

The design was first developed by David Gilbert on his YouTube Channel[1], and later implemented by the PoliSat team as the chosen alternative for communication.

3. Data encoding

To decrease the amount of non useful information sent, the team developed a method of encoding and decoding data. In order to accomplish this, the team uses metadata to encode the data in a way that the information encoded is represented in the least amount of bits possible while being reliable and easy to decode. Both the data and the metadata will be sent in UART to TT&C from the OBC subsystem, however, the metadata of the information must arrive before the information itself since it will allow the ESP32 to determine the length of the transmission.

This encoding of data is made possible using an ESP32 microcontroller module, in which, no matter the type of information or from which subsystem it is from, after being encoded, the encoded message can be be safely stored in an Unsigned Int variable of 32 bits in the ESP32 controller. Multiple messages can be stored at the

same time, acting as a buffer for transmission. Each message represents a package in the RDT communication previously shown above.

Encoded data can be represented with a prefix of a total of 7 bits, the first 4 represents the subsystem from which the information originates, the 3 next bits represent which type of data it holds, which is followed with the data represented as an integer format. This way, while having prefix sizes prone to change due the increases in the amount of different types of information, it allows for float values and negative values to be represented in rather a simple way, and allows telemetries and telecommands to be stored using the same format.

The use of metadata is deeply involved in this, a float number, if its precision has already been determined and stored in the receiver's memory, can be represented using similar to scientific notation while only transmitting the number itself without the order of magnitude, which can be used to recreate the data's original value. For example, to represent an angle of 89.55º, with values ranging from 0º to 360º, with a precision of 10^{-4}, the angle can be represented as $89.55º = 895{,}500*10^{-4}$, while a change between positive and negative numbers can be represented with a single bit.

Using this method allows the reduction of the amount of bits per data sent quite significantly, especially in data that can be represented with few decimal places, allowing for a much faster data transfer rate. For larger payloads like the ADS-B Extended Squitter messages (which span 56 bits for the Message Extended field), the data is chunked across sequential 32-bit words. This also allows the use of multiple bits to ensure the message has error protection by using previous bits that would represent data to dedicate to parity, which is possible since the maximum bits used in standard packages (excluding multi-word frame components) use around 22 to 32 of a total 32 bits (4 bytes).

The following tables describe the chosen encoding for each message:

Encoding prefixes		
Area	**Area prefix**	**Number of bits**
ADCS	"1000"	4
PLD	"1001"	4
EPS	"1010"	4
TCS	"1011"	4
OBC	"1100"	4
TT&C	"1101"	4
	Number of bits to represent datatypes	3
	Total prefix bits	7

Table 5: Encoding prefix per subsystem

ADCS					
Datatype	**Name**	**Telemetry or Telecommand**	**Data max value**	**Precision**	**Number of bits per data**
"000"	Send attitude	Telemetry	180	e-4	3 * 22
"001"	Stabilize azimuth	Telecommand	1	1	8
"010"	Control attitude	Telecommand	180	e-4	3 * 22
"011"	-	-	-	-	-

ADCS					
"100"	**-**	**-**	-	-	**-**

Table 6: Data encoded of ADCS

PLD					
Datatype	**Name**	**Telemetry or Telecommand**	**Data max value**	**Precision**	**Number of bits per data**
"000"	Downlink Format (DF)	Telemetry	31	1	5
"001"	Capability (CA)	Telemetry	7	1	3
"010"	ICAO (Address)	Telemetry	16777215	1	24
"011"	Message Extended (ME)	Telemetry	2^{56}-1	N/A	56 (chunked)
"100"	CRC / Parity (PI)	Telemetry	16777215	N/A	24
“101”	Reception Timestamp	Telemetry	2^{32}-1	1ms	32
“110”	Begin Reading	Telecommand	1	1	8

Table 7: Data encoded of PLD

EPS					
Datatype	**Name**	**Telemetry or Telecommand**	**Data max value**	**Precision**	**Number of bits per data**
"000"	Voltage 3.3v	Telemetry	4	e-3	12
"001"	Current	Telemetry	|100|	e-3	18
"010"	Battery percentage	Telemetry	100	1	7
"011"	-	-	-	-	-
"100"	-	-	-	-	-

Table 8: Data encoded of EPS

TCS					
Datatype	**Name**	**Telemetry or Telecommand**	**Data max value**	**Precision**	**Number of bits per data**
"000"	Battery Temperature	Telemetry	|50|	e-1	10

TCS					
"001"	External Temperature	Telemetry	|50|	e-1	10
"010"	-	-	-	-	-
"011"	-	-	-	-	-
"100"	-	-	-	-	-

Table 9: Data encoded of TCS

OBC					
Datatype	**Name**	**Telemetry or Telecommand**	**Data max value**	**Precision**	**Number of bits per data**
"000"	Time	Telemetry	219660	1	18
"001"	Date	Telemetry	2025	1	3*11
"010"	Power on Subsys	Telecommand	255	1	8
"011"	Deploy	Telecommand	255	1	8

Table 10: Data encoded of OBC

TT&C					
Datatype	**Name**	**Telemetry or Telecommand**	**Data max value**	**Precision**	**Number of bits per data**
"000"	ACK_rdt	Telemetry	255	1	8
"001"	general_telemetry	Telemetry	255	1	8
"010"	general_telecommand	Telecommand	255	1	8
"011"	**-**	**-**	-	-	**-**
"100"	**-**	**-**	-	-	**-**

Table 11: Data encoded of TT&C

To guarantee that the communications with the subsystems of the CubeSat and the ground station don't result in data collisions, the code of ESP32 will use its dual-core capabilities, using 1 core to interface with the subsystems and another core to interface with the LoRa433T module and ground station.

D. Electrical Power Supply (EPS):

1. Introduction

Satellites are complex electromechanical systems, in which each subsystem is composed of several individual components. From sensors and microcontrollers to electric motors and other types of actuators, all devices onboard the satellite need electrical energy to perform their functions, and adequate power supply is essential to guarantee the expected reliability and performance. Thus, the Electrical Power Supply (EPS) plays a crucial role in generating, distributing, and regulating electrical energy to meet the satellite's various energy needs throughout its mission.

Energy is a resource as important as it is scarce in the space environment. As there is no centralized energy generation and distribution infrastructure in space, satellites must be able to generate and store all the energy

necessary for their operation. Currently, there are two main methods of energy generation in space applications, one of which is solar panels.

Solar panels are arrays of photovoltaic cells attached to the surface of satellites or to external structures. The incidence of solar radiation on the panels causes the excitation of electrons and, consequently, the generation of an electrical voltage between the terminals. As advantages, scalability stands out, as they can be used both in large satellites, such as communication satellites and the International Space Station (ISS), and in smaller satellites, such as CubeSats. The main disadvantages of solar panels are their dependence on solar illumination, which is subject to interference from external factors, such as periods of darkness due to light obstruction, and also the need for batteries or other forms of energy storage, so that the satellite remains operational even during these intervals. Radioisotope decay generation systems, on the other hand, explore the temperature difference between space and the heat generated by the decay of radioactive elements to obtain electricity. They are typically used in larger satellites and in orbits further from the Sun, where the incidence of energy per unit area is lower, thus substantially limiting the effectiveness of solar panels.

In view of the characteristics of both generation methods presented, it is worth noting that solar panels are currently the most employed method, especially in CubeSats, which typically operate in low Earth orbit, where luminous energy is satisfactory for the use of solar panels, and also due to limited space, as the panels can be attached to the external structure without compromising large volumes in the internal part of the satellite. As mentioned earlier, however, the use of solar panels is accompanied by some limitations, the most relevant of which is dependence on solar illumination. As a satellite's orbit exposes it to varying periods of light and darkness, it requires a means of accumulating excess electrical energy produced during sunlit phases for use during eclipses or periods when the satellite is in the Earth's shadow. For this, batteries or other forms of storage, such as supercapacitors, for example, are commonly used, in order to be charged when there is abundant solar energy and discharged when additional energy is needed or during the orbital night.

In addition to generation and storage, power distribution is also a central responsibility of the EPS. The conditioning of energy in regulated buses is essential so that the various components of the satellite can operate under the conditions specified by the manufacturers and from the same energy source. Furthermore, the supply of "higher quality" energy, that is, adjusted to the needs of each component, is essential to ensure the reliability and predictability of the systems, so that they always operate in the same way and with a reduced chance of failure. In view of this, it is not advisable to power devices directly from the batteries, since the voltage between the terminals normally does not correspond to the voltage level indicated for all components, in addition to being able to vary with the load. Consequently, the use of voltage converters is necessary, capable of providing a regulated output signal at a specific voltage level from an input whose voltage can vary within a known range.

Redundancy and circuit protection are factors that must be considered in the design of the EPS. In general, systems operating in a space environment must be highly reliable, given the difficulty, cost-ineffectiveness, and even impossibility of repairing satellites after launch. With EPS it is no different; as a critical support subsystem for the operation of all other subsystems, it is essential to implement measures for circuit protection, in order to prevent failures, as well as redundancies, with the aim of mitigating their severity. In addition, monitoring the satellite's consumption, as well as fault detection, are tools that add to both operational reliability and satellite control.

2. Requisites

EPS’s project took into consideration mainly the requisites for satellite installed components powers and voltages, by elaborating a *Power Budget* spreadsheet, in which each subsystem listed the devices that would be employed in the mission. Then, according to each component datasheet, a listing of maximum, average and idle power consumption was done per component. After that, Table 14 was elaborated, which includes data for current, voltage and per-subsystem maximum and average power consumption.

Then, in collaboration with the Onboard Computer subsystem, a listing of components that are active for each operational mode was made, including average and peak power consumption. With that, the table 15 was made, including total average and peak power consumption per state and for the whole satellite.

After power budget analysis, it was concluded that just three voltage converters were necessary, to 5V, 3.3V and 13.8V. The 5V and 13.8V converters were boost-type DC-DC converters and the 3.3V converter was a LDO. For transmission, the subsystem reserved a few pins on the main board connector and tried to even the load, adding two GND and 3.3V connections for different subsystems.

Subsystem	Component	Qty.	Voltage (V)	Idle (mA)	Average (mA)	Peak (mA)	Idle (W)	Average (W)	Peak (W)	Total Idle (W)	Total Active Average (W)	Time active (h/day)	Average Power (W)
ADCS	ESCON 24/2	1	13,6	94	393,48	1666	1,2784	5,43	22,6576	1,2784	5,43	0,0027	0,0063
	MPU-6050	1	5	0	0	3,5	0	0,02	0,02	0	0,02	0,0167	1
	Arduino Nano	1	5	15	15	19	0,075	0,095	0,095	0,075	0,095	24	0,095
TT&C	LoRa E32 433T (IDLE and RECEPTION)	1	3,3	3,000	3,000	14	0,0099	0,0462	0,0462	0,0099	0,0462	21,6	0,04158
	LoRa E32 433T (TRANSMISSION)	1	3,3	0,000	0,000	120	0	0,396	0,396	0	0,396	2,4	0,0396
	ESP32 (Generic data)	1	3,3	20,000	20,000	50	0,066	0,165	0,165	0,066	0,165	24	0,165
Payload	Raspberry Pi Zero W	1	5	100,000	100,000	230	0,5	1,15	1,15	0,5	1,15	12	0,575
	RTL SDR V3	1	5	60	60	250	0,3	1,25	1,25	0,3	1,25	24	1,25
TCS	Resistor 3W	1	13,8	0	217,39	217,39	0	3	3	0	3	0,0167	0,0021
OBC	STM32F103RB	1	3,3	32,000	32,000	150	0,1056	0,495	0,495	0,1056	0,495	24	0,495
	INA3221	3	5	0,350	0,350	0,35	0,00175	0,00175	0,00175	0,00525	0,00525	24	0,00525
	BME280	2	3,3	0,004	0,004	0,0036	0,00001188	0,00001188	0,00001188	0,00002376	0,00002376	24	0,00002376
	DS3231	1	5	0,200	0,200	0,2	0,001	0,001	0,001	0,001	0,001	24	0,001
EPS	MAX17048	2	5	0,001	0,001	0,005	0,000025	0,000025	0,000025	0,000005	0,00005	12	0,000025
	INA3221	2	5	0,350	0,350	0,35	0,00175	0,00175	0,00175	0,0035	0,0035	12	0,00175
	MP2733	2	5	1,000	1,000	4	0,005	0,02	0,02	0,01	0,04	24	0,04
	TOTAL									**2,35997876**	**12,05004196**		**3,675844398**

Table 12: Power Budget

Subsystem	Component	Qty.	Voltage (V)	Idle (W)	Detumbling (W)		Mission (W)		Deploy (W)		Transmit (W)	
					Average	Peak	Average	Peak	Average	Peak	Average	Peak
ADCS	MPU-6050	1	5	off	0,0175	0,0175	0,0175	0,0175	0,0175	0,0175	0,0175	0,0175
	ESCON 24/2	1	13,8	off	5,430024	22,9908	off	off	off	off	off	off
	Arduino Nano	1	5	0,075	0,095	0,095	0,095	0,095	0,095	0,095	0,095	0,095
TT&C	LoRa E32 433T (IDLE and RECEPTION)	1	3,3	0,0099	0,0462	0,0462	0,0462	0,0462	0,0462	0,0462	0,0462	0,0462
	LoRa E32 433T (TRANSMISSION)	1	3,3	off	0,396	0,396	0,396	0,396	0,396	0,396	0,396	0,396
	ESP32 (Generic data)	1	3,3	0,066	0,165	0,165	0,165	0,165	0,165	0,165	0,165	0,165
Payload	Raspberry Pi Zero W	1	5	0,5	0,5	0,5	1,15	1,15	0,5	0,5	off	off
	RTL SDR V3	1	5	0,3	0,3	0,3	1,25	1,25	0,3	0,3	off	off
TCS	Resistor 3W	1	13,8	off	off	off	off	off	2,999982	2,999982	off	off
OBC	STM32F103RB	1	3,3	0	0	0	0	0	0	0	0	0
	INA3221	3	5	0,1056	0,495	0,495	0,495	0,495	0,495	0,495	0,495	0,495
	BME280	2	3,3	0,00175	0,00525	0,00525	0,00525	0,00525	0,00525	0,00525	0,00525	0,00525
EPS	DS3231	1	5	0,00001188	0,00002376	0,00002376	0,00002376	0,00002376	0,00002376	0,00002376	0,00002376	0,00002376
	MAX17048	2	5	0,001	0,001	0,001	0,001	0,001	0,001	0,001	0,001	0,001
	INA3221	2	5	0,0000025	0,00005	0,00005	0,00005	0,00005	0,00005	0,00005	0,00005	0,00005
	TOTAL	**22**		**1,066**	**7,5**	**25,055**	**3,665**	**3,665**	**5,065**	**5,065**	**1,265**	**1,265**

Table 13: Power per operational mode

3. Electronic Project

EPS's project was divided into modules, whose composition can be seen on Fig. 21. The first module consists of 3 solar panels distributed along the sides of the satellite. Their structure was made by a collaboration with the Structure subsystem and it contains a few small-sized solar panels, the KXOB121K04TF, in an optimized disposition. The solar panels are wired to the INA3221 voltage and current sensor, for real time monitoring of the panels. After the sensor, they are connected to the next module.

The second module is the battery management system. Its main component is the MP2733, a single-cell charger and Maximum Power Point Tracking (MPPT) to optimize the solar panels power throughput. This chip also monitors battery voltage, current and temperature, as well as solar panel total power. Also, it allows the system to be fed through an external power source and when operated without the battery, the system voltage is regulated at maximum output without focusing on the charging. The system batteries are two 18650-Q30 from Samsung. They have a 3000mAh capacity and have a 3.6V nominal voltage. Another very important component from this module is the State-of-Charge (SOC) chip, the MAX17048G. It tracks the battery charge using the ModelGauge algorithm, eliminating the need to use a current-sensing resistor. The chip also alerts if anything is out of the ordinary, such as overvoltage.

The third and final module is the voltage regulation and distribution system. First, the two batteries and its systems are interconnected by an ideal diode, to minimize power loss. The chip is a LTC4358. After the diode, the voltage is then driven into the two boost converters, chip name LM2735X, that convert the voltage to 5V and 13.8V. Both of the outputs are monitored by an INA3221 sensor. After the sensor, they go straight to the voltage rails for use by other subsystems. The 3.3V rail is powered by a LDO converter, LD1117V33, that is fed

by the 13.8V converter. Its output is not measured, because the board did not have physical space for another sensor and its voltage source is already measured.

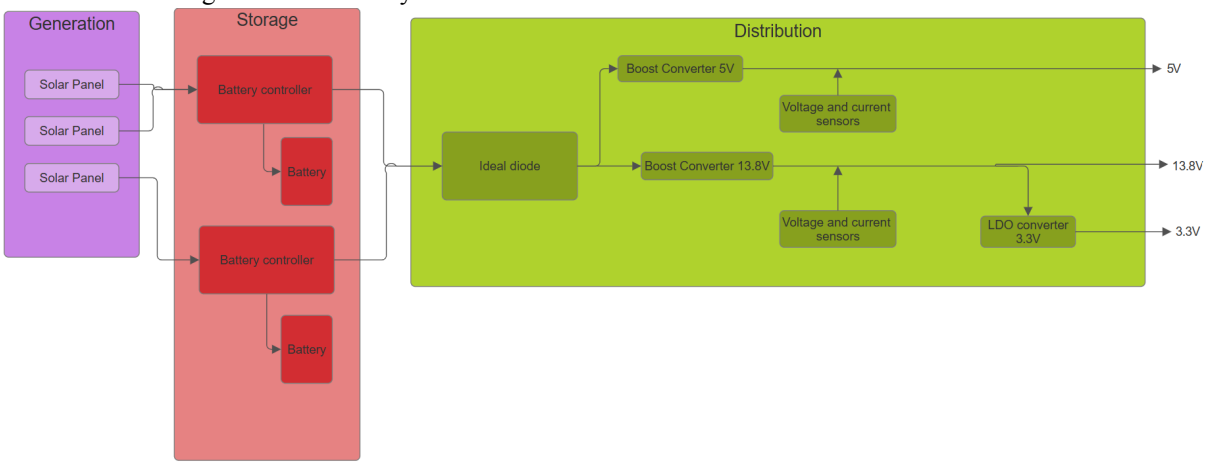
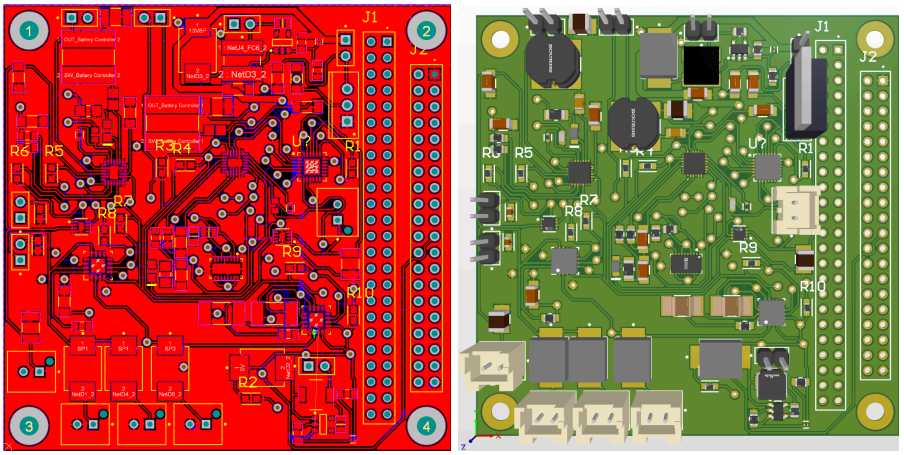


Figure 21: Electronic Block Diagram



a) 2D PCB layout. **b) 3D PCB top-view.**

Figure 22: Different Views of the *Altium* Software PCB Design

E. Payload (PLD):

1. *Main Payload (ADS-B Receiver):*

a) General hardware overview:

The goal of the Payload module is to capture ADS-B signals sent by aircrafts, process it and send the decoded data to the ground station. This is important in adverse environments, where it's not possible to install receiver antennas, such as: mountain regions or open sea; in that scenario, the satellite acts as a transceptor device, responsible for retransmitting the information broadcasted by the aircrafts. Therefore, the hardware chosen for Hermes is:

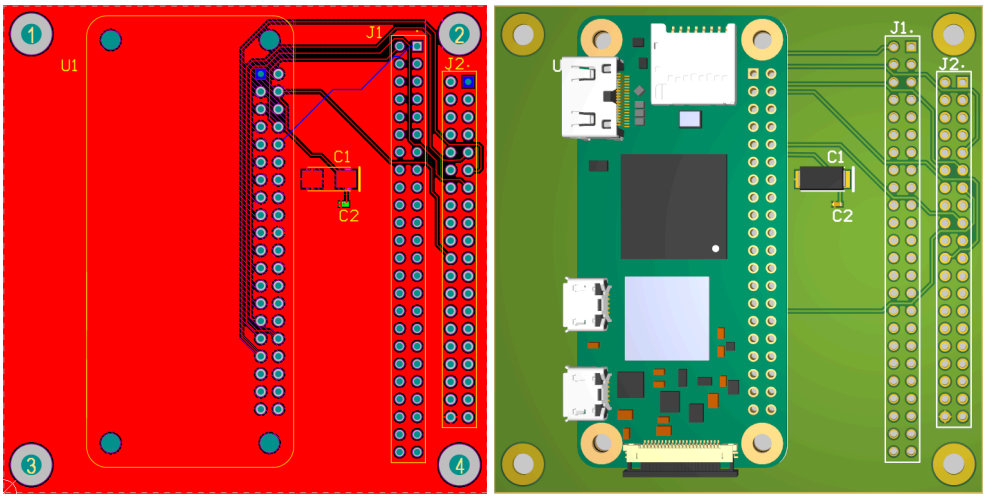
a) ***Raspberry Pi Zero W***: chosen to run the Hermes mission's signal processing algorithm because it possesses one of the lowest power consumptions in the entire Raspberry Pi family, and is also notable for its remarkably small size (66.0mm x 30.5mm x 5.0mm)—essential characteristics for CubeSats. Moreover, the Raspberry interface for developing, programming and debugging is relatively straightforward and there is considerable community support on the internet.

b) ***RTL-SDR V3***: the most popular Software Defined Radio (SDR) on the market, it's a cheap device built from mass produced DVB-T chips that combines a big community support with ease of use. It's responsible for receiving and digitalizing the 1090MHz ADS-B signal, before handling it to the Raspberry.

c) Custom 1090MHz Corner-truncated Patch Antenna: this planar antenna is connected directly to the RTL-SDR and acts as the physical transducer that captures invisible electromagnetic radio waves in the air and converts them into electrical signals that the SDR can process.

To allow for communication with other subsystems, the team developed a PCB that connects the Raspberry Pi to the satellite's general bus, with capacitors to reduce noise and mitigate spikes in power delivery. The GPIO, UART and power pins are connected with headers also placed on the board.

The board's schematics are shown in the appendices, whereas its layout and 3D view are presented below:



a) 2D PCB layout. **b) 3D PCB top-view.**

Figure 23: Payload PCB layout and its 3D representation

The chosen antenna design is a truncated patch antenna with a passthought feed-port. It is printed onto a PCB which will be coupled to the external surface of the cubesat — specifically, the face that'll point towards the ground, aided by the ADCS pointing system.

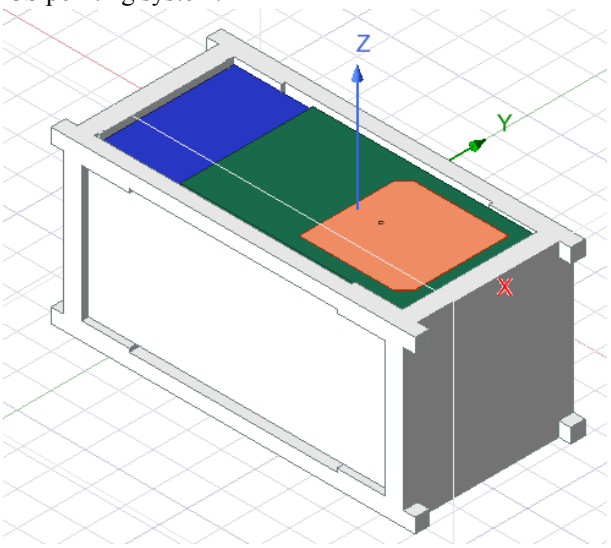


Figure 24: Ansys HFSS model of the antenna and satellite

This model of antenna was chosen because it allows for non-linear polarization, which reduces mismatch loss from the signal received from the aircrafts. The simulated AR for the antenna is < 5.07dB(1.79) for +/- 30°.

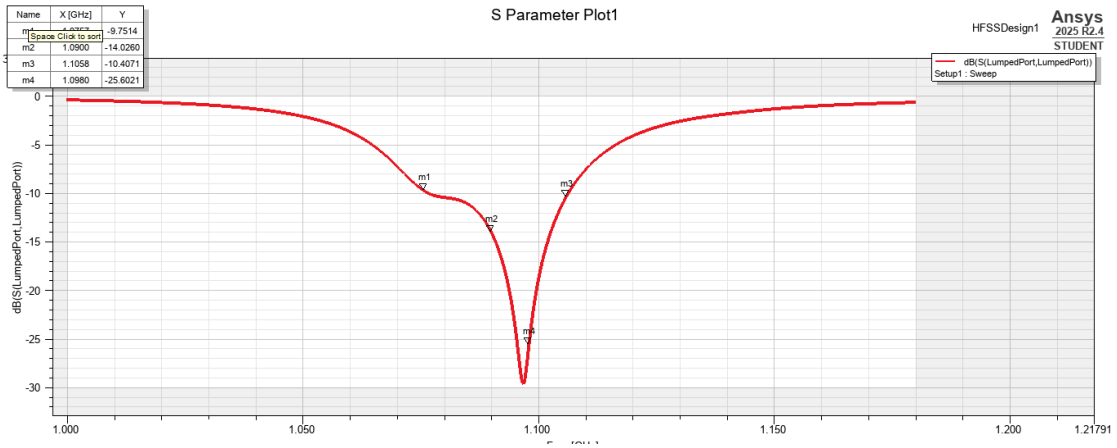


Figure 25: Ansys HFSS graph for S_{11}

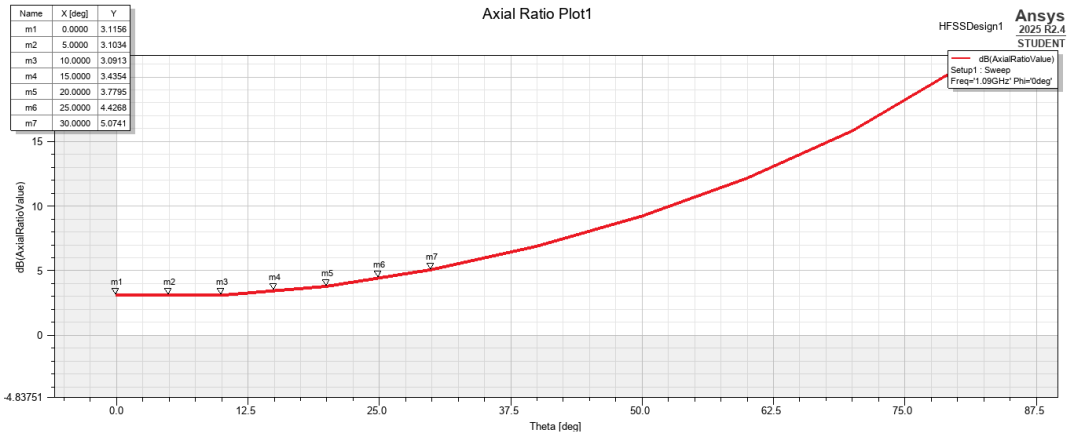


Figure 26: Ansys HFSS graph for the antennas Axial Ratio

In total, with an S_{11} of 14.03dB at 1.09GHz, a Peak Directivity of 6.272dBi and a Radiation Efficiency of -4.827dB (32.9%), the Peak Realized Gain is 1.27dB, with a pattern shown below:

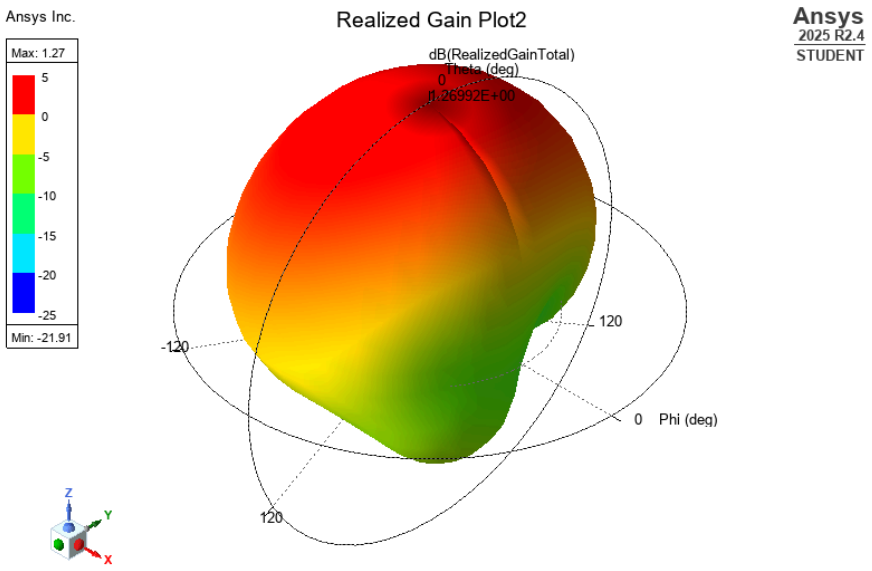


Figure 27: Ansys HFSS's Gain 3D Polar Plot

Using — similarly to TT&C — the Friis transmission equation (in decibel form):

$$P_R^{[dB]} = P_t^{[dB]} + G_{do}^{[dBi]} + G_{di}^{[dBi]} - FSPL^{[dB]} - A_{Pol}^{[dB]} - \sum_{i=0}^{n} A_i^{[dB]}$$

$$FSPL^{[dB]} = 20\,log(\frac{4\pi d}{\lambda}) = 148.75\,dB$$

Where $d = 600\,km$.

For the multiple attenuations the team considered the following values:

- **Impedance mismatch (Mismatch loss):** 0.18 dB (VSWR = 1.5)
- **Loss in the connectors (Connector loss):** 0.5 dB
- **Attenuation atmospheric absorption*:** 0.7 dB
- **Attenuation due to rain*:** 0.3 dB
- **Attenuation due to clouds and fog*:** 0.3 dB
- **Attenuation due to atmospheric reflection*:** 0.1 dB
- **Attenuation effect due to scintillations in the troposphere and ionosphere:** 2.7 dB (~2.0 to 3.0 dB)

***Worst-case estimation. Likely much lower, usually negligible.**

$$\sum_{i=0}^{n} A_i^{[dB]} = 4.78\,dB$$

Considering that the RTL-SDR Blog V3 has a typical sensibility of -133 dB and the power loss due to polarization mismatch (for the worst case, 30°) is calculated below to be ~6.25dBi.

$$PLF = \frac{1}{2}(1 \pm \frac{AR_{linear}^{2} - 1}{AR_{linear}^{2} + 1})$$

For the worst case (30°), we choose the worst alignment axis (smaller elliptical axis):

$$PLF_{min} = \frac{1}{2}(1 - \frac{AR_{linear}^{2} - 1}{AR_{linear}^{2} + 1}) = \frac{1}{2}(1 - 0.5255) \approx 0.2372$$

$$A_{pol}^{[dB]} = -10 \cdot log(PLF_{min}) = -10 \cdot log(0.2372) \approx 6.25\text{dB}$$

The following table shows that the CubeSat to aircraft links are well designed and have a decent margin between the power received and respective receiver sensibilities.

Links of Aircraft to CubeSat (worst case @ 30°)	
$-A_{pol}^{[dB]}$	-6.25
$-FSPL^{[dB]}$	-148.75
$G_{di}^{[dBi]}$	1.27
$G_{do}^{[dBi]}$	~2
$-\sum A_i^{[dB]}$	-8.14
$P_t^{[dBm]}$	51 (estimated ~125W)
$P_R^{[dBm]}$	**-105.51**
$S/N^{[dBm]}$	**-133**
Margin(dBm)	**27.49**

Table 14: Links of CubeSat to ground station

b) General software overview:

(1) RPi PLD:

The software for the satellite's main payload is structured around a receiver program based on the known software “dump1090”, an open-source, lightweight Mode S signal decoder optimized for Software Defined Radio (SDR) front-ends. Because space operation on a CubeSat imposes severe constraints on power budget and thermal dissipation, the software is written natively in C and tuned to operate within the limits of a Raspberry Pi Zero W single-board computer. The hardware stack consists of an RTL-SDR V3 USB dongle coupled with a custom 1090 MHz corner-truncated patch antenna mounted to the nadir-facing face of the satellite and oriented via the ADCS pointing

system. The pipeline operates as a real-time stream processor: it interfaces with the SDR over USB, ingests continuous RF samples, performs preamble matched-filtering, demodulates physical-layer frames, parses high-level payload fields, and queues structured telemetry for downlink.

Bit	No. bits	Abbreviation	Information
1–5	5	DF	Downlink Format
6–8	3	CA	Transponder capability
9–32	24	ICAO	ICAO aircraft address
33–88 (33–37)	56 (5)	ME (TC)	Message, extended squitter (Type code)
89–112	24	PI	Parity/Interrogator ID

Figure 28: Structure of ADS-B message[2]

The ADS-B software pipeline is organized into three primary operational modules:

1. Signal Acquisition and Demodulation:

This module manages lower-level driver interactions with the RTL-SDR V3 tuner and handles initial DSP conversion steps. The radio front-end is configured to a center frequency of 1090 MHz with a fixed quadrature sampling rate of 2.0 MS/s (2.0 MHz). Continuous baseband streaming via asynchronous USB bulk transfers generates interleaved 8-bit unsigned I/Q (In-phase and Quadrature) raw sample blocks.

2. Preamble Detection and Bit Decoding:

The demodulated amplitude array is evaluated sequentially to identify valid Mode S transmissions. A standard Automatic Dependent Surveillance-Broadcast (ADS-B) frame begins with an 8 µs preamble composed of four distinct high-amplitude pulses situated at 0.0 µs, 1.0 µs, 3.5 µs, and 4.5 µs relative to the start of the frame. The engine calculates an adaptive threshold based on the running average noise floor and local peak power. A preamble match is declared only when all four pulse positions exceed the noise floor threshold, match expected relative power levels, and the quiet intervals between pulses remain below prescribed energy levels.

3. Payload Parsing and Error Correction:

Once a 112-bit binary frame is reconstructed, it undergoes rigorous validation before storage. The frame carries a 24-bit Cyclic Redundancy Check (CRC) field generated by a fixed polynomial defined in ICAO Annex 10. The software computes the syndrome matrix across the first 88 bits and compares it against the received CRC bits. Frames with zero errors are passed directly; frames with isolated single-bit errors undergo a single-bit error correction algorithm to maximize data yield in high-noise space environments.

Validated frames are categorized by their Downlink Format (DF). The payload focuses primarily on DF17 and DF18 Extended Squitter messages. By parsing the 5-bit Type Code field within the message payload, the software extracts specialized telemetry sub-types:

1. **Aircraft Identification**: Transmits the callsign and 24-bit ICAO mode S transponder address.
2. **Airborne Velocity**: Delivers East-West/North-South velocity vectors, airspeed, vertical rate, and heading.
3. **Airborne Position**: Encodes barometric/GNSS altitude and position using Compact Position Reporting (CPR). Because CPR compresses spatial coordinates into 17-bit odd and even frame formats to conserve bandwidth, a single message cannot solely resolve global latitude and longitude without prior reference. To optimize onboard resource utilization and make use of ground station computing capabilities, the payload package bypasses onboard global CPR coordinate resolution. Instead, validated frames are stamped with precise onboard timestamps, packaged into lightweight binary records, and streamed directly via a UART pin (USART3) to the OBC subsystem, which forwards them via TT&C to the ground station for global coordinate decoding, multi-position movement verification, and final analysis.

(2) Ground Station Processing and Calculation Pipeline:

Because the satellite payload operates strictly as a lean, real-time stream processor, forwarding raw timestamped Mode S frames via UART and TT&C without heavy onboard compute overhead, the ground station computer handles the resource-intensive analytical calculations. Upon receiving the downlinked telemetry stream, the ground station execution pipeline performs the following computational steps to format and verify the data for final challenge submission:

- Packet Ingestion and Frame Sorting: The ground receiver script ingests the incoming TT&C data stream, parses the binary packet structures, and sorts incoming messages by their 24-bit ICAO transponder addresses. Messages are segregated into dynamic tracking buffers based on

their Downlink Format (DF) and Type Code (TC) headers (identifying them as identification, airborne velocity, or position frames).

- Global Compact Position Reporting (CPR) Decoding: To resolve global geodetic coordinates from the compressed ADS-B data, the ground computer executes global CPR algorithms. Because a single position message contains only local fractional coordinates, the algorithm scans the historical buffer for a valid pairing of consecutive odd and even frame formats received within the active window. Once both frames are matched for a given ICAO address, the computer computes the absolute latitude and longitude in decimal degrees.
- Unit Conversion and Scale Normalization: The ground pipeline converts raw internal register values into the strict measurement units demanded by the competition guidelines:
    - Altitude: Barometric or geometric height registers are converted and scaled into feet.
    - Velocity: East-West and North-South velocity components or raw speed bits are computed into horizontal knots.
    - Heading/Track: Directional vectors are normalized into ground track angles ranging from 0° to 360°.
    - ICAO Address: The 24-bit integer values are formatted into standard six-character hexadecimal strings.
- Trajectory Reconstruction and Movement Tracking: To satisfy the mission objective of tracking targets and reconstructing trajectories in remote environments (such as oceans, deserts, or mountain regions), the ground station script queries its time-indexed logs. It selects and validates multiple position records per aircraft, pairing each coordinate set with its precise reception timestamp to estimate movement over time.
- Final Submission Table Generation:Within the strict post-round deadlines, the ground pipeline aggregates the decoded, converted, and timestamped records into clean, structured output tables containing:
    - ICAO Address (6 hex chars) Aircraft Identification (simulated registration/callsign)
    - Latitude and Longitude (decimal degrees)
    - Altitude (feet)
    - Ground Speed (knots)
    - Ground Track (degrees)
    - Timestamp (reception time for each record)

System Architecture Justification: In accordance with the CubeDesign ADS-B mission parameters for remote air surveillance, which requires handling up to 20 aircraft simultaneously across continuous mission intervals, the system architecture optimizes reliability through a hybrid-streamlined strategy.

- Payload Role (Capture & Stream): The onboard payload software acts as a lean stream processor, handling physical-layer tasks that must execute locally, such as RTL-SDR USB interfacing, 1090 MHz preamble matching, 24-bit CRC error correction, and binary packet timestamping.
- Ground Station Role (Heavy Computing): Resource-intensive tasks, such as global CPR decoding, trajectory reconstruction, and multi-position history tracking, are fully offloaded to the ground station computer. This prevents buffer overflows on the OBC/UART link, ensures robust performance within space constraints, and cleanly satisfies the mission's operational objectives.

2. *Secondary Payload (ADS-B anomaly analiser* ****Theoretical only. Not implemented****):*

a) Overview and Objectives:

The secondary payload is intended to monitor received ADS-B messages and flag data that may be inconsistent or unreliable. It runs on the Raspberry Pi Zero W alongside the primary ADS-B receiver, which uses the RTL-SDR V3 to receive aircraft transmissions at 1090 MHz.

The anomaly-detection software evaluates decoded messages using three types of information: the consistency of reported aircraft motion, the integrity of received messages, and the timing of successive transmissions. These checks are intended to identify unusual data patterns without requiring direct access to the satellite's orbital position.

The main objectives are to:

- Identify ADS-B messages or aircraft tracks that warrant further verification.
- Reduce unnecessary downlink traffic by transmitting relevant anomaly information instead of repeatedly sending flagged data.
- Run within the processing and memory limits of the Raspberry Pi Zero W without interfering with the primary receiver.

b) AI Architecture & Raspberry Pi Zero W Optimization:

The secondary payload runs concurrently with the primary ADS-B receiver on a Raspberry Pi Zero W, which uses a 1 GHz ARM11 processor and has 512 MB of RAM. Since both tasks share the same computing resources, the anomaly-detection pipeline must be designed to limit CPU and memory usage. Computationally

intensive models could compete with the primary SDR processing pipeline for processing time, potentially increasing message-processing latency or causing received frames to be dropped.

To address this constraint, the proposed architecture combines a lightweight Isolation Forest model with deterministic state-tracking rules. The machine-learning component identifies observations that differ from the patterns represented in its training data, while the rule-based component evaluates predefined conditions derived from ADS-B message content and track history. This hybrid approach combines statistical anomaly scoring with explicit checks while keeping the computational requirements suitable for the target hardware.

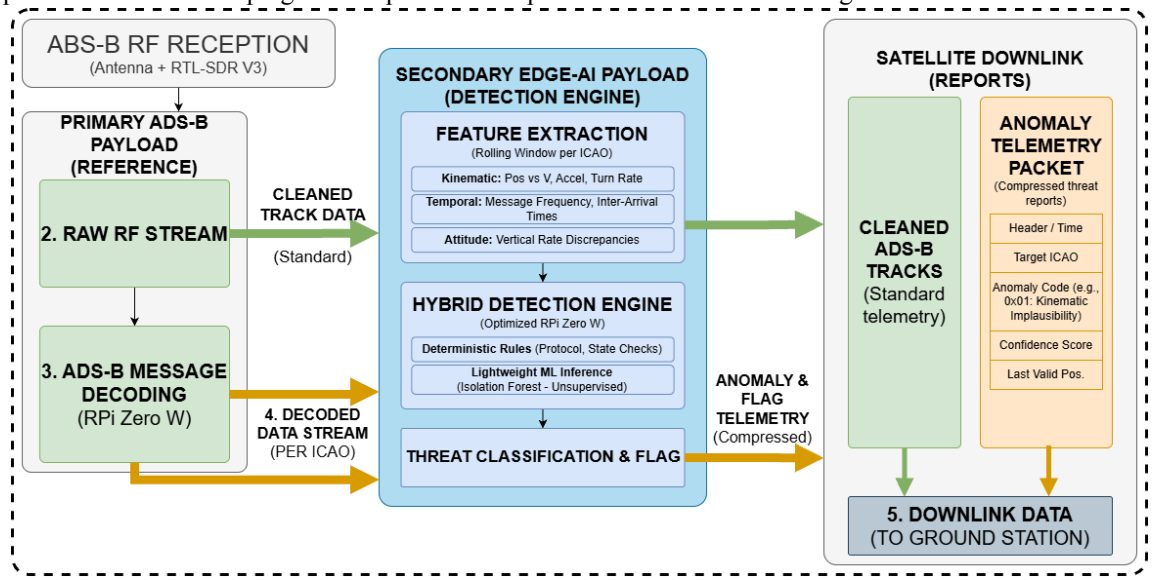


Figure 29: ADS-B Anomaly Detection Payload Diagram

- Feature Vector Generation: Decoded ADS-B messages are organized into sliding time windows associated with individual aircraft, identified by their 24-bit ICAO addresses. For each track, the system extracts features describing the consistency of reported motion, altitude information, and message timing. The feature vector comprises three groups:
  - **Kinematic Vector:** Differences between successive reported positions are compared with the displacement implied by the reported ground velocity and elapsed time. Additional features include estimated acceleration and turn rate, which can help identify inconsistencies in the reported aircraft trajectory.
  - **Altitude Vector:** This group includes changes in reported altitude, vertical rate of climb or descent, and discrepancies between geometric and barometric altitude when both values are available. These features provide additional information for evaluating the consistency of an aircraft's reported vertical motion.
  - **Temporal Vector:** Message arrival intervals, frame repetition rate, and track duration are used to characterize the timing of received transmissions. These features can help identify unusual repetition patterns or interruptions in the observed message sequence.
- **Inference Engine (Isolation Forest):** The Isolation Forest algorithm is used to assign anomaly scores to the generated feature vectors. Rather than requiring an explicit model of every possible anomalous condition, Isolation Forest identifies observations that are relatively easy to isolate within its collection of randomized decision trees. Its suitability for this application depends on the size of the trained model, the dimensionality of the feature vectors, and the computational cost of evaluating each track window. The model is trained offline using historical ADS-B flight data selected to represent valid aircraft operations. After training, the serialized model is loaded into RAM during system startup, avoiding the need to perform model training onboard. The model then evaluates incoming feature vectors as they become available. Its anomaly scores are combined with the deterministic checks to determine which observations should be flagged for further analysis.

c) Detected Threat & Anomaly Taxonomy:

Without needing the satellite's orbital coordinates, the AI identifies four primary signal degradation and attack categories:

1. Kinematic Implausibility (GPS Spoofing / Meaconing): Sudden coordinate jumps ("teleportation") between consecutive message frames or acceleration values exceeding standard commercial flight envelopes.

2. Channel Flooding / Denial-of-Service (DoS): Abnormally high transmission rates from a single ICAO address or repeated identical frame injections indicative of a malicious ground-based spoofer or faulty transponder.
3. Protocol & State Inconsistencies: Mismatched airborne/surface status flags, impossible vertical speed markers, or invalid callsign formatting.
4. Data Corruption & Overlay Interference: Collision of overlapping messages in the 1090 MHz band causing corrupted message payloads or erroneous altitude readings.

d) Anomaly Telemetry Packet Format:

To conserve downlinked bandwidth, detected anomalies do not trigger raw log dumps. Instead, the secondary payload generates a brief, structured Anomaly Telemetry Packet:

Field	Size	Description
Header / Time	4 bytes	Local satellite timestamp (epoch seconds)
Target ICAO	3 bytes	Aircraft 24-bit unique identifier
Anomaly Code	1 bytes	Categorized alert (e.g., 0x01: Kinematic Implausibility, 0x02: Rate Flooding)
Confidence Score	1 bytes	Outlier confidence score generated by Isolation Forest (0-100%)
Last Valid Pos.	6 bytes	Latitude and Longitude immediately prior to anomaly flag

Table 15: Anomaly Telemetry Packet structure

By shifting signal validation directly to the satellite's edge processor, the payload delivers clean, verified ADS-B data and prioritized threat reports to the ground station.

F. Mission Concept of Operations (ConOps) Overview:

A ConOps (Concept of Operations) demonstrates the behavior which the satellite would assume during a real mission's lifecycle: from launch to End of Life (EOL). The operational architecture, comprising the main modes and their respective internal states, is illustrated in Fig. 30.

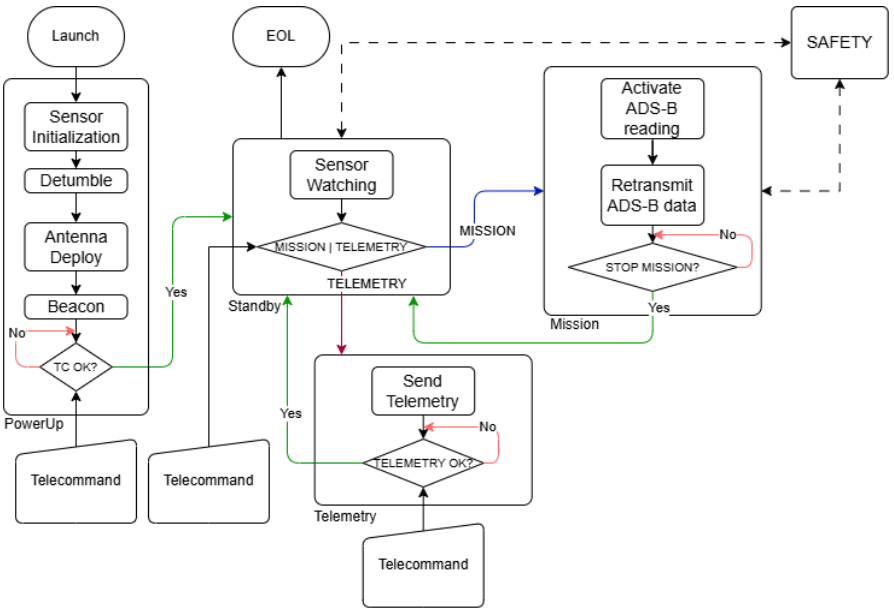


Figure 30: Space mission ConOps diagram

The goal of this ConOps is to ensure the satellite meets all operational demands necessary to enter mission mode. The modes are activated by input and output criteria from different devices, by telecommand (TC), or by operational criteria.

i. POWERUP:

Triggered immediately after ejection from the launch vehicle, this mode ensures the satellite achieves a stable state before initiating communications. It begins with Sensor Initialization to calibrate onboard instruments, followed by the Detumble state to stabilize the satellite's attitude. Subsequently, the Antenna Deploy state is executed, allowing for the activation of the communication systems. Once stabilized, the satellite transmits its Beacon, providing essential health and status telemetry. The transition to Standby mode is contingent upon receiving a valid telecommand (TC) packet; if the TC is not confirmed, the system loops back to re-verify the signal.

ii. STANDBY:

Acting as the central hub of the satellite's operations, this mode is entered after a successful PowerUp. Its primary function is Sensor Watching, where the onboard computer continuously monitors system health and environmental data. From Standby, the system performs a decision-based branching (MISSION | TELEMETRY) to determine the next operational path. If the mission duration is complete, the satellite may eventually transition directly to the EOL (End of Life) state.

iii. TELEMETRY:

Activated via telecommand from the Standby mode, this operational state focuses on data downlink. The satellite enters the **Send Telemetry** state to transmit detailed diagnostic or mission data to the ground station. A **TELEMETRY OK?** decision block is used to ensure data packet integrity; if the link is not confirmed, the system iterates until a successful transmission occurs or a new command is received to return to Standby.

iv. MISSION:

This mode encompasses the primary objective of the Hermes satellite. It follows a sequential operational loop:

- *Activate ADS-B reading: The satellite initiates the collection of ADS-B signals.*
- *Retransmit ADS-B data: The payload processes and downlinks the collected signals to the ground segment.*
- *A STOP MISSION? decision block manages the mission lifecycle; if the mission is not stopped (No), the satellite continues the cycle. If the mission objectives are met or a stop command is*

issued (Yes), the satellite safely exits the mission loop and returns to the Standby mode for further instruction.

v. SAFETY:

This contingency mode is critical for satellite survivability. It is designed to be triggered autonomously if the Sensor Watching process in Standby or the mission processes detect a significant failure or critical anomaly. In this state, the satellite enters a "safe" configuration, isolating non-essential subsystems and powering only what is necessary for attitude control and ground communication, awaiting a recovery telecommand to return to normal operations.

The operational architecture shown in Fig. 39 and described above represents a baseline for a real mission in orbit and some changes were made to adapt for CubeDesign tests. The main planned revisions are:

- **Detumble and Antenna Deploy by telecommand:** instead of being executed automatically in sequence during POWERUP, both states will be triggered by a telecommand (TC) sent from the ground station.
- **Beacon**: since the POWERUP consists now only of initializing the sensors, the beacon will indicate only if the initialization occurred well, instead of Detumble and Deploy too.
- **Telemetry**: The telemetry available will be sent in a fixed interval instead of after a telecommand, which during testing will make it easier to recognize possible problems.
- **Azimuth search and azimuth transmission by telecommand:** the azimuth search and the subsequent transmission of the azimuth value will also be commanded via TC, rather than running autonomously.

These changes are motivated by test and verification: having each of these steps triggered by an explicit telecommand makes each function individually testable and easier to demonstrate, since the response to a single TC can be observed in isolation. Fig. 31 shows the modified CONOPS, and the resulting state machine is described under the On-Board Computer section. It is important to note that all modes go back to STANDBY once the task is done, which was not shown on the diagram in order to make it more intelligible.

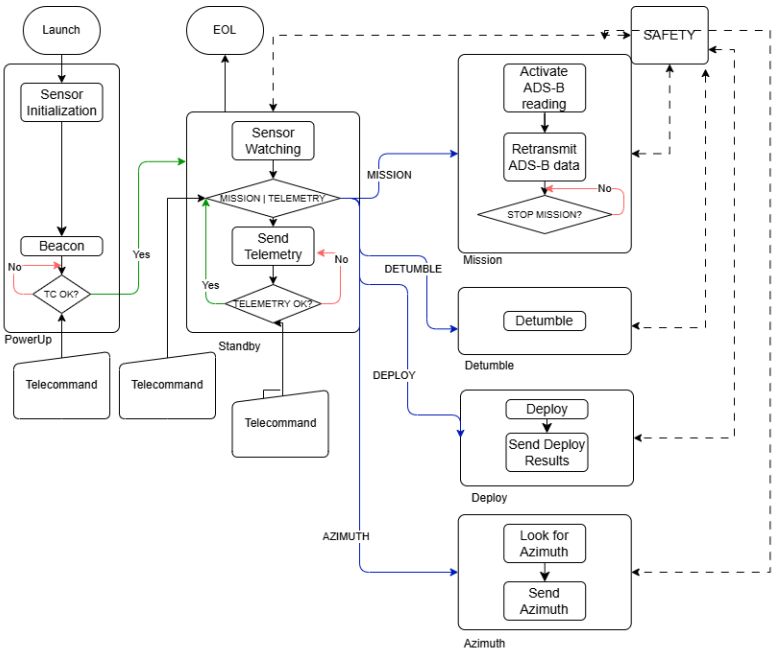


Figure 31: ConOps Diagram for the Cubedesign tests

G. Structure System (SS):

1. System Design

The structural subsystem comprises the primary satellite structure and mechanically engineered components, including the solar panels developed by PoliSat, the radiation shielding, and the antenna deployment mechanism.

1.1 Standards

The structural requirements adopted for this project were drawn from the Cal Poly CubeSat Laboratory specifications. The nominal external envelope was defined for the 2U configuration. Additional requirements used in the design are listed below.

Subject	Standard
Material	Low-outgassing materials with a total mass loss below 1%.
	The structure should use an aluminum alloy such as 7075, 6061, 6082, 5005, or 5052.
Mechanical manufacturing	The ±Z faces contain the standoffs.
	The maximum protrusion normal to the rail plane is 6.5 mm.
	Deployable components must be contained within the CubeSat rather than by the dispenser.
	The rails must have a minimum width of 8.5 mm, measured from the edge to the beginning of the adjacent face.
	At least 75% of the CubeSat rails must be in contact with the dispenser rails.
	Standoffs must extend 0.5–7 mm from the Z-face plane and have a minimum contact area of 6.5 mm × 6.5 mm.
	No part of the CubeSat shall detach from the main structure.

Table 16: Standards considered for the development.

Some requirements remain under development, including the requirement that the center of mass lie within 2 cm of the geometric center along each axis. The following requirements were not incorporated into the present design:

Subject	Standard
Material	The CubeSat must have an ascent-venting ratio of ventable volume to vent area below 50.8 m.
	The structure must be anodized
	The rail surface roughness must be less than 1.6 μm
Mechanical manufacturing	Standoffs must have a minimum chamfer radius of 1 mm
	The CubeSat must use one deployment switch and two springs with initial and final forces of 2.224 N and 6.672 N, respectively, with a minimum ejection stroke of 1.27 mm above the standoff surface.
	The deployment switch must keep the power system off until ejection and, when actuated, must remain centered

	at or below the standoff level.
Electrical Isolation	The rail edges must have a minimum corner radius of 1 mm.

Table 17: Additional requirements.

The current design does not satisfy all international requirements applicable to a satellite launch.

2. Structural Parts

The structural components were designed to facilitate assembly. Quasi-static loading and vibration responses were also assessed through modal, sinusoidal, and random vibration analyses.

2.1 Structural Design

The structure is manufactured from aluminum 7075 and consists of two types of components:

Four lateral beams;

Three supports.

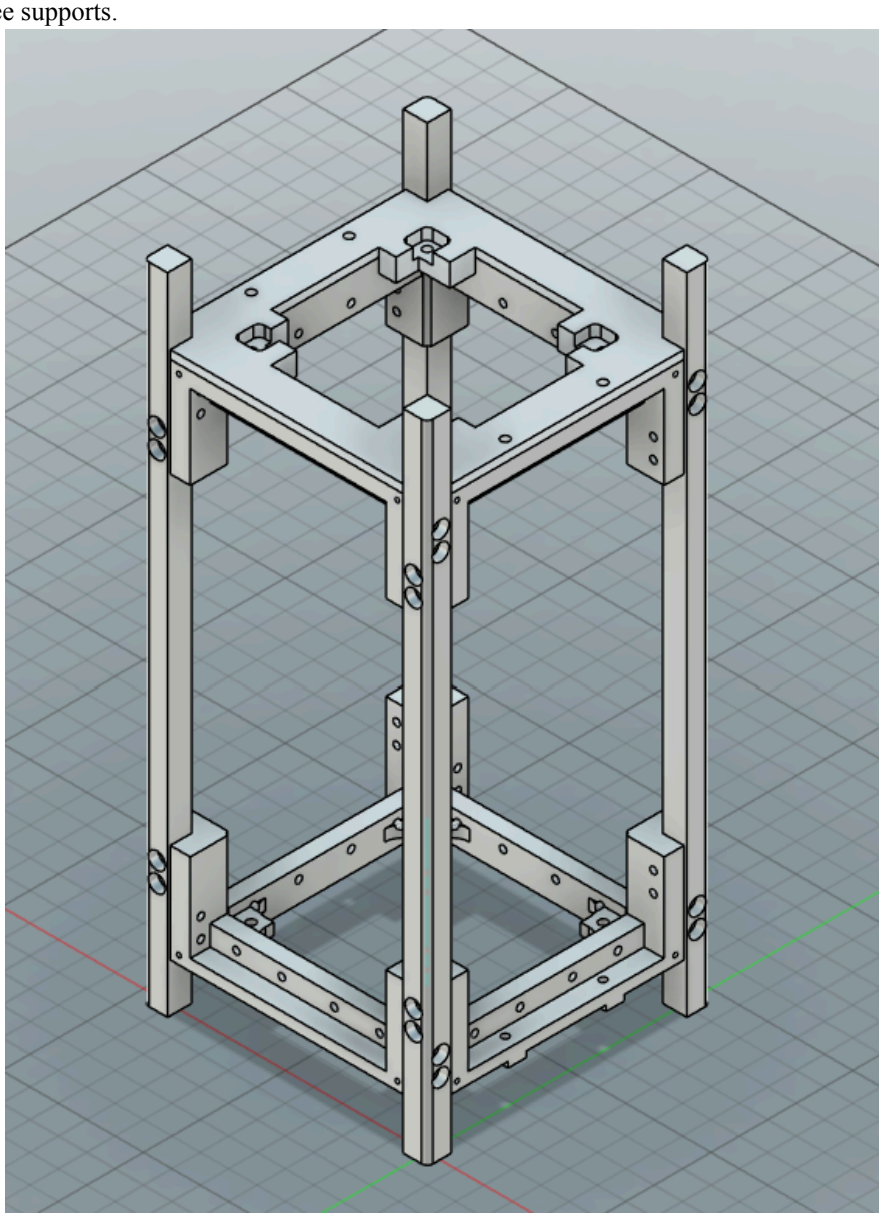


Figure 32: Mechanical structure.

The standoffs secure the printed circuit boards inside the CubeSat structure.

3. Environmental Verification: Vibration Analysis

3.1 Analysis Scope

This section presents the modal and random vibration assessment performed in ANSYS Mechanical. The objective is to identify the relevant structural modes and estimate the acceleration, displacement, and stress responses under the prescribed environmental test excitation.
The applied load case follows the CubeDesign/INPE specification: a power spectral density (PSD) profile from 20 to 2000 Hz, an overall level of 14.1 g RMS, and a duration of 2 minutes in each orthogonal axis. GSFC-STD-7000B (GEVS) was used as a complementary reference for environmental and vibroacoustic verification practices. The assumptions adopted in the numerical model are described together with the corresponding analysis steps.

3.2 Structural Model and Simplifications

The satellite structural model was simplified to reduce computational cost while retaining the principal load paths and the relevant mass and stiffness effects. The printed circuit boards (PCBs), the attitude determination and control system (ADCS), electronic components, and other complex geometries were represented by simplified equivalent geometries. This approach is suitable for a preliminary modal assessment provided that the total mass, center-of-mass location, moments of inertia, and stiffness of the relevant interfaces are adequately preserved. The simplified model is shown in the following figure.

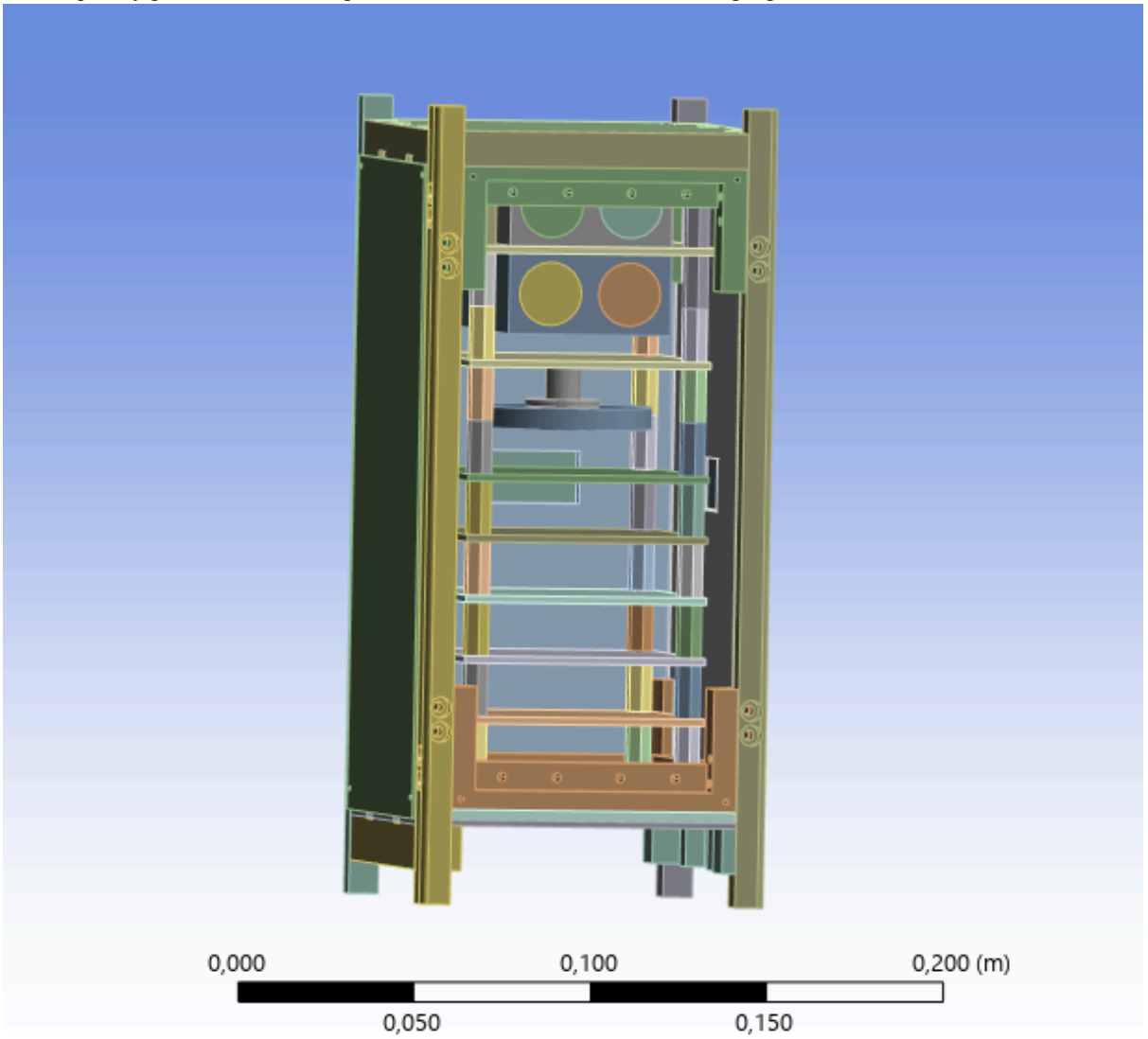


Figure 33: Simplified structural model used in ANSYS.

3.3 Random Vibration Input Spectrum

The simulation used the random vibration qualification profile specified by CubeDesign/INPE: an overall level of 14.1 g RMS over the 20–2000 Hz frequency range, applied for 2 minutes per axis. The competition documentation refers to NASA/GEVS requirements[3].

Frequency (Hz)	ASD Level (g^2/Hz)	
	Qualification	Acceptance
20 20-50 50-800 800-2000 2000	0.026 +6 dB/oct 0.16 -6 dB/oct 0.026	0.013 +6 dB/oct 0.08 -6 dB/oct 0.013
Overall	14.1 G_{rms}	10.0 G_{rms}

Figure 34: NASA/GEVS random vibration PSD qualification profile.

3.4 Boundary Conditions and Deployer Representation

Depending on the deployer geometry and mechanism, nonlinear effects may arise from clearances, intermittent contact, and friction between the CubeSat rails and the deployment device. At this stage, these effects were neglected and the interfaces were represented using linearized boundary conditions. Normal displacement was constrained on the lateral rail surfaces in contact with the deployer in the X and Y directions. At the Z+ rail ends, displacement normal to the contact surface was constrained. To suppress the remaining rigid-body motion, a specific region was constrained in all three translational degrees of freedom, as shown in the following figure. This approach is analogous to that described by Fagerudd in Stress Simulation of the SEAM CubeSat Structure During Launch[4]. The localized constraint may introduce additional stiffness; this effect is included among the limitations of the preliminary model.

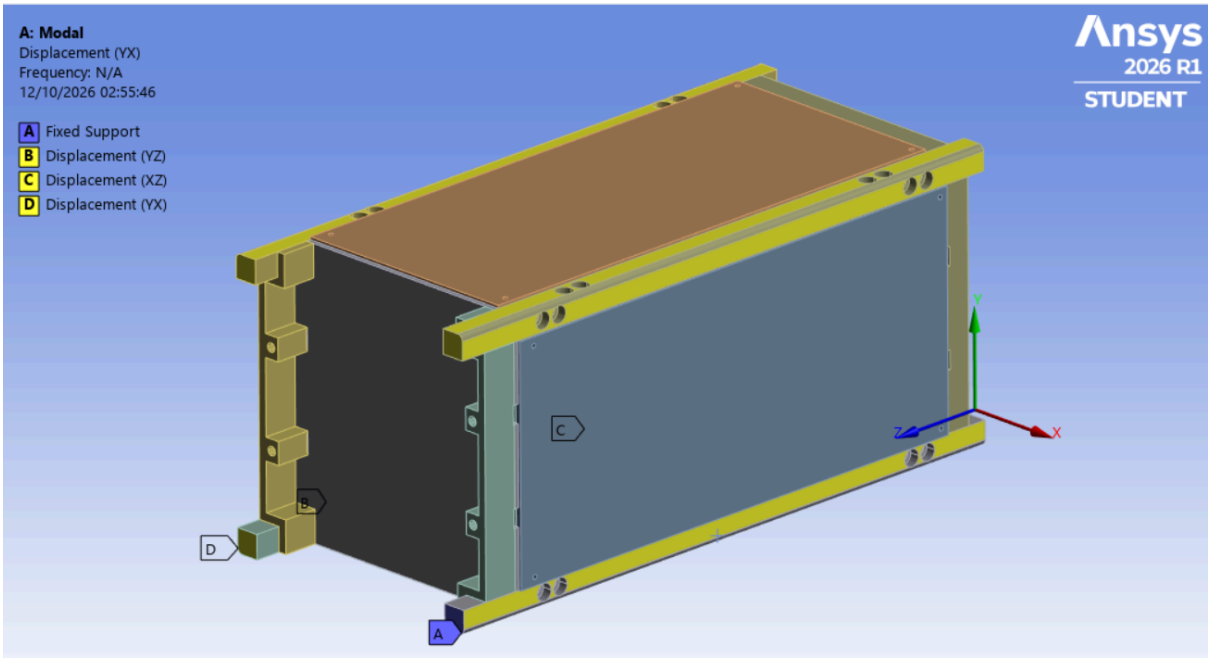


Figure 35: Boundary conditions and deployer-interface representation.

3.5 Modal Analysis

Natural modes were extracted up to 2000 Hz to cover the complete frequency range of the applied excitation spectrum. The 2000 Hz upper limit is consistent with the random vibration control range specified in NASA-STD-7001B[5]. The calculated natural frequencies are presented below.

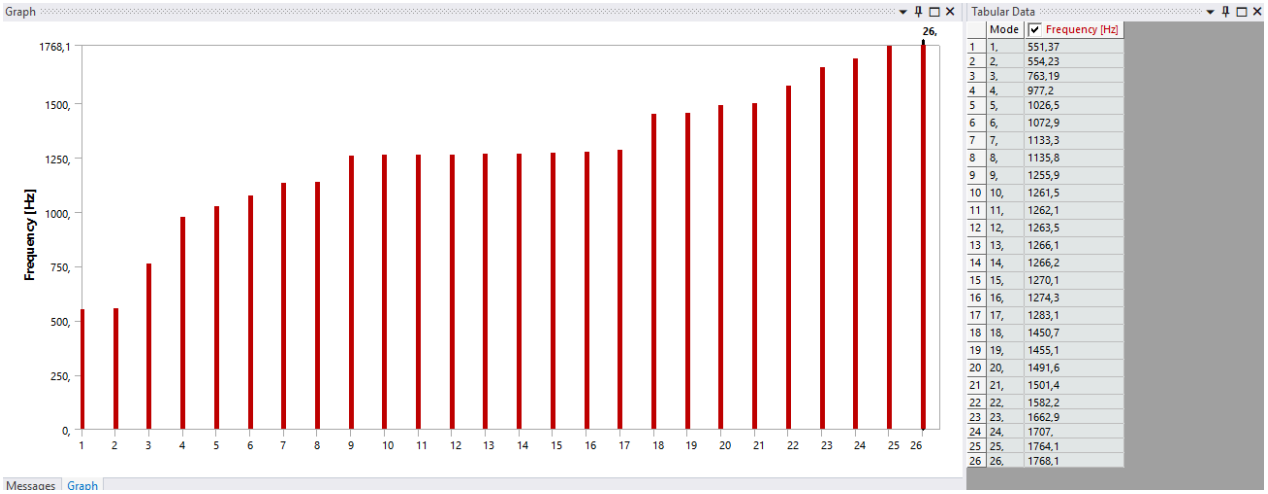


Figure 36: Calculated natural frequencies up to 2000 Hz.

The observed mode shapes are primarily associated with the PCB stack and the ADCS support. The significant contribution of the ADCS support is consistent with the concentrated mass located in this region. The first mode shape is presented below.

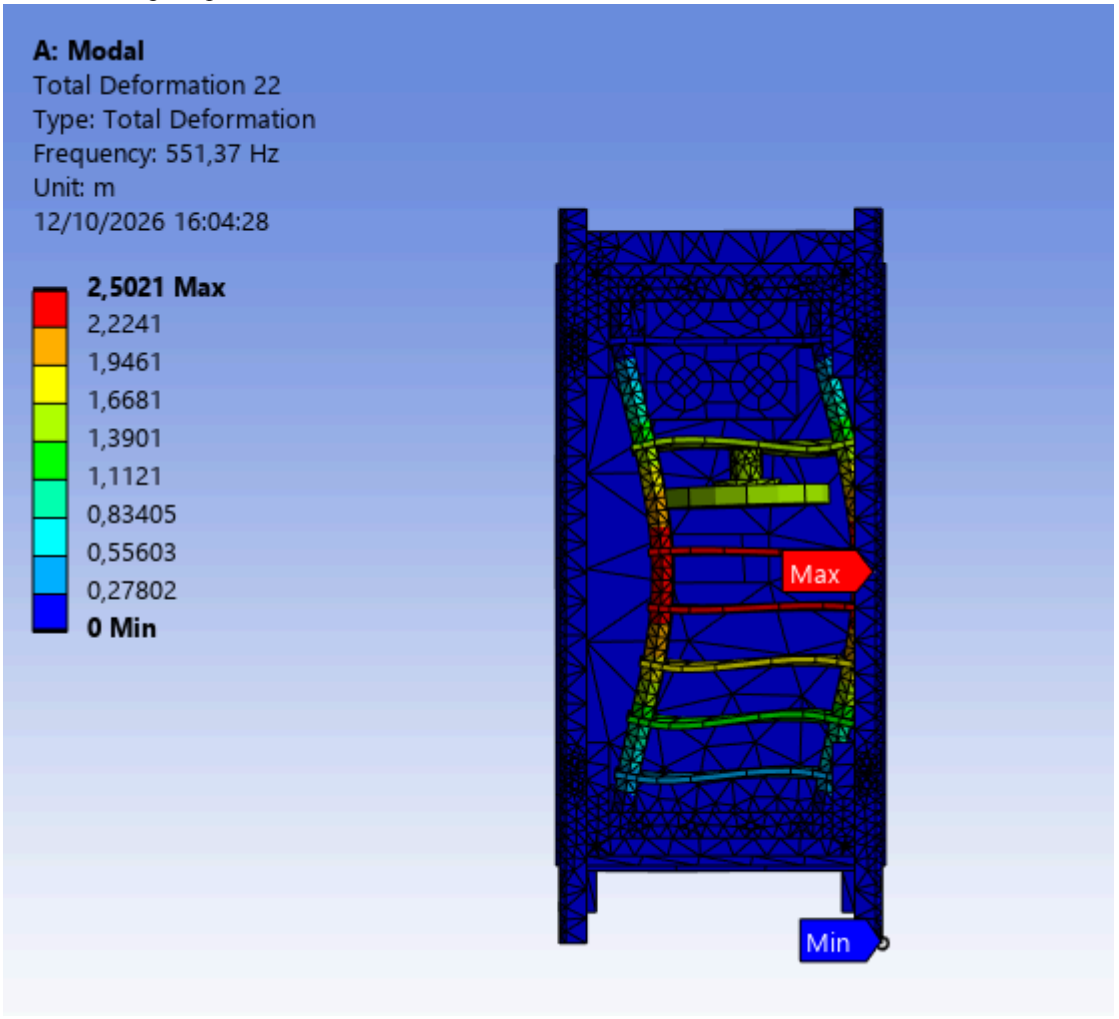


Figure 37: First structural mode shape.

3.6 Interpretation of the Relevant Modes

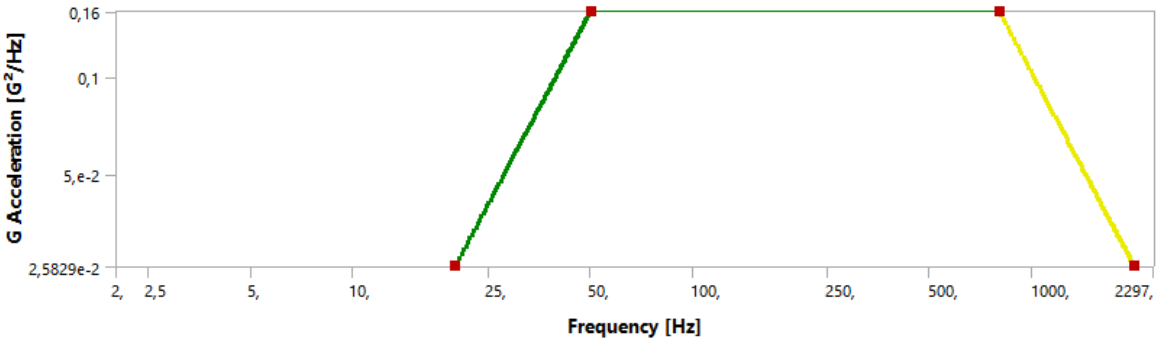
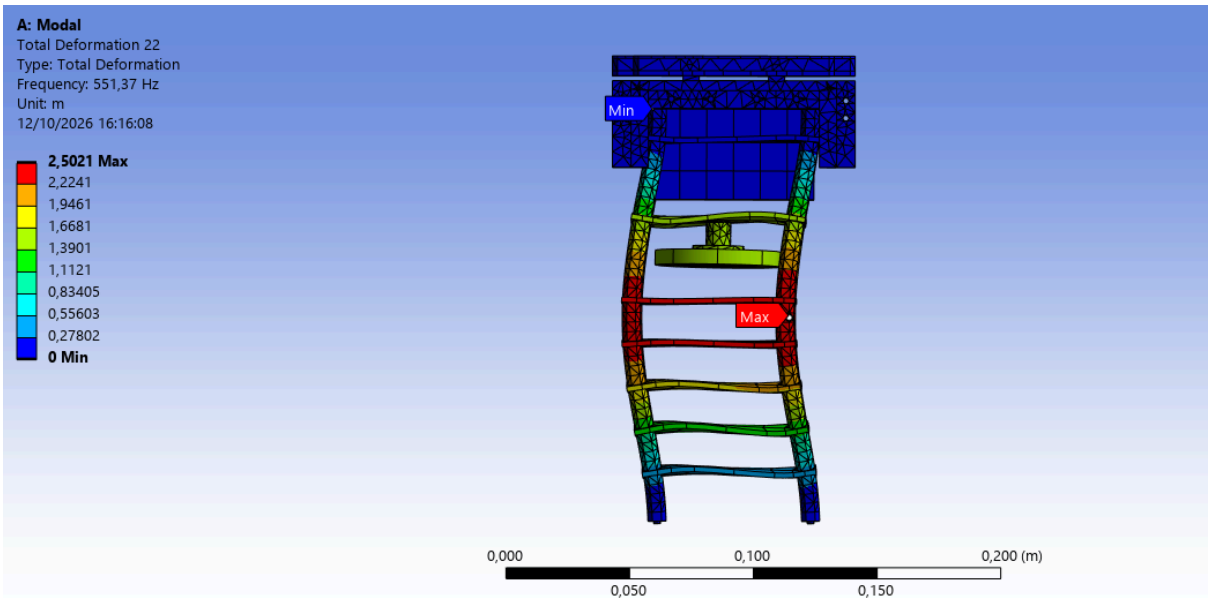


Figure 38: Relevant structural modes within the 50–800 Hz range.

The input spectrum reaches its highest PSD plateau between 50 and 800 Hz. Modes within this interval therefore receive particular attention. Frequency alone, however, does not determine modal relevance; participation factors, effective modal mass in each excitation direction, damping, and the local acceleration, displacement, and stress responses must also be considered. Three modes were identified within the 50–800 Hz range and were examined in greater detail, without excluding modes outside this interval that may contribute to local responses.

Modes 1 and 2 — Translation in the XY Plane

The first two modes exhibit predominantly translational motion in the XY plane. Consequently, stronger coupling is expected for base excitation applied along the X and Y directions. This interpretation is consistent with the observed mode shapes and may be quantified in future work using participation factors and effective modal masses.



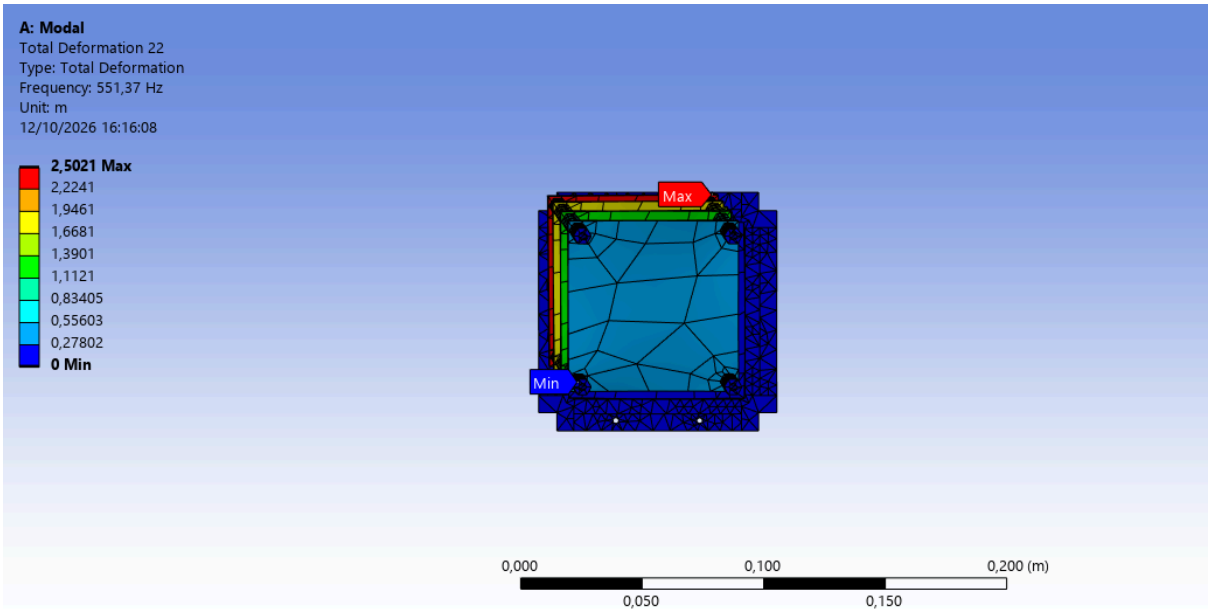
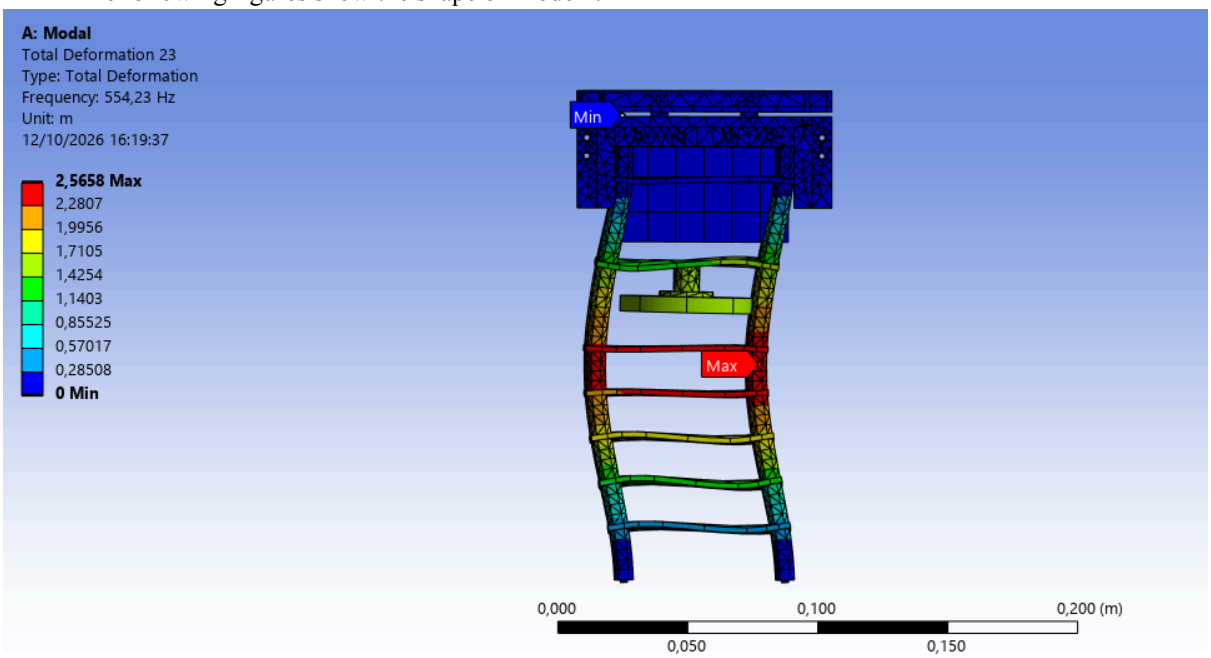


Figure 39: Mode 1 translational mode shape in the XY plane.

The following figures show the shape of Mode 2.



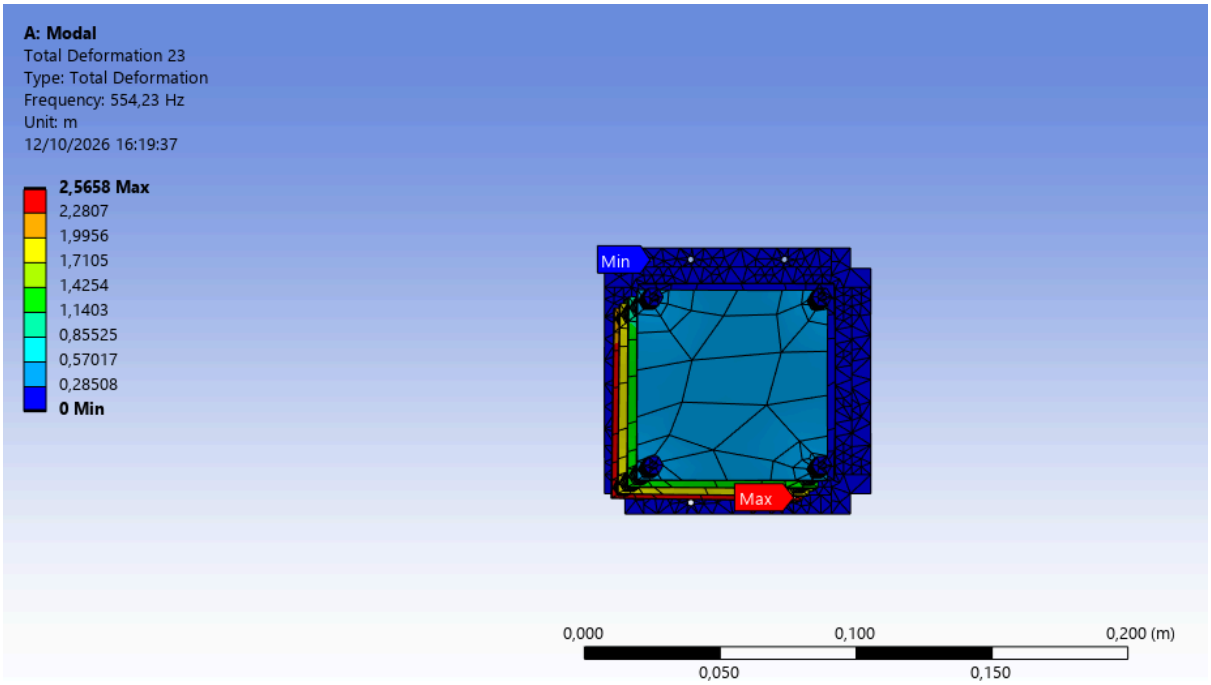
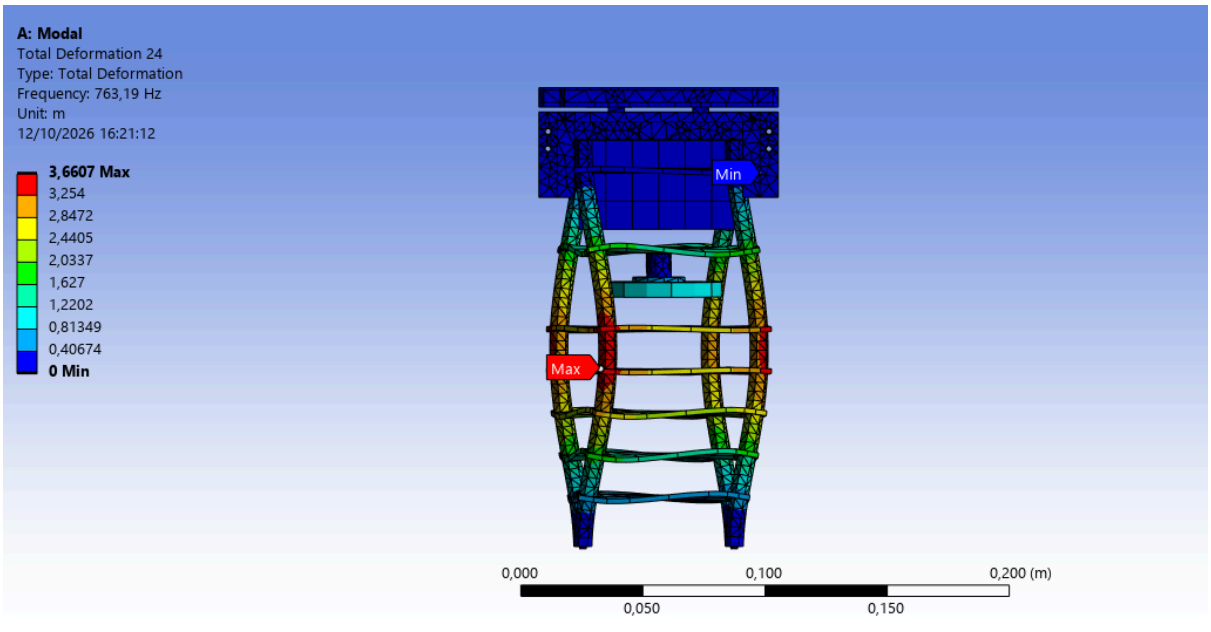


Figure 40: Mode 2 translational mode shape in the XY plane.

Mode 3 — Torsion

The third mode exhibits predominantly torsional behavior, as shown in the following figure. Its contribution is associated with the coupling between the translational base excitation and the asymmetric mass and stiffness distribution of the assembly, particularly at the PCB stack, ADCS support, and attachment interfaces.



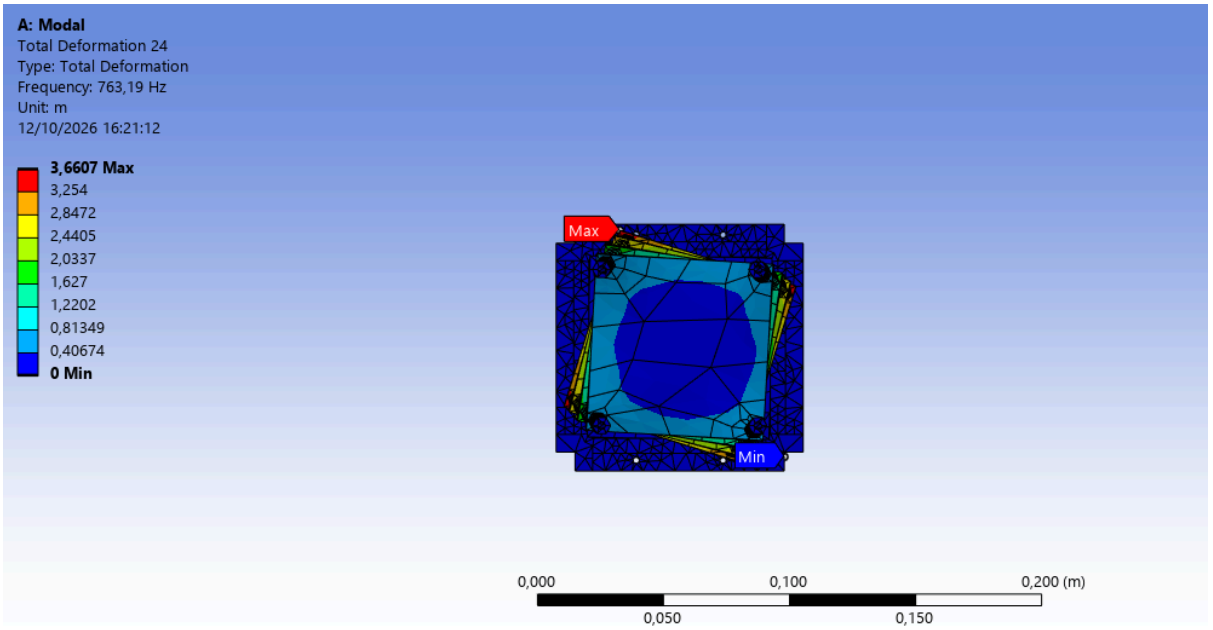


Figure 41: Mode 3 torsional mode shape.

3.7 Random Vibration Analysis Results

3.7.1 Z-Axis Excitation

The von Mises equivalent stress distribution obtained for Z-axis base excitation is shown in the following figure.

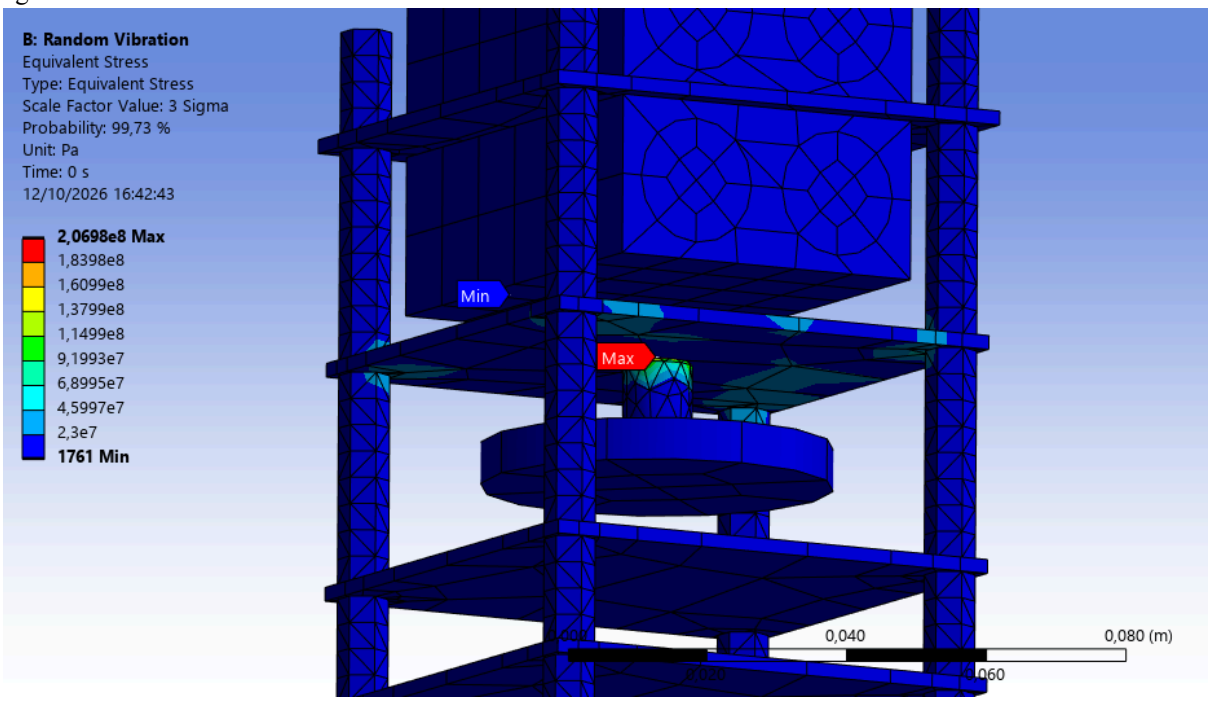


Figure 42: Z-axis von Mises equivalent stress distribution.

The maximum reported value is 2.068×10^8 Pa (206.8 MPa), located at the interface between the flange and the ADCS support. A 3σ statistical multiplier was applied directly in ANSYS; therefore, this value represents the estimated 3σ equivalent stress rather than the one-standard-deviation RMS stress. Because the peak occurs at an interface affected by geometric simplification, contact representation, and mesh discretization, it was retained as a conservative preliminary value.

Using a preliminary yield strength of $S_\gamma = 505$ MPa for the assumed aluminum 7075 condition, the yield factor of safety for the Z-axis case is $FS = S_\gamma/\sigma_{3\sigma} = 505/206.8 \approx 2.44$. This is the lowest factor of safety obtained

among the evaluated metallic critical regions and is considered acceptable for the preliminary CubeDesign random vibration assessment. The calculation applies to the modeled vibration test only and does not include superimposed launch quasi-static loads, gravity, assembly preload, or fatigue damage.

The remaining regions were assessed using the material properties and failure criterion applicable to each component.

The Z-direction deformation distribution is presented in the following figure.

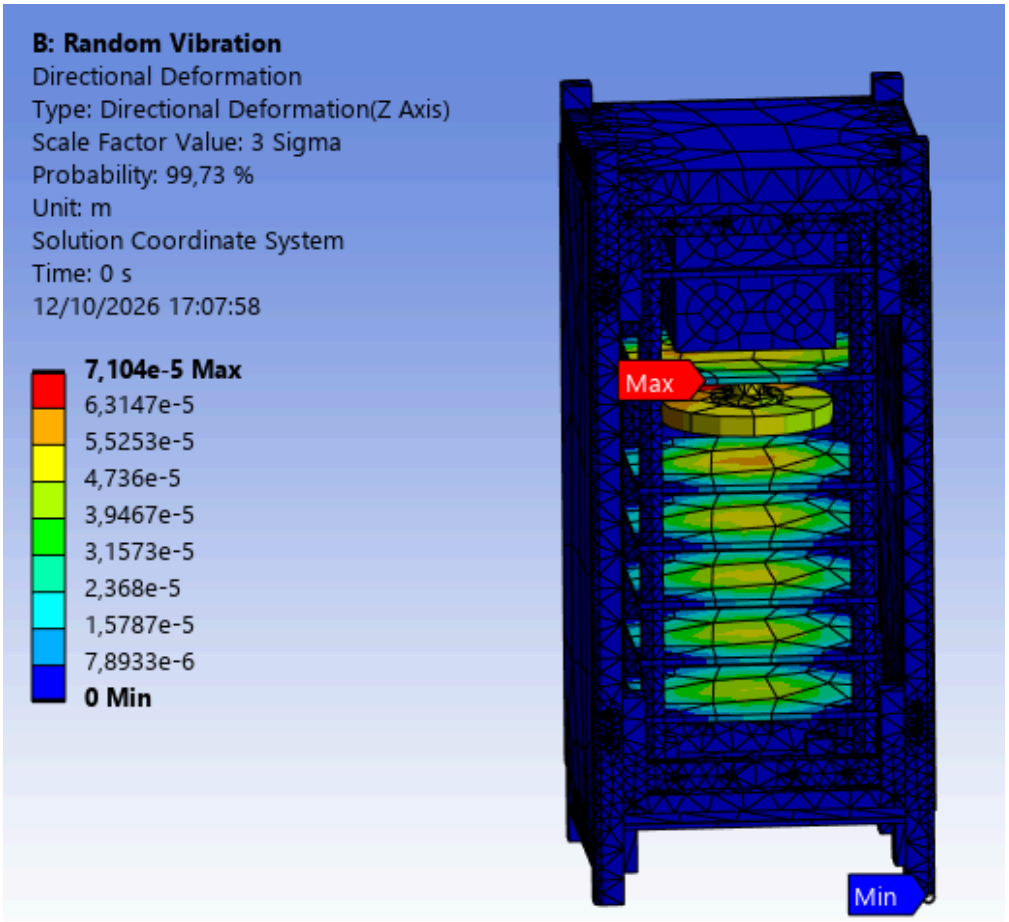


Figure 43: Z-direction deformation distribution.

To inspect the remaining structural regions, the flange and reaction wheel were hidden only in the result visualization; the solved finite element model was not modified. In this view, the maximum equivalent stress at the standoff–ADCS support interface is 6.16×10^7 Pa (61.6 MPa). This value is well below the stress at the governing flange interface and below the preliminary material allowable adopted in this analysis.

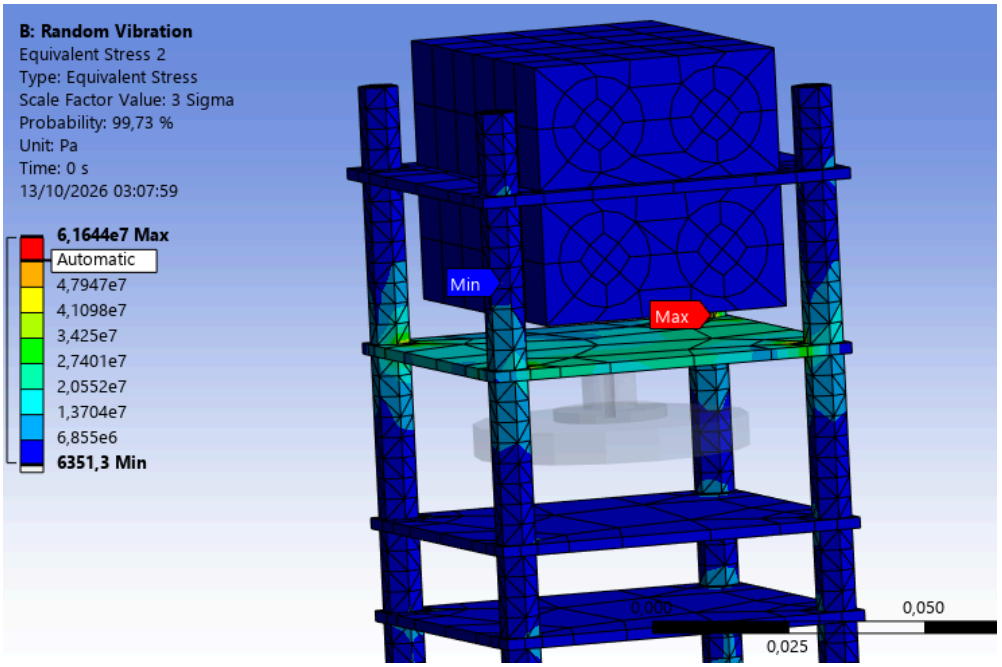


Figure 44: Local stress distribution at the standoff–ADCS support interface under Z-axis excitation.

3.7.2 Y-Axis Excitation

For Y-axis base excitation, the maximum estimated 3σ equivalent stress is 1.26×10^8 Pa (126 MPa). The maximum again occurs at the interface between the flange and the aluminum 7075 ADCS support. Using the same preliminary yield strength of 505 MPa, the corresponding local yield factor of safety is FS ≈ 4.01. This value is above unity with a substantial margin and is considered acceptable for the present preliminary assessment.

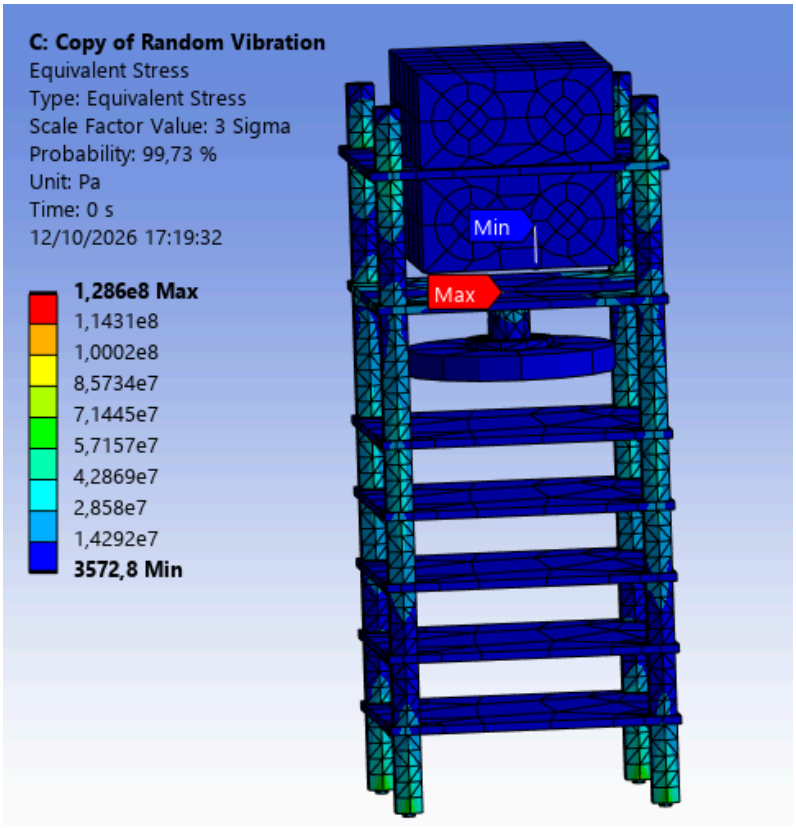


Figure 45: Y-axis equivalent stress distribution.

The corresponding directional deformation distribution is presented in the following figure.

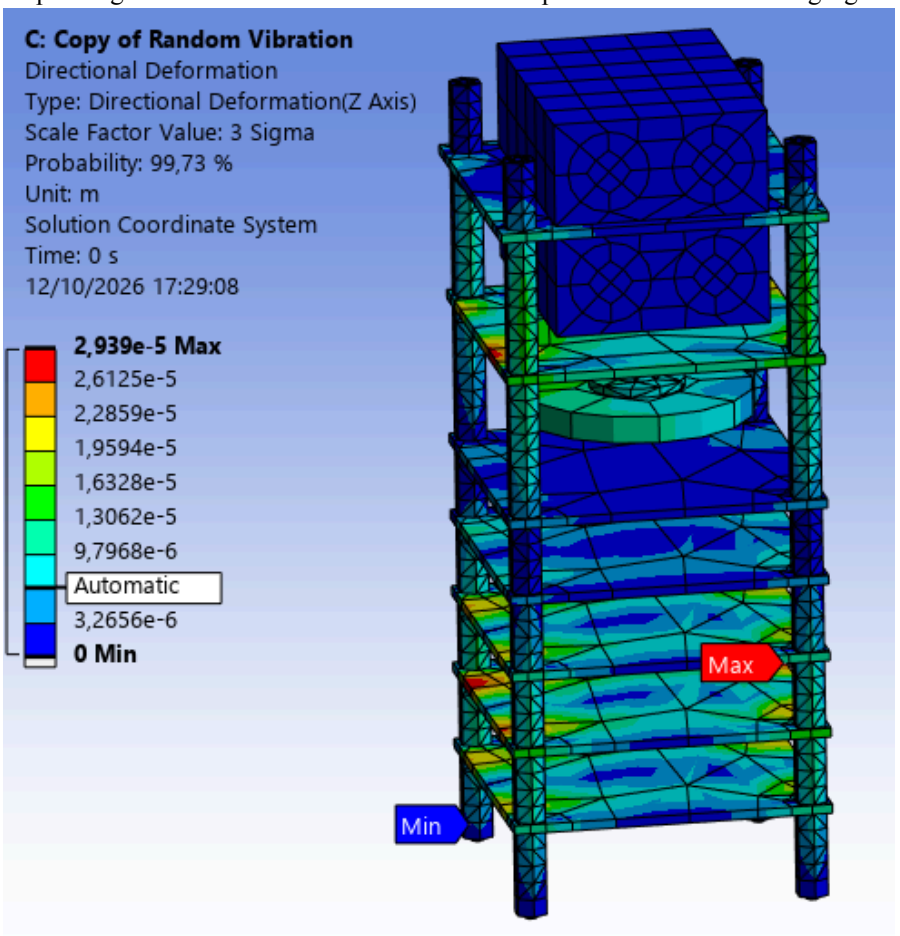


Figure 46: Y-direction deformation distribution.

When the flange–ADCS support interface is excluded from the visualization, local stresses between 5.68×10^7 and 1.021×10^8 Pa (56.8–102.1 MPa) are observed in the PCB-stack standoffs. These values remain below the governing Z-axis stress and within the preliminary acceptable range. The locations coincide with changes in section and PCB attachment interfaces, where local stress concentrations are expected.

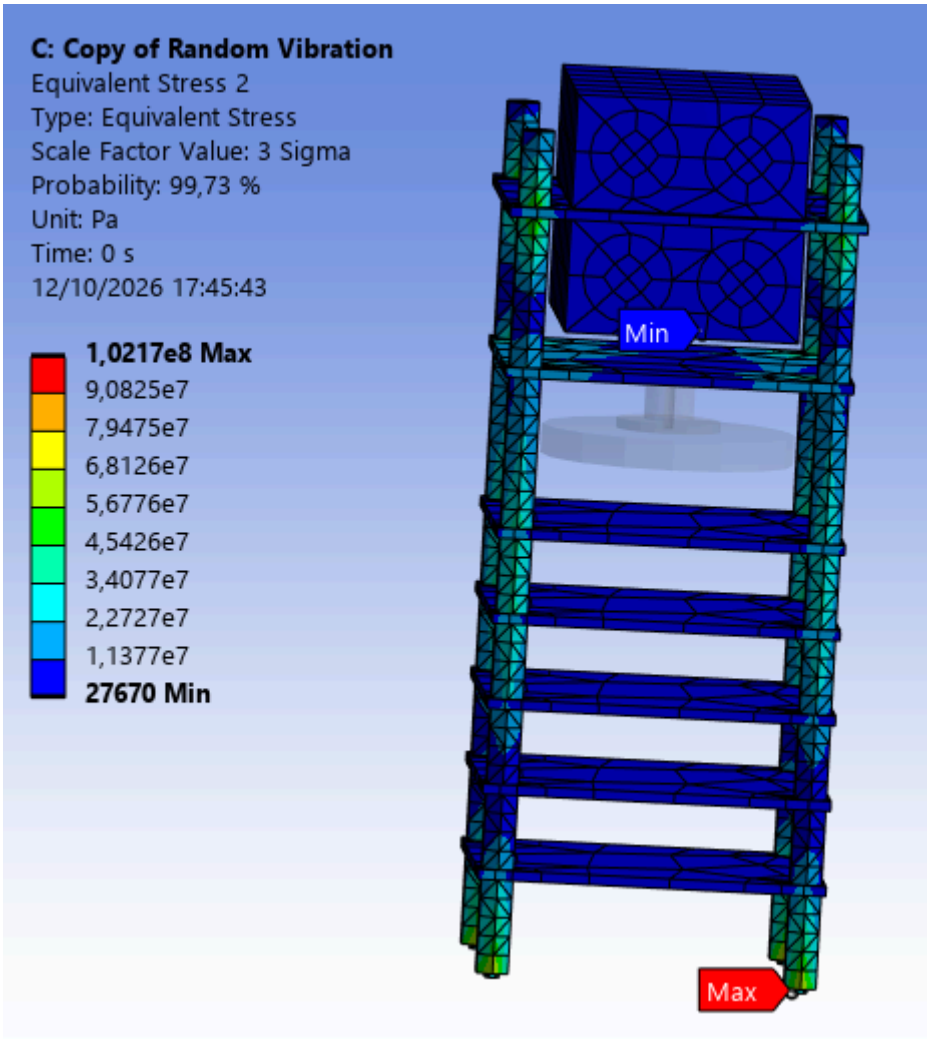


Figure 47: Local stress distribution in the PCB-stack standoffs under Y-axis excitation.

3.7.3 X-Axis Excitation

For X-axis base excitation, the critical location differs from the Y- and Z-axis cases. The maximum occurs at the connection between a standoff and the battery PCB, reaching 1.1551×10^8 Pa (115.51 MPa). The change in critical location is consistent with the different modal participation associated with each excitation direction.

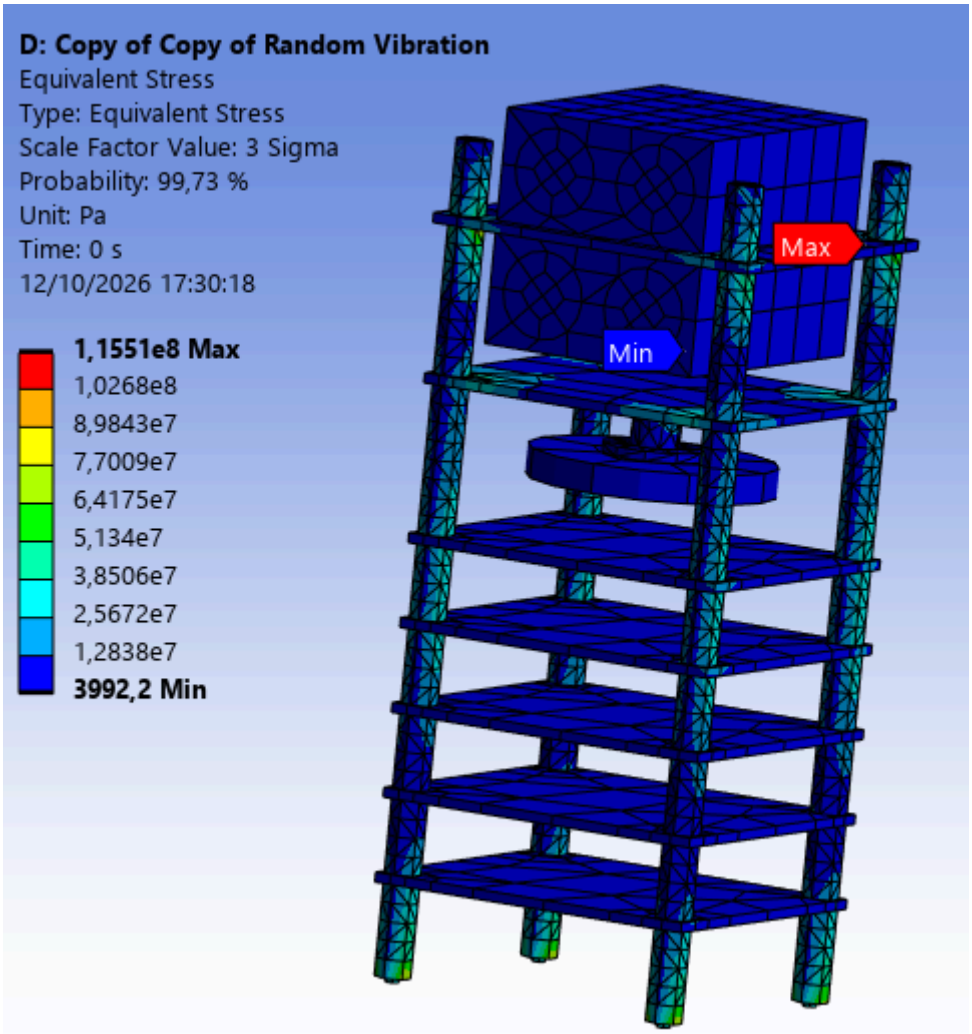


Figure 48: X-axis equivalent stress distribution.

The X-direction deformation distribution is presented in the following figure.

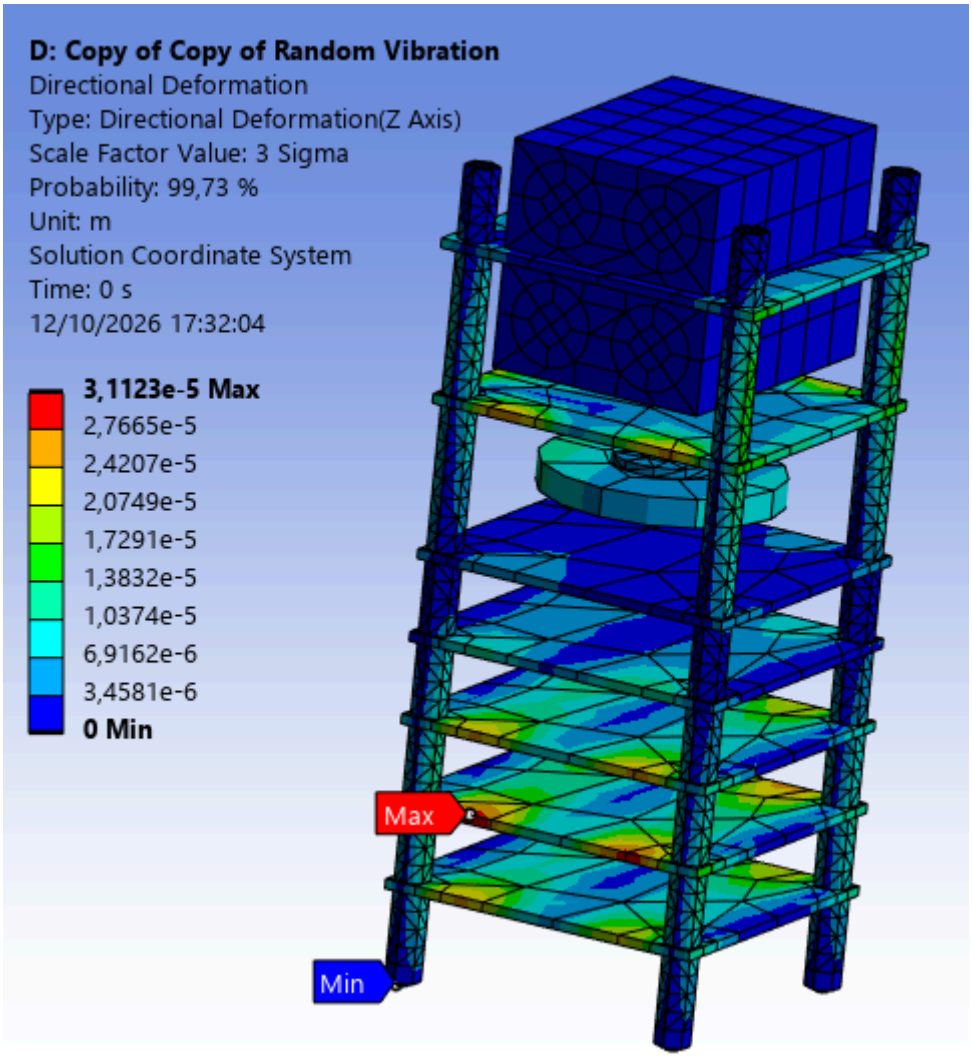


Figure 49: X-direction deformation distribution.

The maximum estimated 3σ stress at the battery PCB–standoff interface is 115.51 MPa and acts predominantly in the plane of the laminate. The Isola IS410 FR-4 datasheet reports in-plane tensile strengths of 420 MPa in the length direction and 316 MPa in the cross direction. Because the laminate orientation relative to the calculated stress has not yet been documented, the lower cross-direction value was adopted conservatively. The resulting preliminary in-plane tensile factor of safety is FS = 316/115.51 ≈ 2.74. This value is acceptable for the present preliminary assessment. Since FR-4 is an orthotropic laminate without a conventional metallic yield point, this ratio is referenced to ultimate tensile strength and should not be interpreted as a yield factor of safety.

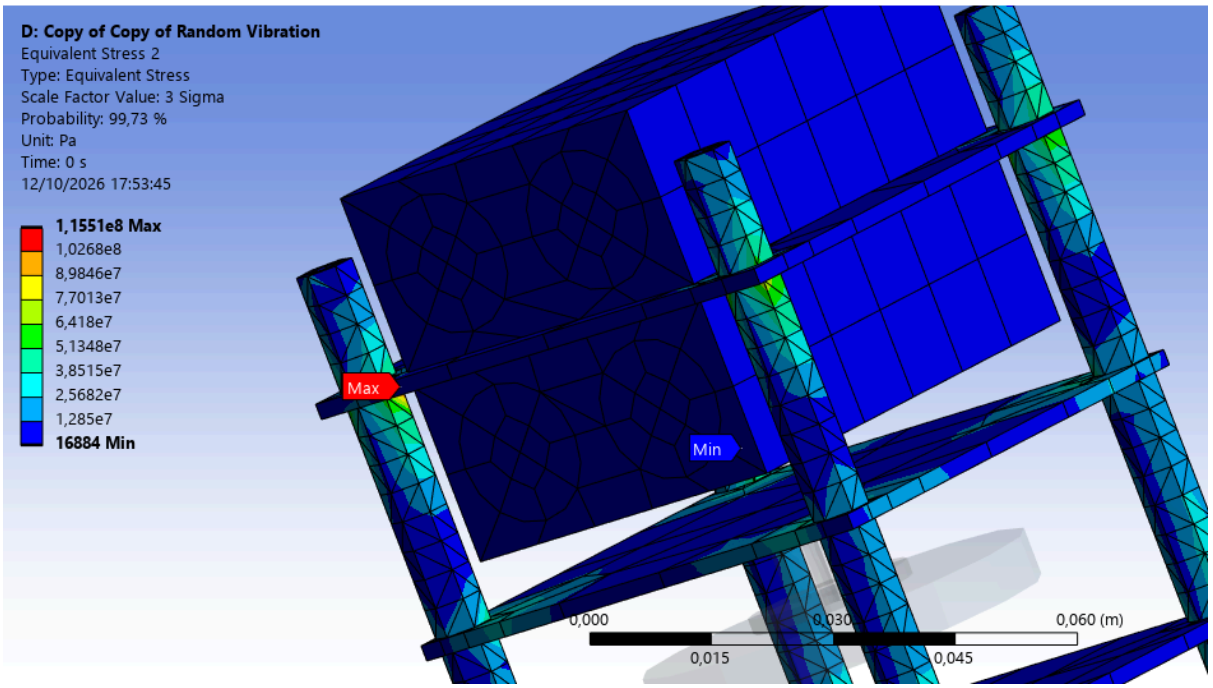


Figure 50: X-axis equivalent stress at the battery PCB–standoff interface.

3.8 Conclusions

All three random vibration load cases produced acceptable preliminary structural responses. The governing metallic case was Z-axis excitation, with a maximum 3σ equivalent stress of 206.8 MPa at the flange–ADCS support interface. Y-axis excitation produced 126 MPa at the same interface, while X-axis excitation produced 115.51 MPa at the battery PCB–standoff connection. Using a preliminary aluminum 7075 yield strength of 505 MPa, the minimum metallic yield factor of safety was 2.44 for the Z-axis case, while the Y-axis case had a factor of safety of approximately 4.01. For the predominantly in-plane FR-4 stress in the X-axis case, the conservative cross-direction tensile strength of 316 MPa resulted in a preliminary ultimate factor of safety of approximately 2.74. Therefore, all evaluated critical regions remained below their adopted material allowables and exhibited acceptable factors of safety.

The results therefore indicate that the structure has an adequate preliminary margin for the CubeDesign random vibration test in all three excitation directions. This conclusion is restricted to the modeled test environment and the stated material assumptions. Launch quasi-static loads, gravity-induced prestress, assembly preload, fatigue accumulation, and nonlinear deployer contact were not superimposed because they are outside the scope of this preliminary test assessment.

3.9 Recommended Future Improvements

A refined assessment could include a local mesh-convergence study at the flange, standoff, and PCB interfaces; confirmation of the exact material tempers and allowables; effective modal mass and participation-factor reporting; laminate-specific verification of the FR-4 PCBs; and response-PSD probes at the PCB-stack center, ADCS support, and critical interfaces. These additions would improve traceability and local-result confidence without changing the principal conclusion of the present preliminary analysis.

4. Radiation Shielding Protection

4.1 Introduction

The NASA presentation on radiation effects and CubeSat mission costs (4) motivated the development of radiation shielding. It discusses how radiation-induced failures can shorten mission life despite the substantial investment required to design and launch a CubeSat.

4.2 Development

The analysis used SPENVIS, developed for ESA's Space Environments and Effects Section. A nominal 4-year low Earth orbit (LEO) lifetime was defined, with simulations conducted over 20 representative orbits at an altitude of approximately 500 km. Trapped radiation was modeled using the AP-8 model for protons (solar minimum) and the AE-8 model for electrons (solar maximum). Solar particle fluences were simulated using the SAPPHIRE model covering ions from H to Fe with a 95% confidence level.

- Orbit lifetime: 4 years
- Representative orbits: 20
- Approximate orbital altitude: 500 km.

Simulations of trapped proton and electron fluxes:

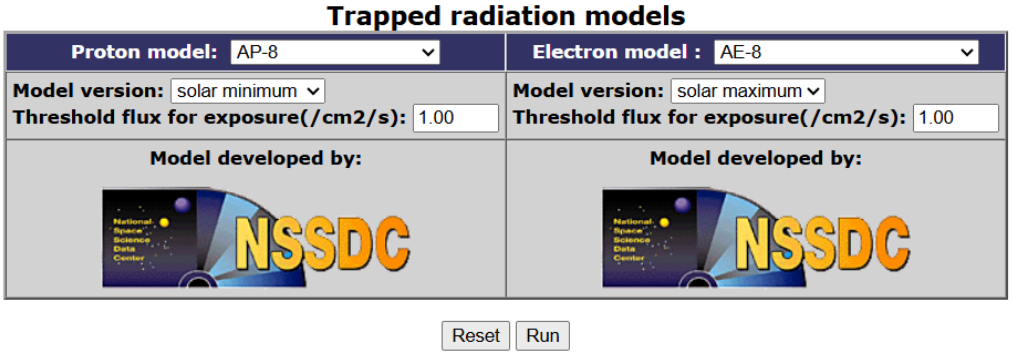


Figure 51: Trapped-radiation models.

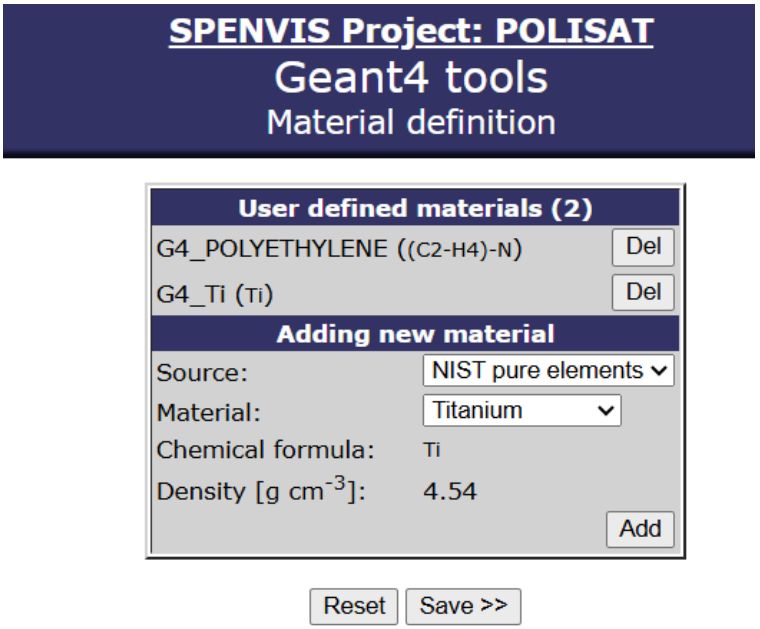


Figure 52: Material definitions in SPENVIS.

Monte Carlo radiation transport simulations were carried out using Geant4/MULASSIS. The shielding configuration evaluated consists of a 3 mm high-density polyethylene (HDPE) outer layer, followed by a 2 mm titanium (Ti) layer, and a 0.5 mm silicon (Si) layer representing the onboard sensitive electronics and silicon substrates. Radiation exposure from independent sources—trapped electrons, trapped protons, and solar proton fluence—was evaluated through total ionizing dose (TID) and displacement damage/non-ionizing energy loss (NIEL) analyses.

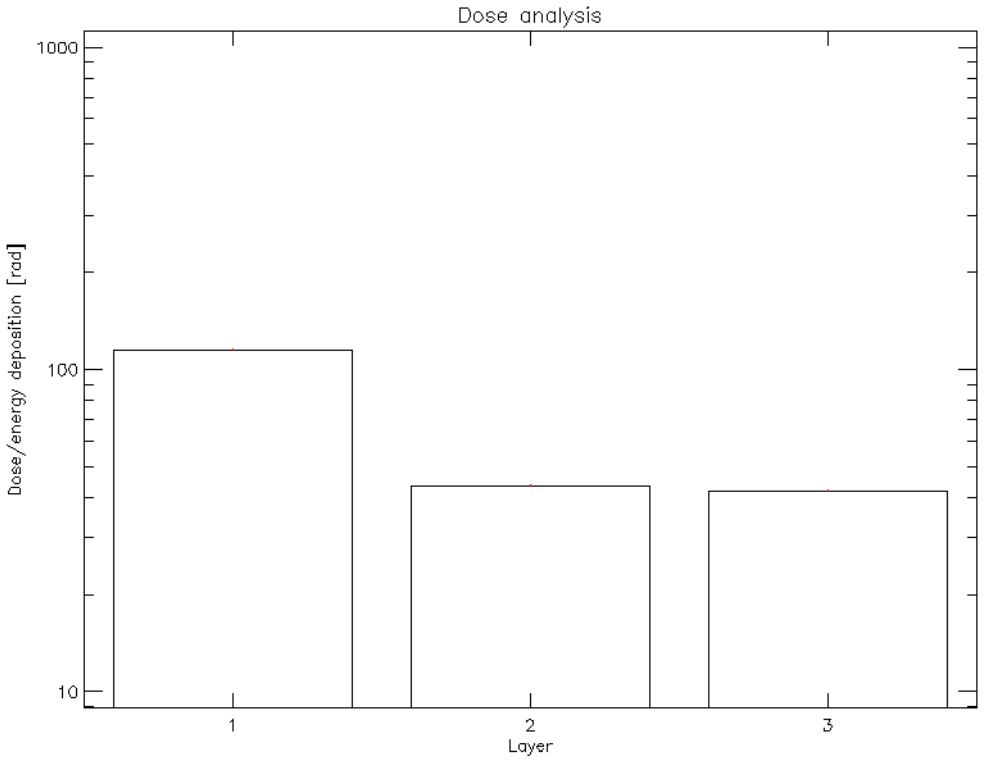


Figure 53: Proton dose deposition by layer in SPENVIS (1: HDPE; 2: Ti; 3: Al; 4: Si detector).

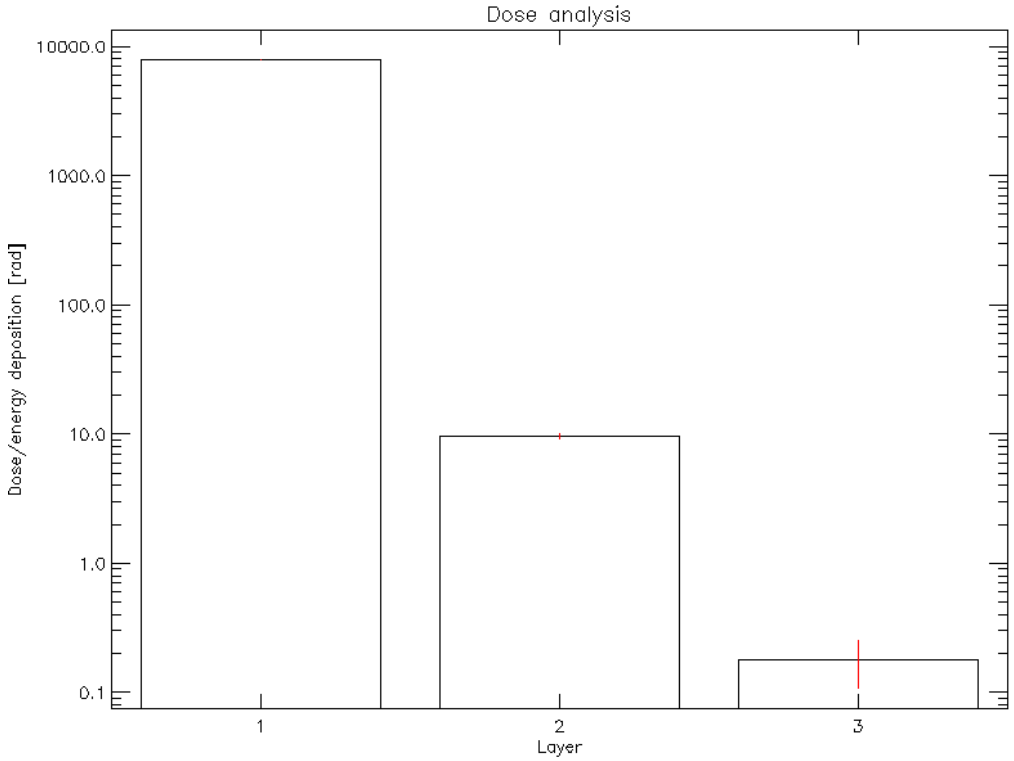


Figure 54: High-energy electron dose deposition by layer in SPENVIS (1: HDPE; 2: Ti; 3: Al; 4: Si detector).

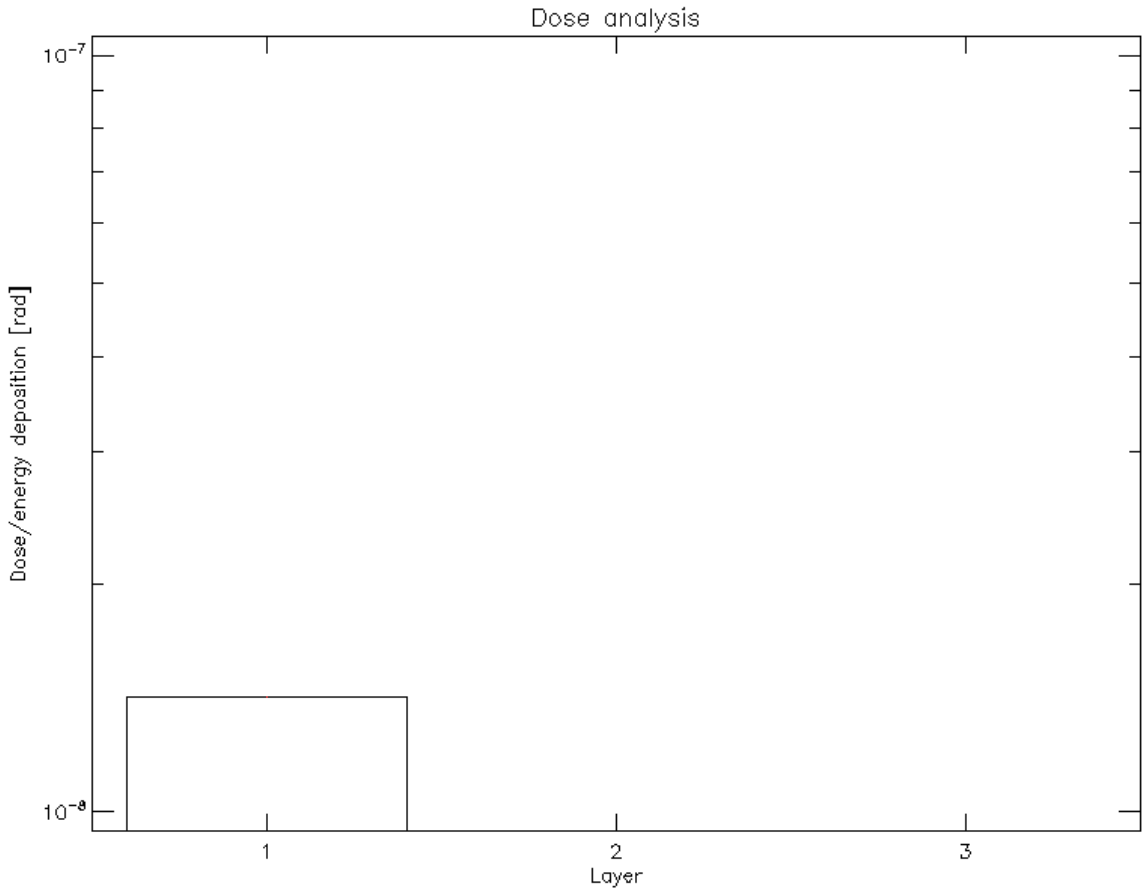


Figure 55: Dose deposition from solar proton fluence by layer in SPENVIS (1: HDPE; 2: Ti; 3: Al; 4: Si detector).

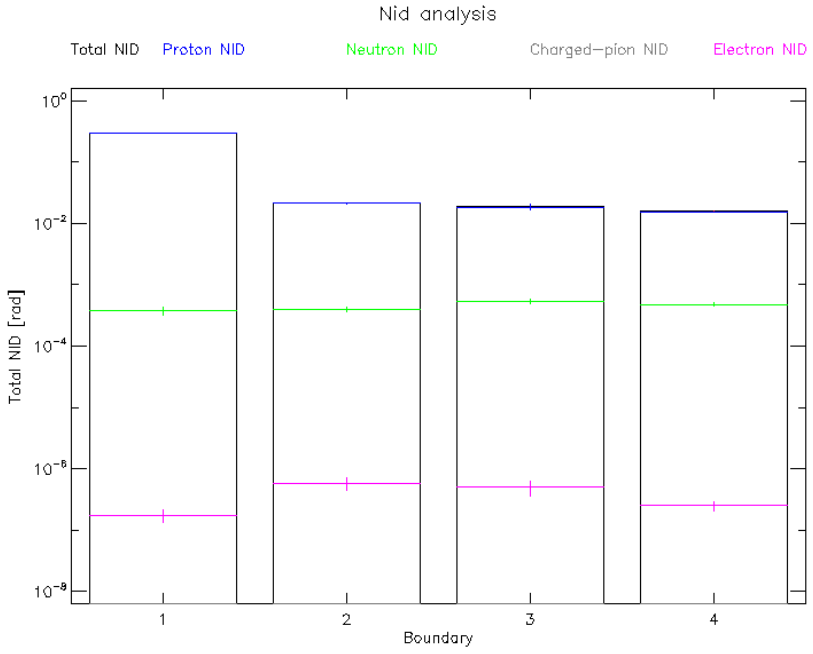


Figure 56: NIEL simulation results.

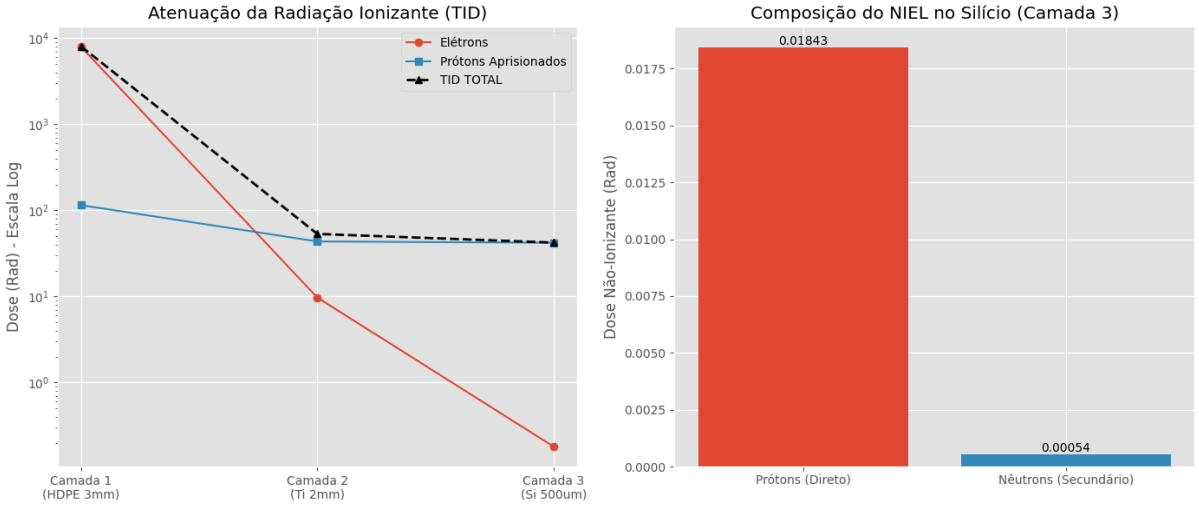


Figure 57: Total ionizing dose (TID) analysis performed using Monte Carlo simulations.

4.3 Conclusions and Design

The total ionizing dose (TID) analysis indicated that the cumulative dose in the active silicon layer, representing the onboard avionics, remains below 100 rad over the mission lifetime, supporting the effectiveness of the proposed shielding architecture. Based on these results, future mass and geometry optimization for this LEO mission should consider increasing the thickness of the hydrogen-rich polymer layer (HDPE) while reducing or eliminating the high-Z titanium layer, given its limited contribution to high-energy proton attenuation. Furthermore, the non-ionizing energy loss (NIEL) analysis quantified proton-induced nuclear spallation and showed that secondary-neutron effects are several orders of magnitude smaller than those caused by primary proton interactions. Applying a standard radiation design margin (RDM = 2×), the design successfully complies with all payload survivability requirements.

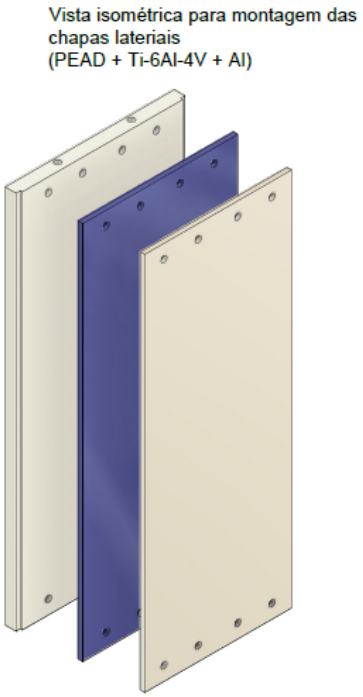


Figure 58: Lateral radiation-shielding assembly.

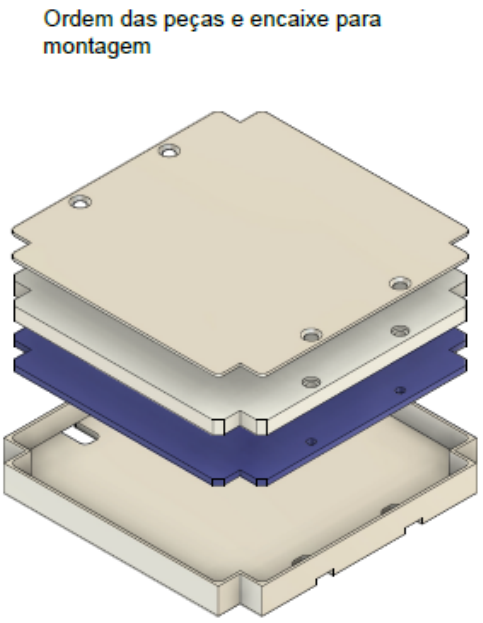


Figure 59: Bottom radiation-shielding assembly.

5. Antenna Deployment Mechanism

5.1 Introduction

Because the antennas developed by the TT&C team were longer than the available CubeSat dimensions, the team developed a dedicated antenna deployment mechanism.

5.2 Standards

The deployment mechanism was designed for installation on the top face of the satellite. Its 100 mm × 100 mm × 16 mm envelope ensures compatibility with the CubeSat structure.

5.3 Project

The first challenge was defining an antenna geometry that would fit within the limited available volume. Based on the design study, the team selected a steel tape-spring antenna with geometric characteristics similar to those of a tape measure.

The deployment system keeps the antenna coiled and locked inside a compartment whose door is secured to the structure by thin Dyneema restraints. When a high voltage is applied to the resistive wire, it melts the Dyneema restraints, releasing the door and allowing the antenna to deploy, as illustrated in the figure below.

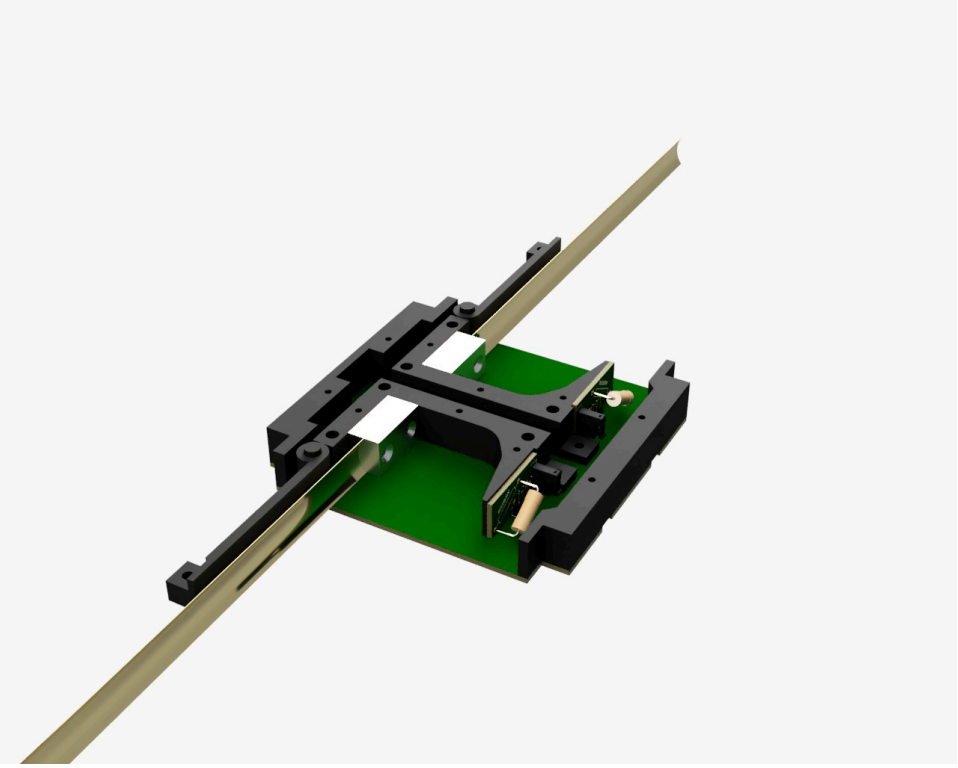


Figure 60: Antenna deployment mechanism CAD model.

6. Outgassing Considerations

One of the team's most recent developments was the detailed selection of satellite materials, considering the outgassing characteristics of each structural component and fastener as well as resistance to radiation and atomic oxygen (ATOX). Of particular relevance to this report is the extensive use of additively manufactured polyphenylene sulfide (PPS) components, because PPS offers high resistance to ultraviolet radiation and low outgassing. This includes parts for the antenna deployment mechanism, as well as secondary structural components such as spacers, ADCS motor mounts, and battery brackets.

In other areas, financial constraints prevented the team from applying strict material-level outgassing limits. Instead, the team plans to perform a thermal-vacuum (TVAC) bake-out to remove volatile compounds. Assembly materials identified as potential outgassing sources include Loctite 242 threadlocker used to secure fasteners, Araldite Professional 24h adhesive used to mount the ADCS motor to a flange, and thermal paste used in the battery-heating system.

B. Thermal Control System (TCS):

Battery thermal control is essential to the safety, performance, and service life of the satellite power system.
The battery will have an enclosure of aluminum 7075 and will be heated by an electrically controlled resistor. A temperature of 25 °C is used for the cold-case Biot assessment. The battery inside the enclosure may exhibit a radial temperature distribution.
The high thermal conductivity of aluminum motivates representing the enclosure by a single temperature, T_Al(t). This assumption is initially assessed using an equivalent Biot indicator evaluated for the following controlled cold case:

$$T_{\mathrm{Al,Bi}} = 25\,^{\circ}\mathrm{C}, \qquad T_{\mathrm{ext,Bi}} = -10\,^{\circ}\mathrm{C}.$$

This case gives a temperature difference of $\Delta T_{Bi} = 35\ K$ between the aluminum and the cold environment. Its designation as critical applies to this preliminary cold-case assessment; the influence of the hot case, orbital conditions, material properties, and thermal-contact uncertainties will not be assessed in this work.
For the simplified model, the enclosure will have a square cross section with a central cylindrical cavity. Two auxiliary hollow cylinders, one inscribed in the square and one circumscribed around it, are used to assess geometric sensitivity. The resulting inequalities bound the defined equivalent Biot indicator under the stated assumptions; they do not directly bound the actual temperature field.

The battery is not assumed to be isothermal. Its flat ends are approximated as adiabatic. With axially and circumferentially uniform conditions and an axially uniform initial temperature, its temperature field is represented by T_bat(r, t). A Bessel-function expansion describes this radial field and couples it to the lumped aluminum node.
The objective is to develop a traceable preliminary model that can:

- evaluate the hypothesis of uniform temperature in aluminum;
- document the thermal case used in the calculation of Biot;
- represent the radial gradient of temperature in the battery;
- incorporate an arbitrary heater-power command;
- support numerical verification and subsequent identification of the assembly's thermal parameters.

1. Vacuum thermal cycle and design case

1.1 Preliminary thermal profile

Figure 61 shows the two consecutive thermal cycles defined by the preliminary requirements. A 60 °C temperature change at 1.5 °C/min requires a 40-minute ramp. The intermediate cold plateau is shared by the two cycles. The plot represents the prescribed environmental temperature, rather than the calculated aluminum or battery response.

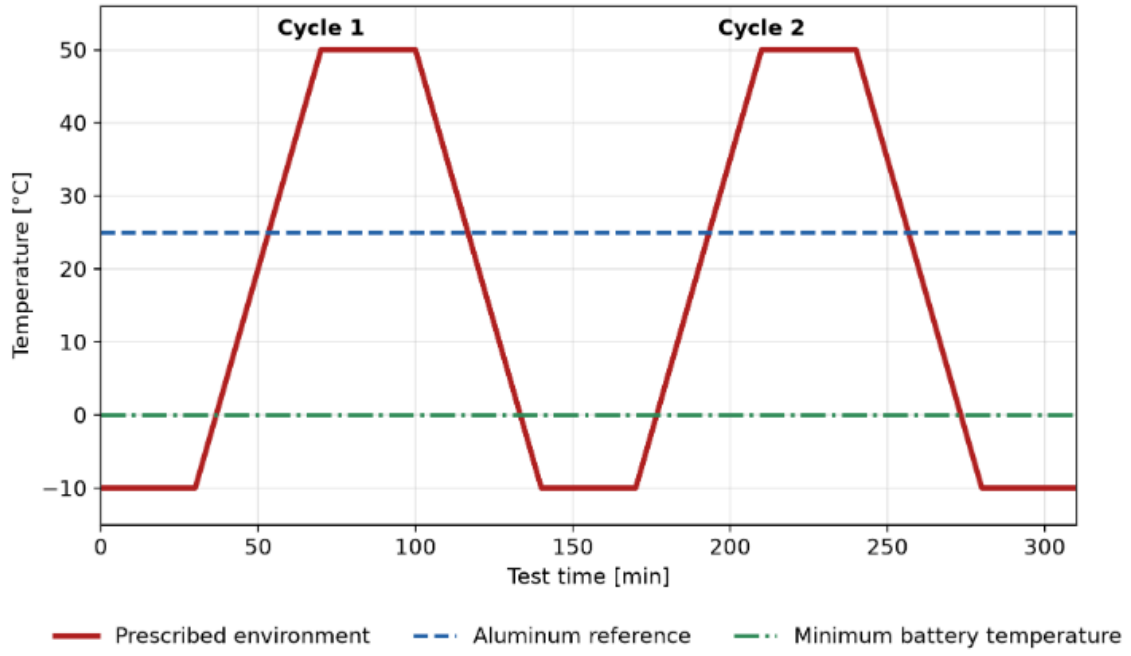


Figure 61: Two preliminary thermal-vacuum cycles. The red curve is the prescribed environment, the blue line is the preliminary aluminum setpoint, and the green line is the minimum battery-temperature requirement.

1.2 Cold case used for the Biot assessment

$$T_{\mathrm{Al,Bi}} = 25\,^{\circ}\mathrm{C}, \qquad T_{\mathrm{ext,Bi}} = -10\,^{\circ}\mathrm{C}, \qquad \Delta T_{\mathrm{Bi}} = T_{\mathrm{Al,Bi}} - T_{\mathrm{ext,Bi}} = 35\ \mathrm{K}.$$

$\Delta T_{Al,Bi}$ denotes the enclosure temperature in the Biot assessment, and $T_{ext,Bi}$ denotes the external temperature at the cold plateau. Here, $T_{ext,Bi}$ also represents the effective radiative surroundings temperature, T_{∞}. These temperatures are used to evaluate the secant radiation coefficient and the equivalent external conductance.
The +50 °C plateau is retained in the transient test profile but is not treated as a separate heater-sizing case. This study develops a preliminary heating-control model to keep the battery above its minimum operating temperature. Active control of the maximum temperature is outside the present scope.

2. Physical and geometric description

2.1 Original geometry

The physical assembly consists of battery cells housed in an aluminum block that conducts and distributes heat.

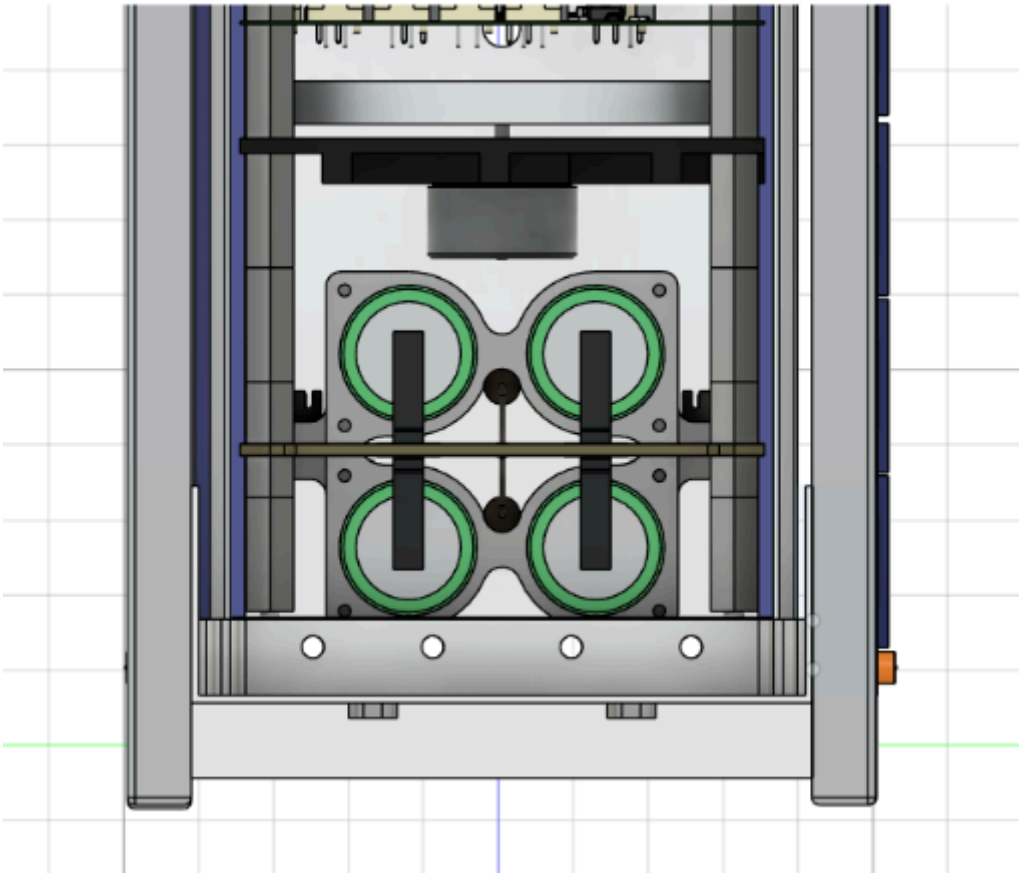


Figure 62: Battery assembly and aluminum enclosure.

A simplified geometry is first introduced to support an approximate analytical formulation.

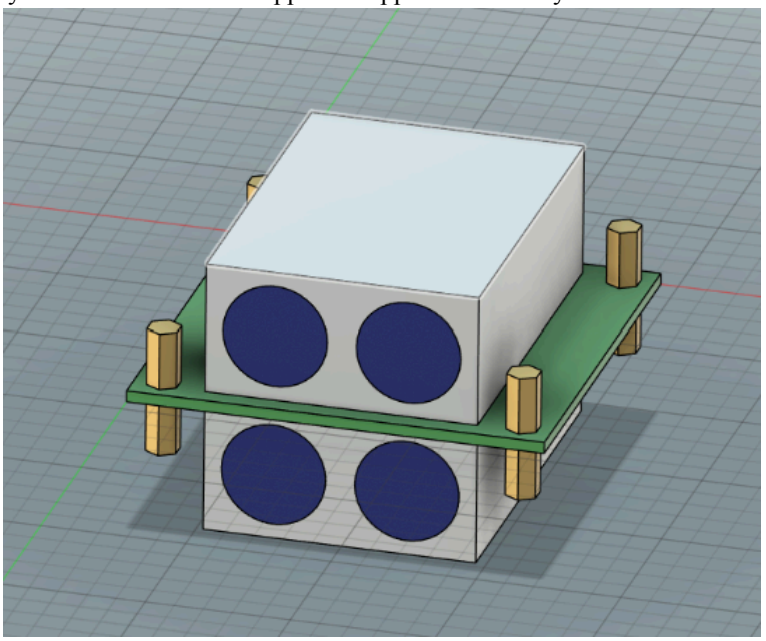


Figure 63: Geometry of the simplified model.

Geometric and thermal symmetry are then used to reduce the model domain.

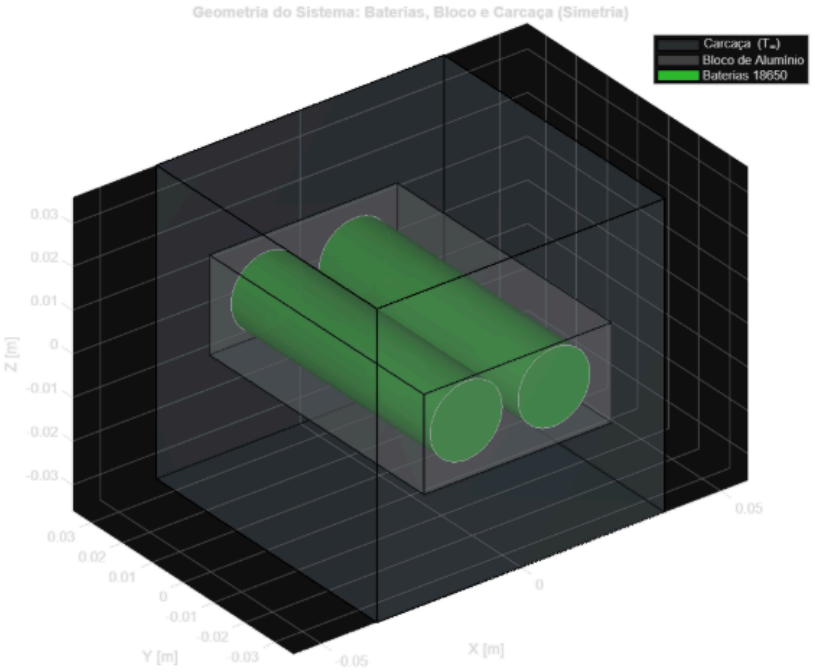


Figure 64: Reduced geometry of the assembly after application of symmetry.

Heat exchange between the PCB and the aluminum–battery assembly is applied to the lower face of the two-battery domain. By symmetry, the total heat-transfer rate is

$$Q_{ctrl,maxPCB,2} = 2Q_{ctrl,maxPCB,1}.$$

A second symmetry plane reduces the domain to a single battery. The cut surface is represented by a zero-normal-heat-flux condition, provided that the materials, heat sources, and other boundary conditions are also symmetric:

$$-k_{\mathrm{Al}} \frac{\partial T}{\partial n}\bigg|_{\Gamma_{\mathrm{sim}}} = 0.$$

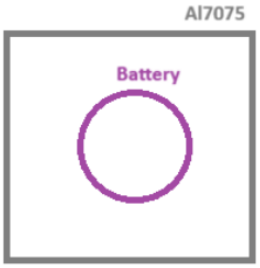


Figure 65: Simplified thermal model: a square-section block containing a central cylindrical cavity.

3. Boundary conditions

The outer lateral area of the square block is

$$A_{o,\mathrm{quad}} = 4aL.$$

Here, “a” is the square-section side length and “L” is the longitudinal length. Each of the four lateral faces has area “aL” and accounts for 25% of $A_{o,quad}$. The external boundary conditions are assigned as follows:

- two radiating faces: $A_{rad} = 2aL = 0.50A_{o,quad}$;
- one face connected to the effective aluminum–PCB–environment path: $A_{PCB} = aL = 0.25A_{o,quad}$;
- one adiabatic face: $A_{ad} = aL = 0.25A_{o,quad}$.

These percentages refer only to the four external lateral faces of the rectangular block. They exclude the cylindrical battery interface and the two flat end faces. The longitudinal ends are approximated as adiabatic; this assumption must be checked against heat transfer through the 3D-printed supports, electrical terminals, and other solid connections.

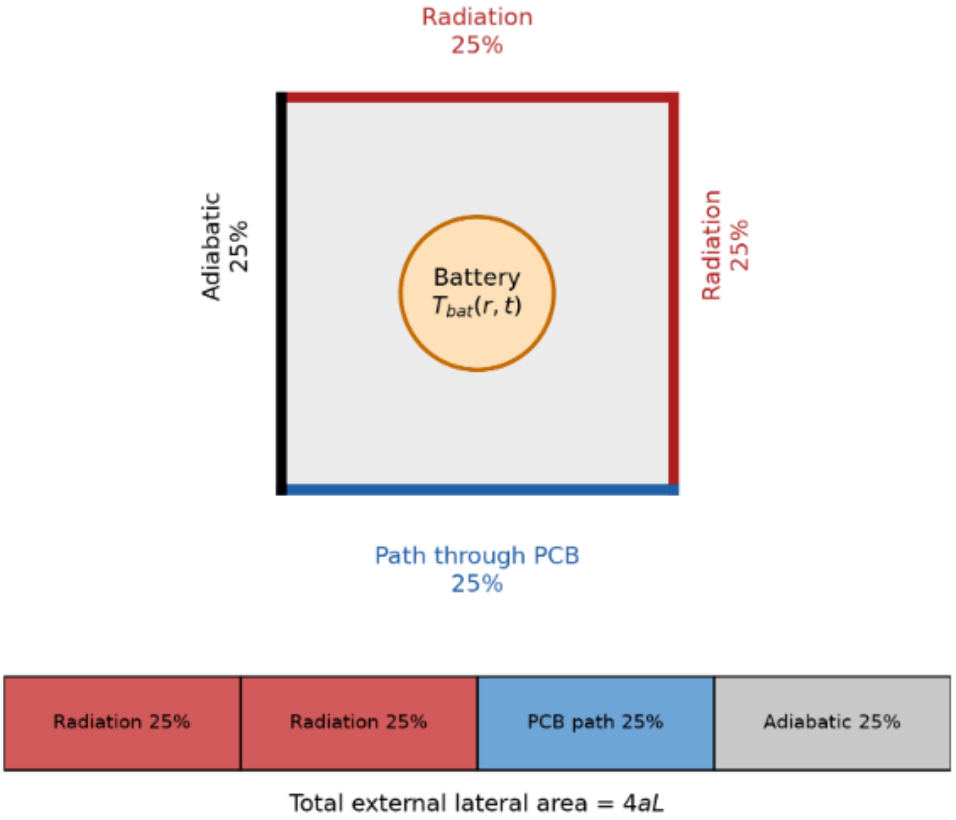


Figure 66: Boundary conditions on the four external lateral faces. Each face has area aL and represents 25% of $A_{o,quad}$

The radiative surroundings seen by the aluminum and by the PCB are represented by the same prescribed temperature, $T_{ext}(t)$. The PCB connection is represented by an effective thermal path that includes conduction through solids and contacts, followed by radiative heat rejection from the PCB.

Symbol	Meaning
$T_{ext}(t)$	external thermal-environment temperature
$T_{Al}(t)$	mean aluminum-enclosure temperature
$T_{Al,ref}(t)$	aluminum control setpoint
$T_{bat}(r,t)$	radial battery temperature
$T_{bat,c}(t)$	battery-center temperature, r=0
$T_{bat,s}(t)$	battery-surface temperature, r=R_{bat}

Symbol	Meaning
$T_{Al,Bi}$	aluminum temperature used in the Biot calculation
$T_{ext,Bi}$	external temperature used in the Biot calculation

Table 18 - Temperatures employed in the model.

For the cold case used in the Biot assessment,

$$T_{\mathrm{Al,Bi},K} = 25 + 273{,}15 = 298{,}15\ \mathrm{K},$$
$$T_{\mathrm{ext,Bi},K} = -10 + 273{,}15 = 263{,}15\ \mathrm{K}.$$

The temperature difference, taken as positive from the aluminum toward the environment, is

$$\Delta T_{\mathrm{Bi}} = T_{\mathrm{Al,Bi}} - T_{\mathrm{ext,Bi}} = 35\ \mathrm{K}.$$

Absolute temperatures are used in the radiation terms. Temperature differences have the same numerical value when expressed in kelvin or degrees Celsius.
The controller model requires only the result of the lumped-capacitance assessment. The complete derivation of the radiative resistance, the constant aluminum--PCB equivalent resistance, the analytical Biot bounds and the purpose for that analysis is retained in Appendix A. Under the stated geometric and boundary-condition assumptions, the result is

$$Bi_{inscribed} < Bi_{square} < Bi_{circumscribed}$$

and all three values remain far below 0.1.

4. Lumped-aluminum justification

4.1 Assessment of the critical cold case adopted

The adopted parameters are a = 24 mm, L = 65 mm, r_i = 9 mm, k_{Al} = 130 W/(m·K), ε = 0.8, and $R_{Al-PCB,eq}$ = 96.946 K/W. The screening case is

$$T_{\mathrm{Al,Bi}} = 25\ ^{\circ}\mathrm{C}, \qquad T_{\mathrm{ext,Bi}} = -10\ ^{\circ}\mathrm{C}.$$

For this pair of temperatures,

$$\boxed{h_{\mathrm{rad,Bi}} = 4{,}026634\ \mathrm{W/(m^2K)}.}$$

Geometry	$R_{int,car}$ (K/W)	$R_{rad,Bi}$ (K/W)	$R_{ext,Bi}$ (K/W)	$Bi_{eq,bi}$
Inscribed	0,0061528	101,348	49,549	1,24177×10^-4
Square	0,0073401	79,598	43,710	1,67927×10^-4
Circumscribed	0,0128748	71,664	41,205	3,12460×10^-4

Table 19: Equivalent Biot indicators for the controlled cold case: $T_{Al,Bi}$ = 25 °C and $T_{ex,Bi}$ = −10 °C.

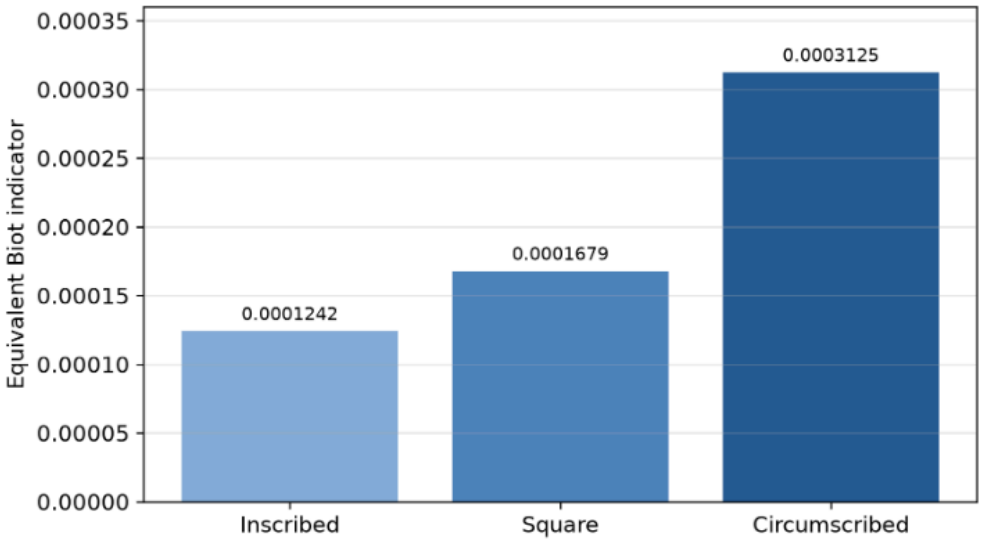


Figure 67: Comparison of the Biot indicators for the selected cold case. The values are specific to the properties and resistances assumed.

All three indicators are well below the conventional criterion Bi < 0.1. This result supports the lumped-aluminum assumption for the external heat-transfer paths included in the screening, but does not independently establish spatial uniformity. Heat exchange with the battery and the localized heater input still require verification. The hot plateau remains part of the transient profile but is not used to size the heater at this stage.

Numerical values confirm the analytically demonstrated ordering:

$$Bi_{\text{ins}} = 1{,}24177 \times 10^{-4} < Bi_{\text{quad}} = 1{,}67927 \times 10^{-4} < Bi_{\text{circ}} = 3{,}12460 \times 10^{-4}.$$

For controller synthesis, the battery is represented by one equivalent thermal node coupled to the lumped aluminum enclosure. The complete radial battery formulation is not discarded: it is retained in Appendix B and is used to verify the internal minimum temperature after the control law has been designed with the reduced-order model. The models from Appendix A and Appendix B will be used for validation of 3d thermal analysis since simplified analytical formulas were found for the problem.

5. Heater Control Model

The control objective is deliberately simple: maintain the aluminum enclosure above a prescribed minimum temperature throughout the thermal- vacuum test. The preliminary requirement is

$$\boxed{T_{\text{Al}}(t) \geq 10\ ^{\circ}\text{C}}$$

The battery is thermally enclosed by the aluminum structure. If the initial battery temperature is admissible, the aluminum boundary remains above the required threshold, and no unmodeled path connects the battery to a colder boundary, then maintaining the enclosure above the threshold is a useful conservative control objective. This statement does not prove that every point inside the battery satisfies the same bound; the previously developed radial model must be used later to verify that conclusion.
Only LQI control is considered in this document and in the MATLAB implementation.

5.1 Digital-twin architecture

The simplified digital twin contains the nonlinear thermal plant, a first-order measurement model, an observer, and an LQI controller. The controller receives only the thermistor temperature and commands a unidirectional heater:

$$\boxed{0 \le \dot{Q}_{\mathrm{ctrl}}(t) \le 3\,\mathrm{W}}$$

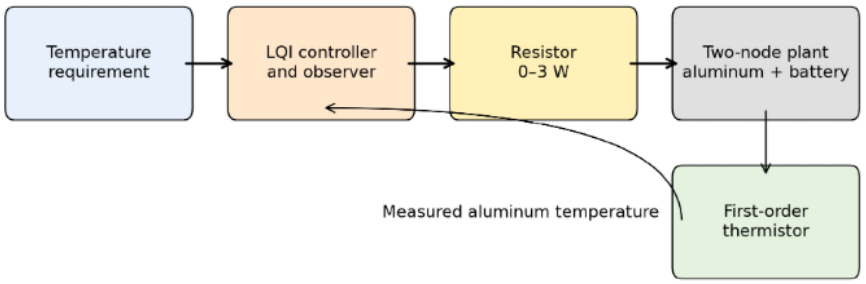


Figure 68: Closed-loop architecture of the simplified digital twin. The battery affects the aluminum dynamics, but only the thermistor attached to the aluminum provides feedback.

5.2 Two-node nonlinear thermal plant

5.2.1 Model assumptions

The aluminum enclosure has a sufficiently small Biot number to be treated as a lumped node. The battery is reduced to one equivalent node only for controller development. Radiative heat transfer remains nonlinear, whereas the identified aluminum--PCB--environment path is represented by the constant resistance $R_{Al-PCB,eq}$. Battery heat generation may be introduced through $Q_{ctrl,maxgen,bat}(t)$.

The thermal capacitances are

$$C_{\mathrm{Al}} = \rho_{\mathrm{Al}} c_{p,\mathrm{Al}} V_{\mathrm{Al}}, \qquad C_{\mathrm{bat}} = \rho_{\mathrm{bat}} c_{p,\mathrm{bat}} V_{\mathrm{bat}}.$$

The effective aluminum--battery resistance is

$$R_{\mathrm{Al-bat}} = \frac{1}{h_c A_c}, \qquad A_c = 2\pi R_{\mathrm{bat}} L$$

5.2.2 Energy balances

With positive heater power entering the aluminum, the coupled balances are

$$\boxed{C_{\mathrm{Al}} \frac{dT_{\mathrm{Al}}}{dt} = \dot{Q}_{\mathrm{ctrl}} - \varepsilon \sigma A_{\mathrm{rad}} \left(T_{\mathrm{Al},K}^4 - T_{\mathrm{ext},K}^4\right) - \frac{T_{\mathrm{Al}} - T_{\mathrm{ext}}}{R_{\mathrm{Al-PCB,eq}}} - \frac{T_{\mathrm{Al}} - T_{\mathrm{bat}}}{R_{\mathrm{Al-bat}}}.}$$

$$\boxed{C_{\mathrm{bat}} \frac{dT_{\mathrm{bat}}}{dt} = \frac{T_{\mathrm{Al}} - T_{\mathrm{bat}}}{R_{\mathrm{Al-bat}}} + \dot{Q}_{\mathrm{gen,bat}}(t).}$$

The battery node provides the dominant coupled thermal load without pretending that the internal battery field is uniform in the final design.

5.2.3 First-order thermistor

The measured temperature is represented by

$$\boxed{\tau_s \frac{dT_{\mathrm{meas}}}{dt} + T_{\mathrm{meas}} = T_{\mathrm{Al}}.}$$

Equivalently,

$$\frac{dT_{\mathrm{meas}}}{dt} = \frac{T_{\mathrm{Al}} - T_{\mathrm{meas}}}{\tau_s}$$

No sensor noise or bias is included at this stage. The time constant τ_s represents the combined response of the sensing element, its attachment, and the local thermal contact.

5.2.4 State-space representation

5.2.4.1 State, input, disturbance, and output

The physical state vector is

$$x = [T_{\mathrm{Al}} \quad T_{\mathrm{bat}} \quad T_{\mathrm{meas}}]^T,$$

The manipulated input is the heater power, $u = Q_{ctrl,max}$; the environmental disturbance is $d = T_{ext}$; and the measured output is

$$y = Cx, \qquad C = [0 \quad 0 \quad 1]$$

5.2.4.2 Linear design model

The controller is designed from a linearization around $T_{Al}=T_{bat}=T_{meas}=T_{ref}$ and $T_{ext}=T_{ext,0}$. Define

$$\begin{aligned} G_{ab} &= \frac{1}{R_{\mathrm{Al-bat}}}, \\ G_{\mathrm{Al}} &= 4\varepsilon\sigma A_{\mathrm{rad}} T_{\mathrm{ref},K}^3 + \frac{1}{R_{\mathrm{Al-PCB,eq}}}, \\ G_{\mathrm{ext}} &= 4\varepsilon\sigma A_{\mathrm{rad}} T_{\mathrm{ext,0},K}^3 + \frac{1}{R_{\mathrm{Al-PCB,eq}}}. \end{aligned}$$

For deviation variables, the linear model is

$$\dot{x} = Ax + Bu + Ed + f_0$$

with

$$A = \begin{bmatrix} -\frac{G_{\mathrm{Al}} + G_{ab}}{C_{\mathrm{Al}}} & \frac{G_{ab}}{C_{\mathrm{Al}}} & 0 \\ \frac{G_{ab}}{C_{\mathrm{bat}}} & -\frac{G_{ab}}{C_{\mathrm{bat}}} & 0 \\ \frac{1}{\tau_s} & 0 & -\frac{1}{\tau_s} \end{bmatrix}, \quad B = \begin{bmatrix} 1/C_{\mathrm{Al}} \\ 0 \\ 0 \end{bmatrix}, \quad E = \begin{bmatrix} G_{\mathrm{ext}}/C_{\mathrm{Al}} \\ 0 \\ 0 \end{bmatrix}.$$

The affine vector f_0 retains the nominal environmental loss that is not eliminated by the selected deviation-coordinate convention. MATLAB prints A, B, E, C, and f_0 explicitly before the transient simulation.

5.3 LQI controller and observer

To remove steady-state error in the measured aluminum temperature, define

$$\dot{\xi} = T_{\mathrm{ref}} - T_{\mathrm{meas}}$$

The augmented matrices are

$$A_a = \begin{bmatrix} A & 0 \\ -C & 0 \end{bmatrix}, \qquad B_a = \begin{bmatrix} B \\ 0 \end{bmatrix}$$

The quadratic cost used to obtain the feedback gain is

$$J = \int_0^\infty \left(x_a^T Q_a x_a + u^T R_u u\right) dt, \qquad x_a = \begin{bmatrix} x^T & \xi \end{bmatrix}^T$$

The unconstrained command is

$$u^* = -K_x \hat{x} - K_i \xi$$

and the physical command applied to the heater is

$$\boxed{u = \min\left[3\,\mathrm{W}, \max(0, u^*)\right].}$$

Conditional integration prevents the integral state from accumulating in a direction that would drive the command further into saturation.

5.4 Parameters and evaluation metrics

Parameter	Value	Meaning
$T_{Al,min}$	10 °C	enclosure requirement
T_{ref}	12 °C	preliminary setpoint
$Q_{ctrl,max}$	3 W	resistor power limit
T_s	15 s	thermistor time constant
$T_{ext,0}$	-10 °C	linearization disturbance
$R_{Al-PCB,eq}$	96.946 K/W	identified external path

Table 20: Preliminary values configured in the MATLAB model.

The script reports heater energy, minimum aluminum temperature, minimum thermal margin, time below the aluminum requirement, minimum equivalent battery temperature, thermistor root-mean-square error, peak heater power, and time spent at the 3 W limit.
The results window contains one tab, labeled LQI. Its upper panel shows the environment, aluminum, equivalent battery, and thermistor temperatures. Its lower panel shows resistor power together with radiative, PCB-path, and aluminum--battery heat-transfer rates.

6. Reduced-order simulation results

The current LQI tuning uses $Q_{ctrl,maxx}$ = diag(1000, 0.01, 100), $q_{integral}$= 2.0e-6, and R_u = 0.01. In the reduced-order simulation, this configuration produced an estimated heater-energy consumption of 0.145 Wh over the complete cycle and a peak heater power of approximately 0.48 W.

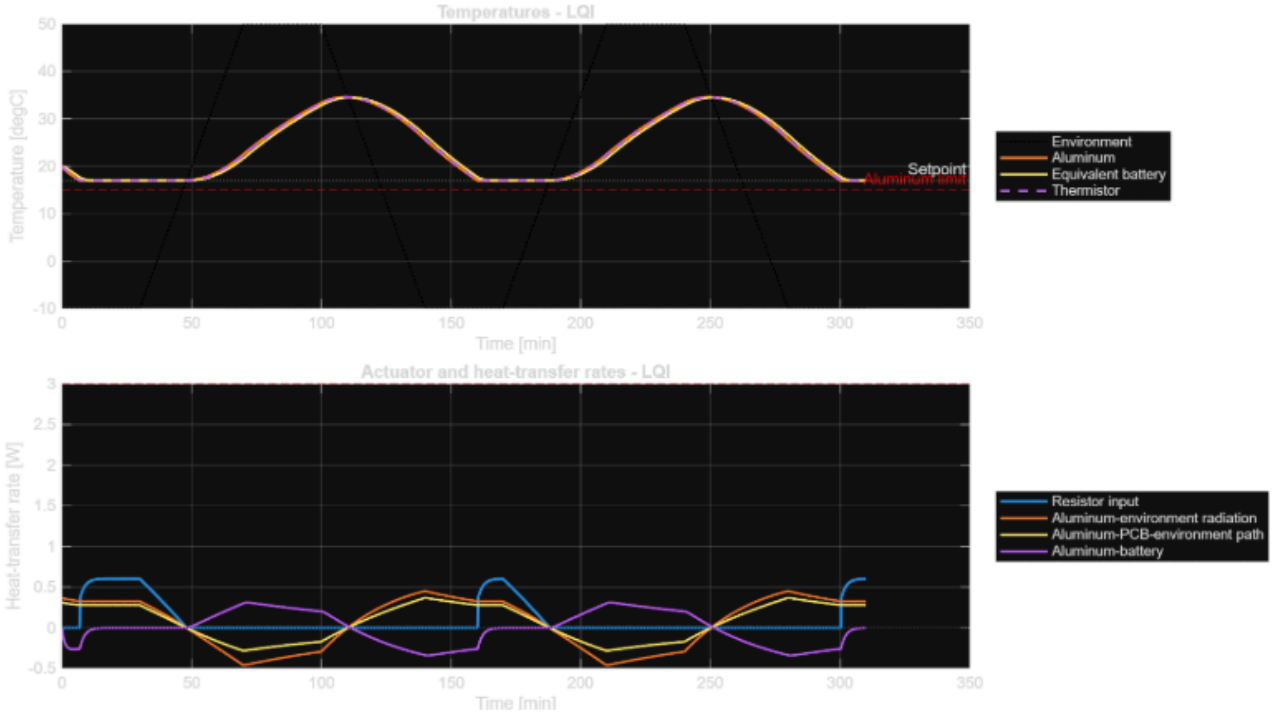
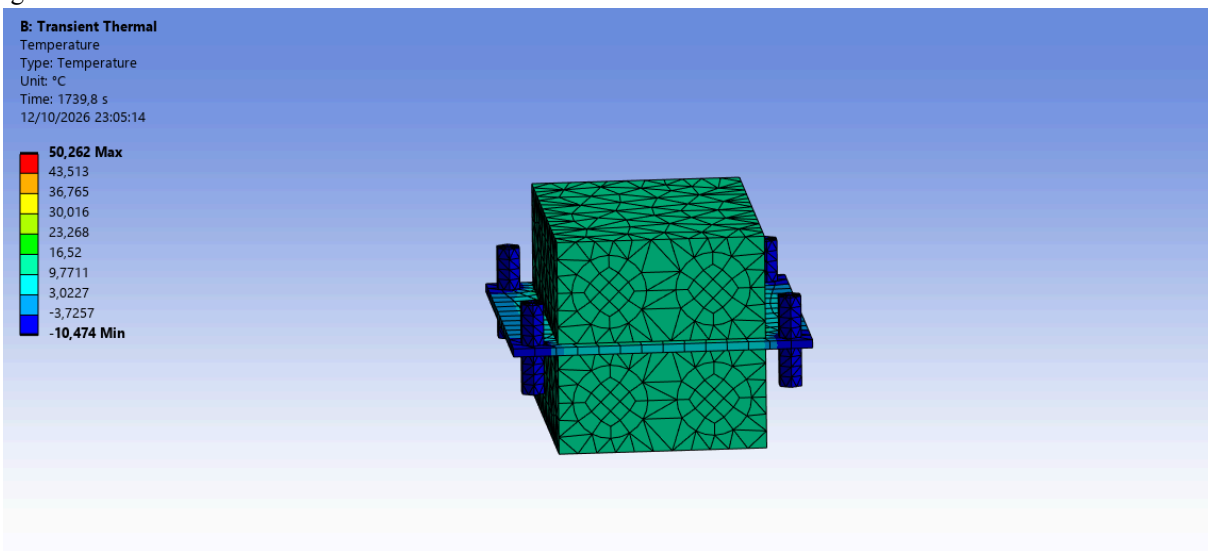


Figure 69: MATLAB simulation results.

Transient thermal simulations were performed in ANSYS using the heater-power history calculated in MATLAB and the prescribed environmental temperature profile. These simulations were used to assess agreement between the reduced-order thermal model and the finite element model.



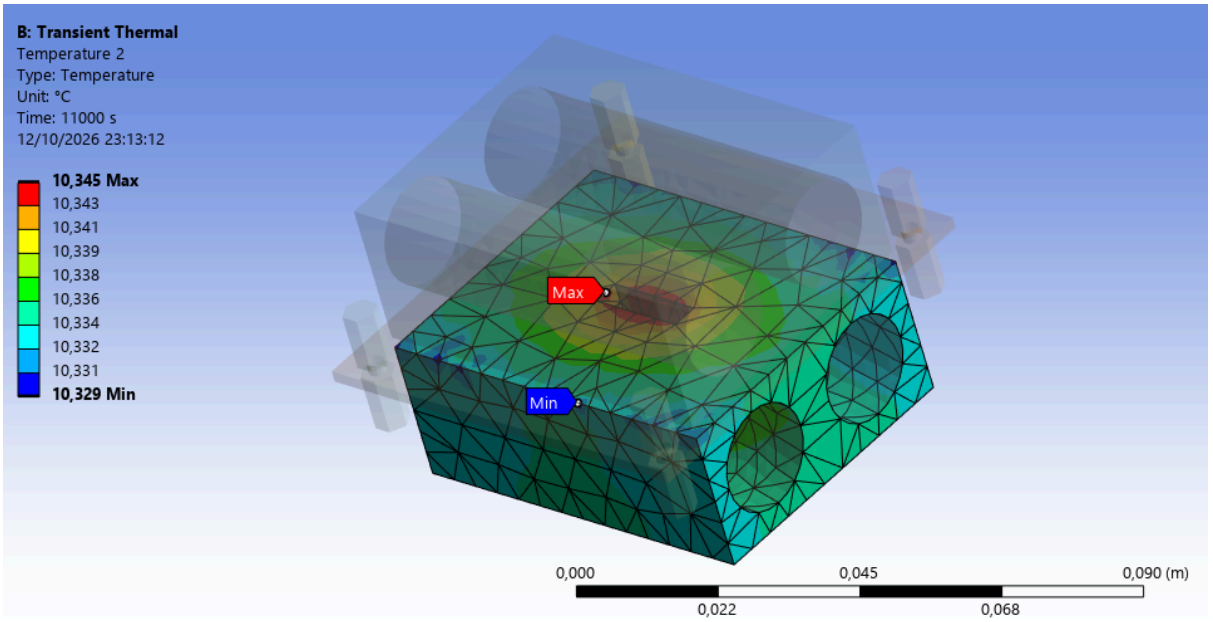


Figure 70: ANSYS transient thermal simulation results.

The lumped MATLAB model and the ANSYS transient thermal model produced comparable temperature responses. The MATLAB controller used a reference of 12 °C, while the reported ANSYS temperature was 10.329 °C. This comparison supports the preliminary usefulness of the simplified model, although the ANSYS simulation replayed the heater history in open loop. Closed-loop performance, in which the ANSYS temperature is fed back to the controller, remains to be evaluated.

Appendix A Equivalent Resistances and Biot Bounds

This appendix contains the complete resistance formulation, geometric comparison, and analytical inequalities supporting the lumped-aluminum assumption used in the control model.

A.1 Equivalent thermal resistances

A.1.1 Radiation

A positive signal is adopted for the heat rate that leaves aluminum towards the environment. In the case used for Biot,

$$\dot{Q}_{\mathrm{rad,Bi}} = \varepsilon\sigma A_{\mathrm{rad}} \left(T^4_{\mathrm{Al,Bi},K} - T^4_{\mathrm{ext,Bi},K}\right)$$

Factoring the difference of fourth powers, linearization is defined secant between the two temperatures of the case analyzed:

$$\boxed{h_{\mathrm{rad,Bi}} = \varepsilon\sigma \left(T_{\mathrm{Al,Bi},K} + T_{\mathrm{ext,Bi},K}\right)\left(T^2_{\mathrm{Al,Bi},K} + T^2_{\mathrm{ext,Bi},K}\right).}$$

Consequently,

$$\dot{Q}_{\mathrm{rad,Bi}} = h_{\mathrm{rad,Bi}} A_{\mathrm{rad}} \left(T_{\mathrm{Al,Bi}} - T_{\mathrm{ext,Bi}}\right),$$

and the secant radiative resistance is

$$\boxed{R_{\mathrm{rad,Bi}} = \frac{1}{h_{\mathrm{rad,Bi}} A_{\mathrm{rad}}}.}$$

In the transient model, radiation is not linearized: it is used directly every instant. A formulation presupposes a factor of unit form and a radioactive environment effective; these hypotheses should be replaced by form factors and the corresponding radiative temperatures in the evolution of the project.

A.1.2 Equivalent conductive path through PCB

In the cold case adopted, the heat rate that leaves aluminum along the way conductive until PCB is represented, in the reduced model, by

$$\boxed{\dot{Q}_{\mathrm{PCB,Bi}} = \frac{T_{\mathrm{Al,Bi}} - T_{\mathrm{ext,Bi}}}{R_{\mathrm{Al-PCB,eq}}}.}$$

$R_{Al-PCB,eq}$,eq in K/W is not a property intrinsic to a single material. This is the equivalent thermal resistance of the conductive path between aluminum and PCB, identified in the cold case of project. Radiative rejection of PCB participates in the boundary condition of ANSYS model used in this identification, but does not transform $R_{Al-PCB,eq}$ in a resistance per area unit. By choice methodologically, its total value is kept constant in the comparison between the studies of the geometries.

A.1.2.1 Conservative identification of equivalent resistance in ANSYS

The ANSYS four battery assembly model uses three conditions main thermals:

1. aluminum maintained at 25 °C;
2. -10 °C radiative PCB environment;
3. other external surfaces of aluminum treated as adiabatic in this identification, because its direct radiation is calculated separately. in the analytical formulation.

So the temperatures used in the identification are

$$T_{\mathrm{Al,Bi}} = 25\,^{\circ}\mathrm{C}, \qquad T_{\mathrm{ext,Bi}} = -10\,^{\circ}\mathrm{C}.$$

Both temperatures are prescribed and constant. PCB rejects heat by radiation to the cold environment, while direct radiation from aluminum is omitted from the ANSYS to avoid double counting. The aim is to identify, in a conservative condition, the total thermal resistance equivalent of aluminum path--PCB--environment.

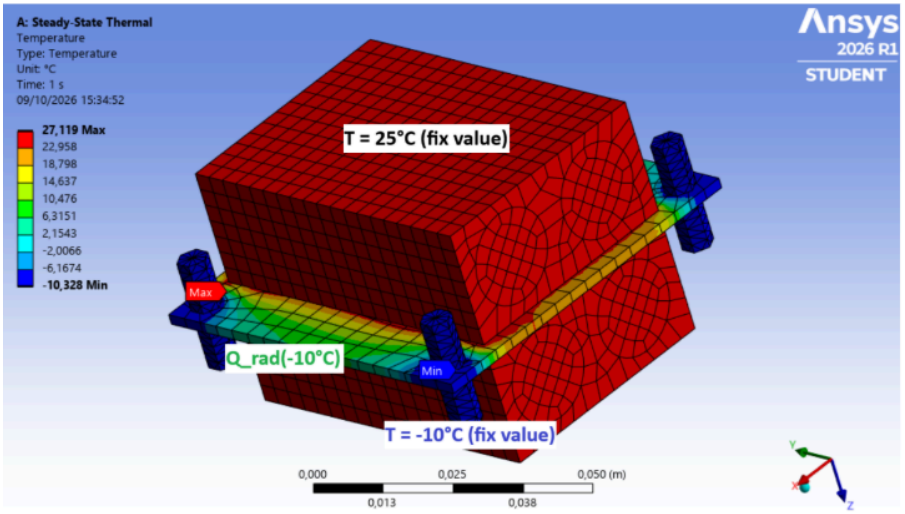


Figure 71: ANSYS model used to identify the equivalent thermal resistance of the complete aluminum--PCB--environment path.

At this point of operation, the total heat rate was obtained. $Q_{ctrl,maxAl\rightarrow PCB,4}$=1.4441 W for the four housings, distributed in two aluminum blocks. Because the four enclosures are thermally equivalent by symmetry, the rate associated with one enclosure is

$$\dot{Q}_{\mathrm{Al\rightarrow PCB,1}} = \frac{1{,}4441}{4} = 0.361025\,\mathrm{W}.$$

With ΔT_{Bi}=25-(-10)=35 K, the result is

$$R_{\mathrm{Al-PCB,eq}} = \frac{\Delta T_{\mathrm{Bi}}}{\dot{Q}_{\mathrm{Al\rightarrow PCB,1}}} = \frac{35}{0{,}361025} = 96.946\,\mathrm{K/W}. \quad (5.8)$$

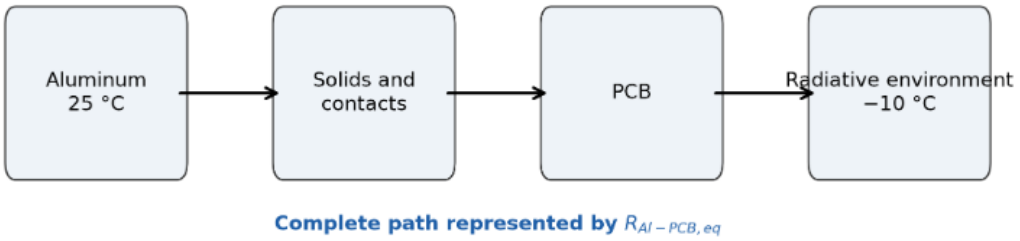


Figure 72: Interpretation of the resistance identified in ANSYS. The equivalent resistance incorporates conduction to the PCB and the PCB's ability to reject heat by radiation at the identification operating point.

Using the lowest plausible value of $R_{Al-PCB,eq}$ maximizes the predicted loss for the same temperature difference. It is therefore conservative for sizing the heater at the cold plateau and also raises the Biot indicator.

In geometric comparison, $R_{Al-PCB,eq}$ is maintained as a constant total thermal resistance; it is not converted into resistance by area unit. The treatment of the radiative branch is different. For the same pair of temperatures, $h_{rad,Bi}$ remains constant, but resistance radiative depends directly on the exposed area:

$$R_{\mathrm{rad,Bi}}(A_{\mathrm{rad}}) = \frac{1}{h_{\mathrm{rad,Bi}} A_{\mathrm{rad}}}, \qquad \frac{\partial R_{\mathrm{rad,Bi}}}{\partial A_{\mathrm{rad}}} = -\frac{1}{h_{\mathrm{rad,Bi}} A_{\mathrm{rad}}^2} < 0.$$

Therefore, radiative resistance decreases when the radiative area increases and It increases when this area decreases. This geometric dependence is preserved in the models inscribed, square and circumscribed.

A. 2 Association of external branches

In the simplified case, direct aluminum radiation and path via PCB connect the aluminum surface to the same thermal node T_ext,Bi. Under the approach of a common surface temperature, the two mechanisms are represented by parallel conductance:

$$\boxed{R_{\text{ext,Bi}} = \left(h_{\text{rad,Bi}} A_{\text{rad}} + \frac{1}{R_{\text{Al–PCB,eq}}} \right)^{-1}.}$$

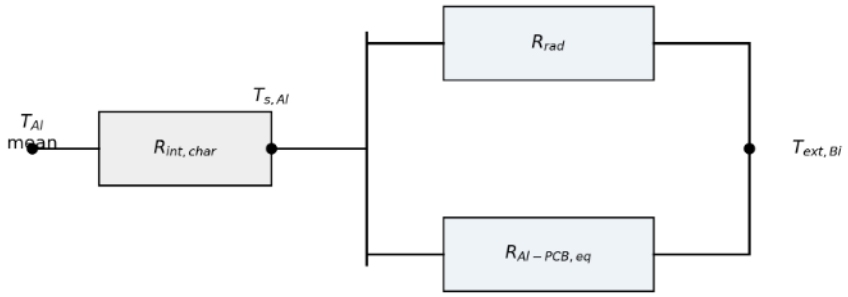


Figure 73: Circuito usado exclusivamente para construir o indicador de Biot the cold case.

A.3 Biot number equivalent in cold case

The external area is defined as active

$$A_{\text{ativa}} = A_{\text{rad}} + A_{\text{PCB}},$$

and the preliminary characteristic length as

$$L_c = \frac{V_{\text{Al}}}{A_{\text{ativa}}}.$$

External conductance of the cold case is

$$G_{\text{ext,Bi}} = h_{\text{rad,Bi}} A_{\text{rad}} + \frac{1}{R_{\text{Al–PCB,eq}}},$$

and the average external coefficient is equivalent

$$h_{\text{eq,Bi}} = \frac{G_{\text{ext,Bi}}}{A_{\text{ativa}}}.$$

The indicator adopted is:

$$\boxed{Bi_{\text{eq,Bi}} = \frac{h_{\text{eq,Bi}} L_c}{k_{\text{Al}}} = \frac{V_{\text{Al}}}{k_{\text{Al}} A_{\text{ativa}}^2} \left(h_{\text{rad,Bi}} A_{\text{rad}} + \frac{1}{R_{\text{Al–PCB,eq}}} \right).}$$

Equivalently, defining the internal resistance scale

$$R_{\text{int,car}} = \frac{V_{\text{Al}}}{k_{\text{Al}} A_{\text{ativa}}^2},$$

leads to

$$Bi_{\mathrm{eq,Bi}} = \frac{R_{\mathrm{int,car}}}{R_{\mathrm{ext,Bi}}}.$$

This number is a global engineering indicator for the cold case $T_{Al,Bi}$=25 °C and $T_{ext,Bi}$=-10 °C. The internal resistance feature is not the exact solution of the three-dimensional field with conditions with non-uniform contour.

In addition, the above indicator considers the external paths used in initial screening. Before concluding definitively that aluminum is isothermal, the exchange with the battery should also be checked, the distribution spatial resistor and thermal diffusion times in aluminum by means of a more detailed simulation.

A.4 Geometric comparison

A.4.1 Physical and mathematical basis of comparison

The number of Biot, understood as the ratio between a resistance scale internal conduction and an external heat exchange resistance scale, can be defined for general geometries. However, the direct relationship between a single value of Bi, internal temperature distribution and analytical transient solutions is more rigorous in canonical geometries one-dimensional: extensive flat plate, cylinder long enough and sphere. In these cases, the field depends, respectively, only on the coordinate through the radial thickness or coordinate, and the eigenvalues of the eigenvalue equation driving are known for uniform boundary conditions.

The actual section of this work is square and contains a circular cavity. Therefore, its four external faces are subjected to different conditions. Consequently, the temperature field in the section is generally two-dimensional: it depends on x and y or equivalently on r and θ. A single Biot number calculated directly for the parallelepiped remains useful as a global screening indicator, but does not alone describe the position or magnitude of the highest internal gradient. In particular, it does not replace the solution of the conduction field in the close to transitions between radiative faces, the face connected to PCB and the adiabatic face.

To give the indicator a canonical geometric reference and allow an analytical comparison, the square section is associated with two cylinders. auxiliary hollows. In these equivalent models, external exchanges are gathered in an average conductance based on active areas. This one. homogenization is the modeling step; it does not transform the real field of a block into an exact radial field. Once you accept the hypotheses stated, the algebraic expressions of the indicators and the inequalities demonstrated in this section are exact within the equivalent model.

To investigate the sensitivity of the Biot indicator to the external dimension, replace the square section with two auxiliary hollow cylinders. The first is inscribed in the square and has the smallest external radius compatible with the side a. The second is circumscribed squared and has radius equal to half of the diagonal. Thus, the two models represent geometrical extremes. within the cylindrical family considered:

$$\boxed{r_{o,\mathrm{ins}} = \frac{a}{2}}, \qquad \boxed{r_{o,\mathrm{circ}} = \frac{a\sqrt{2}}{2} = \frac{a}{\sqrt{2}}}.$$

Consequently,

$$r_{o,\mathrm{ins}} < r_{o,\mathrm{circ}}.$$

This construction is not just used as a visual comparison of volumes. Constantly maintaining the cavity, length, physical area of the path via PCB and the total resistance of this path, it becomes possible to compare term-to-term indicators of the three geometries. The demonstration presented below establishes that the circumscribed cylinder provides a limit higher for the equivalent Biot indicator of the square block. Stopping the dimensions of this design, the inscribed cylinder also provides a lower limit. These limits should not be confused with specific limits for the temperature field or the maximum gradient of the two-dimensional solution.

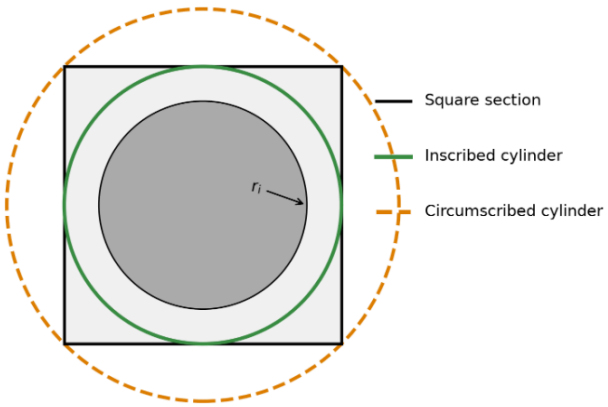


Figure 84: Square section and hollow cylinders inscribed and circumscribed.

A.4.2 Comparison of aluminum volumes

L longitudinal length and r_i cavity radius, maintained equal in all three models. The volumes of aluminum are

$$V_{\text{ins}} = \pi \left(\frac{a^2}{4} - r_i^2 \right) L,$$
$$V_{\text{quad}} = (a^2 - \pi r_i^2) L,$$
$$V_{\text{circ}} = \pi \left(\frac{a^2}{2} - r_i^2 \right) L.$$

Since the cavity fits into the inscribed cylinder, i.e. $2r_i < a$, it works

$$\boxed{V_{\text{ins}} < V_{\text{quad}} < V_{\text{circ}}}.$$

This inequality has an immediate interpretation: the cylinder inscribed removes the regions near the vertices of the square, while the cylinder circumscribed adds material beyond the faces of the square. However, Biot number not only depends on volume; it also depends on area active and external conductance. Therefore, the ordering of volumes, by itself, just doesn't show the sorting of Biot numbers. The proof requires the full expressions presented in the following subsections, in which they are considered simultaneously the volume, the variable radiative area and the branch via PCB kept constant.

A.5 Formulation for a generic hollow cylinder

Consider an external ray hollow cylinder r_o, internal radius r_i and length L. Its aluminum volume and its total external lateral area are

$$V = \pi(r_o^2 - r_i^2)L, \qquad A_o = 2\pi r_o L.$$

In all auxiliary cylinders, half of the lateral surface area is assigned to direct radiation. The thermal path connecting the enclosure to the PCB, however, remains the same as in the reduced square-section model. Therefore, its reference area and total thermal resistance are kept constant:

$$A_{\text{rad}} = \frac{1}{2} A_o = \pi r_o L,$$
$$A_{\text{PCB}} = aL,$$
$$A_{\text{ad}} = A_o - A_{\text{rad}} - A_{\text{PCB}} = (\pi r_o - a)L.$$

Since $r_o \geq$ a/2 for the cylinders considered, the adiabatic area A_{ad}>0. Only direct radiation and heat transfer through the PCB path are included in the external heat exchange with the environment at temperature $T_{ext,Bi}$.

$$A_{\text{ativa}} = A_{\text{rad}} + A_{\text{PCB}} = (\pi r_o + a)L.$$

The assumption A_{PCB}=aL does not imply that the PCB path occupies 25% of the lateral area of each auxiliary cylinder. Instead, it preserves the same physical conductive path as in the reduced square-section model. Consequently, and, for the same temperature difference, the corresponding heat-transfer rate through the PCB path remains constant. In contrast, the radiating area A_{rad} =π r_oL varies with r_o thereby changing the radiative resistance.

The corresponding characteristic length is given by:

$$\begin{aligned} L_c &= \frac{V}{A_{\text{ativa}}} \\ &= \frac{\pi(r_o^2 - r_i^2)L}{(\pi r_o + a)L}, \end{aligned}$$

The total external thermal conductance is the sum of the conductances of the parallel heat-transfer paths:

$$\boxed{L_c = \frac{\pi(r_o^2 - r_i^2)}{\pi r_o + a}}.$$

Total external conductance is the sum of the conductances of the branches in parallel:

$$\frac{1}{R_{\text{ext,Bi}}} = h_{\text{rad,Bi}} \pi r_o L + \frac{1}{R_{\text{Al-PCB,eq}}}.$$

Replacing V, $A_{ativa}$and external conductance in the definition from the equivalent biot, we obtain

$$\boxed{Bi_{\text{eq,Bi}}(r_o) = \frac{\pi(r_o^2 - r_i^2)}{k_{\text{Al}} L(\pi r_o + a)^2} \left(h_{\text{rad,Bi}} \pi r_o L + \frac{1}{R_{\text{Al-PCB,eq}}} \right).}$$

A.6 Exact analytical comparison with square section

Battery diameter is defined as d_B=2r_i and geometric ratio

$$\boxed{\nu = \frac{d_B}{a} = \frac{2r_i}{a}}, \qquad 0 \le \nu < 1.$$

The <1 condition ensures that the cylindrical cavity fits in the section square and inscribed cylinder. To simplify writing, it is still defined

$$G_{\text{PCB}} = \frac{1}{R_{\text{Al-PCB,eq}}}.$$

To the square section,

$$\boxed{Bi_{\text{quad}} = \frac{1 - \pi\nu^2/4}{9k_{\text{Al}}L}(2h_{\text{rad,Bi}}aL + G_{\text{PCB}}).}$$

A.7 Circumscribed cylinder

For $r_{o,circ}$=a/2,

$$A_{\mathrm{rad,circ}} = \frac{\pi a L}{\sqrt{2}},$$
$$A_{\mathrm{ativa,circ}} = \left(1 + \frac{\pi}{\sqrt{2}}\right) a L,$$
$$V_{\mathrm{circ}} = \frac{\pi a^2}{4}(2 - \nu^2) L.$$

Substituting the corresponding geometric expressions gives:

$$\boxed{Bi_{\mathrm{circ}} = \frac{\pi(2-\nu^2)}{4k_{\mathrm{Al}} L(1+\pi/\sqrt{2})^2}\left(h_{\mathrm{rad,Bi}}\frac{\pi a L}{\sqrt{2}} + G_{\mathrm{PCB}}\right).}$$

$$Bi_{\mathrm{circ}} - Bi_{\mathrm{quad}} = \frac{h_{\mathrm{rad,Bi}} a}{k_{\mathrm{Al}}} F_c(\nu) + \frac{G_{\mathrm{PCB}}}{k_{\mathrm{Al}} L} H_c(\nu),$$

where

$$F_c(\nu) = 0{,}1140216 + 0{,}0064110\nu^2,$$
$$H_c(\nu) = 0{,}0402518 + 0{,}0115850\nu^2.$$

Since $F_c(v)$>0 and $H_c(v)$>0 throughout the admissible range (0<v<1), the resulting inequality holds even when the PCB heat-transfer path is included.

$$\boxed{Bi_{\mathrm{circ}} > Bi_{\mathrm{quad}}.}$$

Therefore, under the assumptions adopted in this analysis, the circumscribed-cylinder model provides a more conservative estimate of the equivalent Biot indicator than the square-section model.

A.8 Cylinder inscribed

For $r_{o,ins}$=a/2,

$$A_{\mathrm{rad,ins}} = \frac{\pi a L}{2},$$
$$A_{\mathrm{ativa,ins}} = \left(1 + \frac{\pi}{2}\right) a L,$$
$$V_{\mathrm{ins}} = \frac{\pi a^2}{4}(1 - \nu^2) L.$$

Therefore,

$$\boxed{Bi_{\mathrm{ins}} = \frac{\pi(1-\nu^2)}{4k_{\mathrm{Al}} L(1+\pi/2)^2}\left(h_{\mathrm{rad,Bi}}\frac{\pi a L}{2} + G_{\mathrm{PCB}}\right).}$$

$$Bi_{\mathrm{ins}} - Bi_{\mathrm{quad}} = \frac{h_{\mathrm{rad,Bi}} a}{k_{\mathrm{Al}}} F_i(\nu) + \frac{G_{\mathrm{PCB}}}{k_{\mathrm{Al}} L} H_i(\nu),$$

with

$$F_i(\nu) = -0{,}0355523 - 0{,}0121370\nu^2,$$
$$H_i(\nu) = \quad 0{,}0077267 - 0{,}0315713\nu^2.$$

$F_i(v)$ is always negative. Besides,

$$H_i(\nu) < 0 \quad \text{se} \quad \nu > \nu_0, \qquad \nu_0 = 0{,}494709.$$

In this project,

$$\nu = \frac{d_B}{a} = \frac{18}{24} = 0{,}75 > \nu_0.$$

Consequently, the radiative and conductive parcels of the difference are negative, regardless of the relative magnitude between them. So,

$$\boxed{Bi_{\text{ins}} < Bi_{\text{quad}}.}$$

For 0≤v≤_0, the sign of the difference is not determined by geometry alone; it depends on the relative magnitudes of the PCB-path conductance and the radiative conductance. This limitation does not affect the geometry considered in the present design, for which (v=0.75).
Combining the preceding equations yields the following result for the dimensions adopted in this project:

$$\boxed{Bi_{\text{ins}} < Bi_{\text{quad}} < Bi_{\text{circ}}.}$$

A.9 Interpretation and scope of inequality

$$Bi_{\text{quad}} < Bi_{\text{circ}}$$

The inequality established above does not follow solely from the fact that the circumscribed cylinder encloses the largest volume. Rather, it results from the complete algebraic difference between the equivalent Biot indicators, accounting for changes in radiating area and the constant conductance of the PCB heat-transfer path.

Since the coefficients $F_c(v)$ and $H_c(v)$ are positive throughout the admissible geometric domain v≤1, are positive throughout the admissible geometric domain (v\leq1), the result holds for positive values of h_{rad} ,Bi, k_{Al}, L and $R_{Al-PCB,eq}$ provided that the assumptions of the model are satisfied.

Therefore, the circumscribed-cylinder model establishes an analytical upper bound on the equivalent Biot indicator defined in this study. In particular,

$$\boxed{Bi_{\text{circ}} < 0{,}1 \quad \Longrightarrow \quad Bi_{\text{quad}} < 0{,}1.}$$

This result provides a mathematically traceable, conservative criterion for assessing the lumped-temperature assumption for the aluminum enclosure. The converse does not necessarily hold: if Bi_{circ} ≥0.1, this result alone is insufficient to conclude that the square-section model also violates the criterion.

The physical interpretation of this result must be clearly defined. The proof establishes an ordering of the equivalent Biot indicators defined in this document; it does not constitute a comparison theorem for the heat-conduction solutions in the three geometries.

Consequently, it does not establish that the maximum temperature, maximum temperature gradient, or center-to-surface temperature difference in the actual enclosure must lie between the corresponding values for the two cylindrical models. Such conclusions require an analysis of the two-dimensional or three-dimensional temperature field.

The purpose of the cylindrical comparison is to provide a conservative upper bound for the screening indicator under explicitly stated assumptions. The derivation is analytical and can be independently verified.

Appendix B Distributed Radial Battery Model

This appendix retains the complete Bessel-series formulation used to verify the radial battery temperature after controller synthesis.

B.1 Hybrid transient model: concentrated aluminum and distributed battery

B.1.1 Physical structure

The low value of the Biot indicator supports, in this preliminary stage, the representation of the external aluminum housing by a single temperature average $T_{Al(t)}$. This assumption does not imply that the battery is isothermal. Battery temperature may vary in radial direction, even if it is equal for all angles in the same radius:

$$T_{\mathrm{bat}} = T_{\mathrm{bat}}(r,t), \qquad \frac{\partial T_{\mathrm{bat}}}{\partial \theta} = 0.$$

Thus, the appropriate model is hybrid: a lumped node for aluminum and a one-dimensional radial field for the battery. The angular uniformity of surface temperature results from geometric symmetry, uniformity of the contact, and the approximately uniform aluminum temperature. It does not imply that the battery center and surface have the same temperature.

The electrical heater supplies a controllable heating power, Q̇_ctrl,maxctrl(t), to the aluminum enclosure. , to the aluminum enclosure. The enclosure exchanges heat with the external environment at temperature T_ext(t) through direct radiation and an effective thermal path through the PCB. It also exchanges heat with the battery surface. The battery may additionally generate heat internally, represented by the volumetric heat-generation term q'''_bat(r,t).

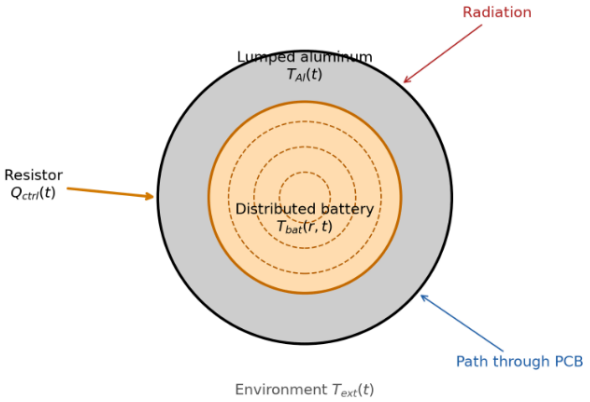


Figure 75: Radial physical model. Aluminum is a lumped node, and the dashed lines are illustrative isotherms of the continuous solution

B.1.2 Battery heat-conduction equation

Admitting isotropic properties and axial and angular symmetry, conduction transient in the battery is governed by

$$\boxed{\rho_{\mathrm{bat}} c_{p,\mathrm{bat}} \frac{\partial T_{\mathrm{bat}}}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left(k_{\mathrm{bat}} r \frac{\partial T_{\mathrm{bat}}}{\partial r}\right) + \dot{q}'''_{\mathrm{bat}}(r,t).}$$

On the battery axis the symmetry condition applies

$$\boxed{\left.\frac{\partial T_{\mathrm{bat}}}{\partial r}\right|_{r=0} = 0.}$$

On the surface $r = R_{bat}$, the exchange with aluminum is represented by a contact coefficient h_c:

$$\boxed{-k_{\text{bat}} \left.\frac{\partial T_{\text{bat}}}{\partial r}\right|_{R_{\text{bat}}} = h_c \left[T_{\text{bat}}(R_{\text{bat}}, t) - T_{\text{Al}}(t)\right].}$$

The cylindrical contact area is

$$A_c = 2\pi R_{\text{bat}} L, \qquad R_c = \frac{1}{h_c A_c}.$$

R_c may be replaced by a measured contact resistance or may include an interface layer. For a homogeneous cylindrical layer between r_1 and r_2,

$$R_{\text{camada}} = \frac{\ln(r_2/r_1)}{2\pi k_{\text{camada}} L}$$

B 1.3 Aluminum energy balance

The concentrated thermal capacity of aluminum is

$$C_{\text{Al}} = \rho_{\text{Al}} V_{\text{Al}} c_{p,\text{Al}}$$

Environmental losses are calculated by

$$\dot{Q}_{\text{rad}} = \varepsilon \sigma A_{\text{rad}} \left(T_{\text{Al},K}^4 - T_{\text{ext},K}^4\right),$$
$$\dot{Q}_{\text{PCB}} = \frac{T_{\text{Al}} - T_{\text{ext}}}{R_{\text{Al-PCB,eq}}}.$$

The total flow of aluminum to the battery can be written on the interface as

$$\dot{Q}_{ab} = h_c A_c \left[T_{\text{Al}} - T_{\text{bat}}(R_{\text{bat}}, t)\right]$$

The balance of the aluminum node is therefore

$$\boxed{C_{\text{Al}} \frac{dT_{\text{Al}}}{dt} = \dot{Q}_{\text{ctrl}}(t) - \dot{Q}_{\text{rad}} - \dot{Q}_{\text{PCB}} - \dot{Q}_{ab}.}$$

B1.4 Bessel function spatial analytical solution

As the two flat ends of π * D^2/4 area are adiabatic, there is no axial flow. If the properties, generation and condition of contour are also uniform in θ, so

$$\frac{\partial T_{\text{bat}}}{\partial z} = 0, \qquad \frac{\partial T_{\text{bat}}}{\partial \theta} = 0.$$

Under these conditions, the finite battery has the same radial equation of the Infinite cylinder. Therefore, it is not necessary to create a radial mesh. A spatial dependence can be represented analytically by eigenfunctions Bessel.

The difference between battery temperature and temperature is defined Aluminum uniform:

$$u(r, t) = T_{\text{bat}}(r, t) - T_{\text{Al}}(t).$$

The surface contact condition becomes homogeneous:

$$-k_{\text{bat}} \left.\frac{\partial u}{\partial r}\right|_{R_{\text{bat}}} = h_c u(R_{\text{bat}}, t).$$

The solution is expanded into cylindrical eigenfunctions,

$$\boxed{u(r,t) = \sum_{n=1}^{\infty} a_n(t) J_0\left(\lambda_n \frac{r}{R_{\text{bat}}}\right),}$$

where J_0 and J_1 are the first species of Bessel. Eigenvalues positive λ_n are the roots of

$$\boxed{\lambda_n J_1(\lambda_n) = Bi_c J_0(\lambda_n), \qquad Bi_c = \frac{h_c R_{\text{bat}}}{k_{\text{bat}}}.}$$

The Bi_c number of this equation refers specifically to the exchange between the battery and aluminum; it is not the same calculated global indicator previously for the aluminum block.

B1.4.1 Modal amplitude equations

Replacing $u = T_{bat-Tal}$, obtained

$$\frac{\partial u}{\partial t} = \alpha_{\text{bat}} \frac{1}{r} \frac{\partial}{\partial r}\left(r \frac{\partial u}{\partial r}\right) + \frac{\dot{q}'''_{\text{bat}}(t)}{\rho_{\text{bat}} c_{p,\text{bat}}} - \frac{dT_{\text{Al}}}{dt},$$

with

$$\alpha_{\text{bat}} = \frac{k_{\text{bat}}}{\rho_{\text{bat}} c_{p,\text{bat}}}.$$

For uniform volumetric generation in the radius, modal projection provides

$$\boxed{\frac{da_n}{dt} + \mu_n a_n = \beta_n \left[\frac{\dot{q}'''_{\text{bat}}(t)}{\rho_{\text{bat}} c_{p,\text{bat}}} - \frac{dT_{\text{Al}}}{dt}\right],}$$

where

$$\mu_n = \alpha_{\text{bat}} \left(\frac{\lambda_n}{R_{\text{bat}}}\right)^2,$$
$$\beta_n = \frac{2J_1(\lambda_n)}{\lambda_n \left[J_0^2(\lambda_n) + J_1^2(\lambda_n)\right]}.$$

The temperature in any radius is then recovered by

$$\boxed{T_{\text{bat}}(r,t) = T_{\text{Al}}(t) + \sum_{n=1}^{N_m} a_n(t) J_0\left(\lambda_n \frac{r}{R_{\text{bat}}}\right),}$$

where N_m is the number of retained modes. N_m does not represent a mesh: It just truncates the analytical series. The result converges to the solution continuous when N_m is increased.
In particular,

$$T_{\text{centro}}(t) = T_{\text{Al}}(t) + \sum_{n=1}^{N_m} a_n(t),$$
$$T_{\text{surf}}(t) = T_{\text{Al}}(t) + \sum_{n=1}^{N_m} a_n(t) J_0(\lambda_n).$$

B1.5 Coupling with aluminum

The flow from aluminum to battery is

$$\dot{Q}_{ab} = h_c A_c \left[T_{\mathrm{Al}} - T_{\mathrm{surf}}\right] = -h_c A_c \sum_{n=1}^{N_m} a_n J_0(\lambda_n).$$

Consequently, the aluminum balance remains

$$\boxed{C_{\mathrm{Al}} \frac{dT_{\mathrm{Al}}}{dt} = \dot{Q}_{\mathrm{ctrl}}(t) - \dot{Q}_{\mathrm{rad}} - \dot{Q}_{\mathrm{PCB}} + h_c A_c \sum_{n=1}^{N_m} a_n J_0(\lambda_n).} \quad (9.23)$$

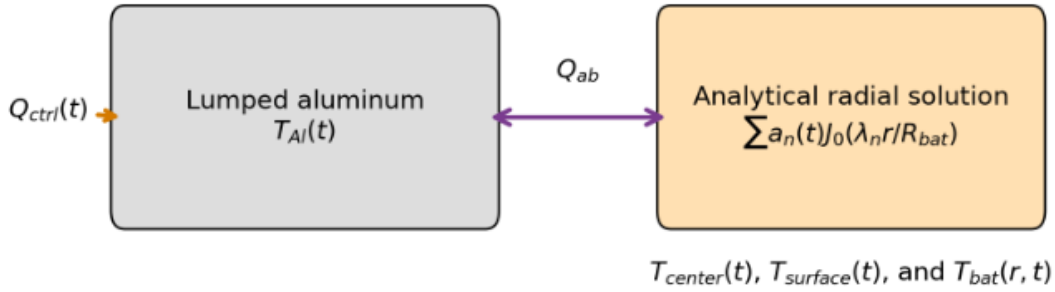


Figure 76: Coupling between the lumped aluminum node and the continuous modal battery solution. There is no spatial mesh.

B1.6 Initial conditions

For a general initial profile, the coefficients are obtained by orthogonality:

$$a_n(0) = \frac{\int_0^{R_{\mathrm{bat}}} r \left[T_{\mathrm{bat}}(r,0) - T_{\mathrm{Al}}(0)\right] J_0(\lambda_n r/R_{\mathrm{bat}})\, dr}{\int_0^{R_{\mathrm{bat}}} r J_0^2(\lambda_n r/R_{\mathrm{bat}})\, dr}.$$

If the battery starts with a uniform temperature T_bat,0,

$$\boxed{a_n(0) = \beta_n \left(T_{\mathrm{bat},0} - T_{\mathrm{Al},0}\right).}$$

B1.7 Control and implementation input

The $\dot{Q}_ctrl, maxctrl(t, T_{Al})$ power may represent step, pulse, experimental sequence or output of a controller. In code for example, we use a saturated proportional control that triggers only the heating when $T_{Al} < T_{Al,ref}$. Radiation is kept in nonlinear form.

A resistor does not provide active cooling. Thus, during the external level of +50 °C, 25 °C can only be maintained if there is a path passive for a colder knot or an additional cooling actuator.

In MATLAB a radial mesh is not created. After integration, the Bessel series provides the profile in any set of rays chosen only for viewing.

A modal convergence study should be conducted, increasing N_m until central and superficial temperatures no longer vary in shape significantly.

Appendix C Distributed Model MATLAB Implementation

```
%% REVISED THERMAL MODEL - ALUMINUM 7075
% The case of Biot uses TAl Bi C and Text Bi C. The transient uses TAl0 C,
% Tbat0 C, TAl ref C and the environmental profile Text C(t).
% The code calculates the equivalent Biot for three geometries and simulates a
% hybrid model: concentrated aluminum and radial battery by Bessel modes.
% Units: SI (m, kg, s, K, W).

clear; clc; close all;

%% 1. USER INPUTS

% Geometry
a  = 24e-3;       % square cross-section side [m]
L  = 65e-3;       % axial length [m]
ri = 9e-3;        % cavity radius [m]

% Aluminum 7075 properties (adjust for the actual alloy temper)
k_Al   = 130;     % aluminum thermal conductivity [W/(m.K)]
rho_Al = 2810;    % aluminum density [kg/m3]
cp_Al  = 960;     % aluminum specific heat [J/(kg.K)]

% Radially distributed battery and aluminum-battery contact
% The values below are examples and should be replaced by real data.
Rbat    = ri;      % battery outer radius [m]
k_bat   = 1.0;     % effective radial conductivity [W/(m.K)]
rho_bat = 2500;    % effective density [kg/m3]
cp_bat  = 1000;    % effective specific heat [J/(kg.K)]
h_c     = 100;     % aluminum-battery contact coefficient [W/(m2.K)]
Nm      = 12;      % number of Bessel modes (not mesh cells)

% Radiation
epsilon = 0.80;               % emissivity [-]
sigma   = 5.670374419e-8;     % Stefan-Boltzmann constant [W/(m2.K4)]

% Cold case used exclusively in the calculation of Biot
TAl_Bi_C  =  25;  % controlled aluminum temperature [degC]
Text_Bi_C = -10;  % external temperature at the cold plateau [degC]

% Initial conditions and reference of the transient model
TAl0_C    = 25;   % initial aluminum temperature [degC]
Tbat0_C   = 5;    % initial uniform battery temperature [degC]
TAl_ref_C = 25;   % preliminary control setpoint [degC]
```

```
% Effective path aluminum-PCB-environment, identified in cold case.
% ReqPCB is maintained as a constant total thermal resistance in geometries.
Qpcb_4_Bi = 1.4441;                % heat flow for four enclosures [W]
Qpcb_1_Bi = Qpcb_4_Bi/4;           % heat flow per enclosure [W]
ReqPCB = (TAl_Bi_C-Text_Bi_C)/Qpcb_1_Bi; % 96.946 K/W

% Proportional illustrative control of resistor (heating only)
Pmax = 3.0;       % maximum resistor power [W]
Kp   = 1.0;       % proportional gain [W/K]
TAl_ref = TAl_ref_C + 273.15;
Qctrl = @(t,TAl) max(0,min(Pmax,Kp*(TAl_ref-TAl)));

% Prescribed external temperature: two vacuum thermal cycles
Text_C = @(t) perfilDoisCiclosC(t); % [degC]
Text   = @(t) Text_C(t) + 273.15;   % [K]

% Uniform battery volumetric generation; use zero when there is no
qgen_bat = @(t) 0.*t; % [W/m3]

% Modal transient simulation numerical parameters
nPontos = 1800;
tFinal = 310*60;   % two consecutive cycles [s]

%% 2. INPUT VALIDATION

assert(a > 0 && L > 0 && ri > 0, ...
    'The dimensions should be positive.');
assert(2*ri < a, ...
    'The cavity shall be fitted to the inscribed cylinder: 2*ri < a is required.');
assert(k_Al > 0 && rho_Al > 0 && cp_Al > 0, ...
    'k Al, rho Al and cp Al must be positive.');
assert(Rbat > 0 && Rbat <= ri, ...
    'Rbat must be positive and must not exceed the radius of the cavity.');
assert(k_bat > 0 && rho_bat > 0 && cp_bat > 0 && h_c > 0, ...
    'Battery and h c properties should be positive.');
assert(Nm >= 1 && Nm == floor(Nm), ...
    'Nm must be a positive integer.');
assert(ReqPCB > 0, 'ReqPCB should be positive.');
assert(epsilon >= 0 && epsilon <= 1, ...
    'Emissivity must belong to the interval [0,1].');
assert(TAl_Bi_C > -273.15 && Text_Bi_C > -273.15 && ...
       TAl0_C > -273.15 && Tbat0_C > -273.15, ...
```

```
    'Temperatures shall be greater than absolute zero.');
assert(nPontos >= 2 && tFinal > 0, ...
    'The numerical parameters must be positive.');

%% 3. TEMPERATURES AND RADIATION LINEARIZATION

TAl_Bi  = TAl_Bi_C  + 273.15;  % [K]
Text_Bi = Text_Bi_C + 273.15;  % [K]
deltaT_Bi = TAl_Bi_C-Text_Bi_C;

% Linearization secant between the aluminum and the environment in cold case.
if abs(TAl_Bi-Text_Bi) > 1e-12
    hrad = epsilon*sigma*(TAl_Bi+Text_Bi) * ...
        (TAl_Bi^2+Text_Bi^2);
else
    hrad = 4*epsilon*sigma*TAl_Bi^3;
end

fprintf('\nCOLD CASE USED FOR THE BIOT CALCULATION\n');
fprintf('T Al,Bi = %.2f degrees C\n',TAl_Bi_C);
fprintf('T ext,Bi = %.2f degrees C\n',Text_Bi_C);
fprintf('Delta T_Bi          = %.2f K\n',deltaT_Bi);
fprintf('h_rad,Bi secante    = %.6f W/(m2.K)\n',hrad);

%% 4. GEOMETRIES AND ACTIVE AREAS

ro_ins  = a/2;
ro_circ = a/sqrt(2);

% Cylinder inscribed
V_ins     = pi*(ro_ins^2-ri^2)*L;
Ao_ins    = 2*pi*ro_ins*L;
Arad_ins  = 0.50*Ao_ins;
Apcb_ins  = a*L;             % caminho fisico via PCB mantido constante
Aativ_ins = Arad_ins + Apcb_ins;

% Original square geometry
V_quad     = (a^2-pi*ri^2)*L;
Ao_quad    = 4*a*L;
Arad_quad  = 0.50*Ao_quad;   % duas faces: 2*a*L
Apcb_quad  = 0.25*Ao_quad;   % uma face ligada ao caminho via PCB: a*L
Aativ_quad = Arad_quad + Apcb_quad;
```

```
% Circumscribed cylinder
V_circ     = pi*(ro_circ^2-ri^2)*L;
Ao_circ    = 2*pi*ro_circ*L;
Arad_circ  = 0.50*Ao_circ;
Apcb_circ  = a*L;            % caminho fisico via PCB mantido constante
Aativ_circ = Arad_circ + Apcb_circ;

%% 5. EQUIVALENT BIOT RESISTANCES AND NUMBERS

V     = [V_ins; V_quad; V_circ];
Arad  = [Arad_ins; Arad_quad; Arad_circ];
Apcb  = [Apcb_ins; Apcb_quad; Apcb_circ];
Aativ = [Aativ_ins; Aativ_quad; Aativ_circ];

Lc   = V./Aativ;
heq  = (hrad.*Arad + 1/ReqPCB)./Aativ;
Rint = Lc./(k_Al.*Aativ);           % igual a V/(k_Al*Aativ^2)
Rrad = 1./(hrad.*Arad);
Rext = 1./(hrad.*Arad + 1/ReqPCB);
Bi   = heq.*Lc./k_Al;               % igual a Rint/Rext

nomes = {'Inscribed cylinder'; 'Square section'; ...
         'Circumscribed cylinder'};
criterio = repmat("Bi >= 0.1",3,1);
criterio(Bi < 0.1) = "Bi < 0.1";

TabelaResultados = table(nomes,V,Arad,Apcb,Aativ,Lc,heq,Rint,Rrad,Rext,Bi,criterio, ...
    'VariableNames',{'Geometria','V_m3','Arad_m2','Apcb_m2', ...
    'Aativa_m2','Lc_m','heq_W_m2K','Rint_K_W','Rrad_K_W', ...
    'Rext_K_W','Biot','Criterio'});

fprintf('\nRESULTADOS\n');
disp(TabelaResultados);

% Independent verifications of cylindrical formulas with Apcb = a*L
nu = 2*ri/a;
Gpcb = 1/ReqPCB;
Bi_ins_analitico = pi*(1-nu^2) / ...
    (4*k_Al*L*(1+pi/2)^2) * ...
    (hrad*pi*a*L/2 + Gpcb);
Bi_circ_analitico = pi*(2-nu^2) / ...
    (4*k_Al*L*(1+pi/sqrt(2))^2) * ...
    (hrad*pi*a*L/sqrt(2) + Gpcb);
```

```
assert(abs(Bi(1)-Bi_ins_analitico) < 1e-12, ...
    'Inconsistency in Biot of the inscribed cylinder.');
assert(abs(Bi(3)-Bi_circ_analitico) < 1e-12, ...
    'Inconsistency in the biot of the circumscribed cylinder.');
assert(Bi(1) < Bi(2) && Bi(2) < Bi(3), ...
    'A ordenacao Bi_ins < Bi_quad < Bi_circ nao foi satisfeita.');

%% 6. HEAT RATES IN INTERVAL LIMITS

% Convention: Positive values represent loss of light.
Qrad_Bi  = epsilon*sigma.*Arad.*(TAl_Bi^4-Text_Bi^4);
Qpcb_Bi = (TAl_Bi-Text_Bi)/ReqPCB*ones(3,1);
Qtotal_Bi = Qrad_Bi+Qpcb_Bi;

TabelaFluxos = table(nomes,Qrad_Bi,Qpcb_Bi,Qtotal_Bi, ...
    'VariableNames',{'Geometria','Qrad_perda_W', ...
    'Qpcb_perda_W','Qtotal_perda_W'});
fprintf('LOSES IN THE CONTROLD COLD CASE');
disp(TabelaFluxos);

%% 7. BIOT GRAPH FOR THE EXTERNAL RAY

ro = linspace(ro_ins,ro_circ,300);
V_ro = pi*(ro.^2-ri^2)*L;
Arad_ro = pi*ro*L;
Aativ_ro = Arad_ro + a*L;
Bi_ro = V_ro./(k_Al*Aativ_ro.^2) .* ...
    (hrad.*Arad_ro + 1/ReqPCB);

figure('Color','w','Name','Sensibilidade geometrica');
plot(ro*1e3,Bi_ro,'LineWidth',2,'Color',[0.15 0.40 0.75]);
hold on; grid on; box on;
plot(ro_ins*1e3,Bi(1),'o','MarkerSize',8,'LineWidth',1.5);
plot(ro_circ*1e3,Bi(3),'s','MarkerSize',8,'LineWidth',1.5);
xlabel('External radius r o [mm]');
ylabel('Number of Biot equivalent [-]');
title(sprintf('Cold case: T {Al} = %.1f ^oC, T {ext} = %.1f ^oC', ...
    TAl_Bi_C,Text_Bi_C));
legend('Bi(r_o)','Inscribed','Circumscribed','Location','best');

%% 8. COMPARATION BETWEEN GEOMETRIA

figure('Color','w','Name','Comparacao de Biot');
```

```
bar(Bi,'FaceColor',[0.30 0.55 0.85]);
grid on; box on;
set(gca,'XTick',1:3,'XTickLabel',{'Inscribed','Square','Circumscribed'});
ylabel('Number of Biot equivalent [-]');
title('Comparacao dos indicadores de Biot');
hold on;
yline(0.1,'--r','Bi = 0,1','LineWidth',1.2);

%% 9. MODEL HYBRIDE WITHOUT BOTTLE: LUNCH + BESSEL MODES

% The aluminum and one in the concentrate. The battery is solved by expansion
% T bat(r,t) = T Al(t) + sum n a n(t)*J0(lambda n*r/Rbat).
% The flat ends are adiabatic; therefore, there is no dependence
% Axial. Nor trunca the analytical series and does not represent a radial mesh.

alpha_bat = k_bat/(rho_bat*cp_bat);
Ac = 2*pi*Rbat*L;
Bi_c = h_c*Rbat/k_bat;
V_bat = pi*Rbat^2*L;

% Autovalues: lambda*J1(lambda) = Bi c*J0(lambda)
lambda = raizesRobinCilindro(Bi_c,Nm);
J0lambda = besselj(0,lambda);
J1lambda = besselj(1,lambda);

mu = alpha_bat*(lambda/Rbat).^2;
beta = 2*J1lambda ./ ...
    (lambda.*(J0lambda.^2 + J1lambda.^2));
gamma_media = 2*J1lambda./lambda;

% Thermal capacity of the aluminium
m_Al = rho_Al*V_quad;
C_Al = m_Al*cp_Al;

% Parameters assembled for system function
p.C_Al = C_Al;
p.epsilon = epsilon;
p.sigma = sigma;
p.Arad = Arad_quad;
p.ReqPCB = ReqPCB;
p.hcAc = h_c*Ac;
p.J0lambda = J0lambda;
p.mu = mu;
```

```
p.beta = beta;
p.rhoCp_bat = rho_bat*cp_bat;

% Initial conditions
TAl0 = TAl0_C + 273.15;
Tbat0 = Tbat0_C + 273.15;
a0 = beta*(Tbat0-TAl0);
Y0 = [TAl0; a0];

tSpan = linspace(0,tFinal,nPontos);

rhs = @(t,Y) rhsAluminioBateriaModal( ...
    t,Y,p,Qctrl,qgen_bat,Text);
opcoes = odeset('RelTol',1e-7,'AbsTol',1e-9);
[t,Y] = ode15s(rhs,tSpan,Y0,opcoes);

TAl_K = Y(:,1);
a_hist = Y(:,2:end);

% Temperatures of interest
Tcentro_K = TAl_K + sum(a_hist,2);
Tsuperficie_K = TAl_K + a_hist*J0lambda(:);
Tmedia_K = TAl_K + a_hist*gamma_media(:);

TAl_C = TAl_K-273.15;
Tcentro_C = Tcentro_K-273.15;
Tsuperficie_C = Tsuperficie_K-273.15;
Tmedia_C = Tmedia_K-273.15;

% Continuous profile reconstruction only for viewing
r_plot = linspace(0,Rbat,160);
Phi = besselj(0,lambda(:)*(r_plot/Rbat));
Tperfil_C = TAl_K + a_hist*Phi - 273.15;

% Environment, control and heat rates
Text_C_t = arrayfun(Text_C,t);
Text_K_t = Text_C_t+273.15;
Qctrl_t = arrayfun(Qctrl,t,TAl_K);
Qrad_t = epsilon*sigma*Arad_quad.*(TAl_K.^4-Text_K_t.^4);
Qpcb_t = (TAl_K-Text_K_t)/ReqPCB;
Qab_t = h_c*Ac.*(TAl_K-Tsuperficie_K);
Qgen_t = arrayfun(qgen_bat,t)*V_bat;
```

```
figure('Color','w','Name','Modal model temperatures');
plot(t/60,TAl_C,'LineWidth',2.0); hold on; grid on; box on;
plot(t/60,Tsuperficie_C,'LineWidth',1.8);
plot(t/60,Tcentro_C,'LineWidth',1.8);
plot(t/60,Tmedia_C,'--','LineWidth',1.6);
plot(t/60,Text_C_t,':k','LineWidth',1.2);
yline(0,'--','Battery limit','LineWidth',1.1);
xlabel('Time [min]');
ylabel('Temperature [^oC]');
title('Concentrated aluminium and radial battery');
legend('Aluminium','Battery surface','Battery centre', ...
    'Battery volumetric media','External environment', ...
    'Battery limit','Location','best');

figure('Color','w','Name','Radial battery profiles');
indices = unique(round(linspace(1,numel(t),5)));
cores = lines(numel(indices));
hold on; grid on; box on;
for j = 1:numel(indices)
    id = indices(j);
    plot(r_plot*1e3,Tperfil_C(id,:), ...
        'LineWidth',1.8,'Color',cores(j,:), ...
        'DisplayName',sprintf('t = %.2f min',t(id)/60));
end
xlabel('Battery radius [mm]');
ylabel('Temperature [^oC]');
title('Perfil radial reconstruido pela serie de Bessel');
legend('Location','best');

figure('Color','w','Name','Fluxos de calor');
plot(t/60,Qctrl_t,'LineWidth',1.8); hold on; grid on; box on;
plot(t/60,Qrad_t,'LineWidth',1.8);
plot(t/60,Qpcb_t,'LineWidth',1.8);
plot(t/60,Qab_t,'LineWidth',1.8);
plot(t/60,Qgen_t,'--','LineWidth',1.5);
xlabel('Time [min]');
ylabel('Heat-transfer rate [W]');
title('Control power and heat-transfer rates');
legend('Q̇_ctrl,max{ctrl}','Q̇_ctrl,max{rad}','Q̇_ctrl,max{PCB}','Q̇_ctrl,max{Al-bat}', ...
    'Q̇_ctrl,max{geracao,bat}','Location','best');

fprintf('HYBRID MODEL: ALUMINUM + RADIAL BATTERY\n');
fprintf('Battery Contact Bi = %.6f\n',Bi_c);
```

```
fprintf('Mode number = %d\n',Nm);
fprintf('Aluminium capacity = %.6f J/K\n',C_Al);
fprintf('Battery volume = %.9e m3\n',V_bat);
fprintf('T Al final = %.4f degrees C\n',TAl_C(end));
fprintf('T final area = %.4f degrees C\n',Tsuperficie_C(end));
fprintf('T centre final = %.4f C\n',Tcentro_C(end));

%% 10. NOTES

% 1) The flat ends of the battery are adiabatic. With symmetry
% angular, the problem is equivalent to the infinite radial 1D cylinder.
% 2) Nm and the number of Bessel modes, not the number of mesh elements.
% 3) Increase Nm and check the Tcentro and Tsurface convergence.
% 4) TAl0 C and the uniform initial aluminum temperature.
% 5) Tbat0 C and the battery uniform initial temperature. For a profile
% initial not uniform, design the profile in the Bessel autofunctories.
% 6) The illustrative control only uses heating. He can't do it.
% limit T Al to 25 degrees C when the warm environment requires cooling.
% 7) k bat, rho bat, cp bat and h c are effective parameters and should be
% measured, reported by the manufacturer or identified experimentally.
% (8) ReqPCB and an equivalent resistance of the Al-PCB-environment path,
% identified in the cold case; it is not a resistance of material.
% 9) Qpcb 4  Bi must be the heat that crosses the Al-PCB path. If it is the
% total heating power, remove direct light radiation before
% and any other losses to avoid double counting.

%% LOCAL FUNCTIONS

function dY = rhsAluminioBateriaModal(t,Y,p,Qctrl,qgen_bat,Text)
    TAl = Y(1);
    a = Y(2:end);
    Text_K = Text(t);

    % Losses of aluminium
    Qrad = p.epsilon*p.sigma*p.Arad*(TAl^4-Text_K^4);
    Qpcb = (TAl-Text_K)/p.ReqPCB;

    % u s = T surface - T Al
    u_s = sum(a.*p.J0lambda);
    Qab = -p.hcAc*u_s; % positive from aluminum to battery

    dTAl = (Qctrl(t,TAl)-Qrad-Qpcb-Qab)/p.C_Al;
```

```
    % Modal equations coupled to aluminum temperature movement
    fonte = qgen_bat(t)/p.rhoCp_bat;
    da = -p.mu.*a + p.beta*(fonte-dTAl);

    dY = [dTAl; da];
end

function T_C = perfilDoisCiclosC(t)
    % Two cycles: -10 to +50 degrees C, 1.5 degrees C/min ramps and
    % 30-minute stabilizers. The intermediate and shared cold plateau.
    tempo_min = [0 30 70 100 140 170 210 240 280 310];
    temp_C = [-10 -10 50 50 -10 -10 50 50 -10 -10];
    T_C = interp1(tempo_min*60,temp_C,t,'linear','extrap');
end

function lambda = raizesRobinCilindro(Bi,N)
    % Solve lambda*J1(lambda)-Bi*J0(lambda)=0 for the first N
    % positive roots without creating space mesh on the battery.
    f = @(x) x.*besselj(1,x)-Bi.*besselj(0,x);
    xmax = (N+3)*pi;
    lambda = [];

    while numel(lambda) < N
        x = linspace(1e-10,xmax,50000);
        fx = f(x);
        idx = find(fx(1:end-1).*fx(2:end) < 0);

        candidatos = zeros(numel(idx),1);
        for j = 1:numel(idx)
            candidatos(j) = fzero(f,[x(idx(j)),x(idx(j)+1)]);
        end

        candidatos = unique(round(candidatos*1e12)/1e12);
        candidatos = candidatos(candidatos > 1e-9);
        lambda = candidatos(:);

        if numel(lambda) < N
            xmax = 1.5*xmax;
        end
    end

    lambda = lambda(1:N);
end
```

Appendix D LQI Control MATLAB Implementation

```
%% SIMPLIFIED THERMAL DIGITAL TWIN - LQI CONTROL
% Two-node plant: lumped aluminum enclosure + equivalent battery node.
% The controller uses only the aluminum thermistor measurement.
% The internal battery temperature distribution is not resolved at this stage.
%
% Requires MATLAB R2016b or later and Control System Toolbox.
% Internal units: m, kg, s, K, and W.

clear; clc; close all;

%% 1. PHYSICAL PARAMETERS

% Reduced geometry for one battery enclosure
p.a    = 24e-3; p.L = 65e-3; p.ri = 9e-3; p.Rbat = 9e-3;

% Aluminum 7075
p.rhoAl = 2810; p.cpAl = 960;
p.VAl = (p.a^2-pi*p.ri^2)*p.L;
p.CAl = p.rhoAl*p.cpAl*p.VAl;                % [J/K]

% Equivalent battery node: preliminary values to be identified
p.rhoBat = 2500; p.cpBat = 1000;
p.VBat = pi*p.Rbat^2*p.L;
p.CBat = p.rhoBat*p.cpBat*p.VBat;            % [J/K]

% Aluminum-to-battery thermal contact
p.hc = 100;                                  % [W/(m2.K)]
p.Ac = 2*pi*p.Rbat*p.L;                      % [m2]
p.RAlBat = 1/(p.hc*p.Ac);                    % [K/W]

% External heat transfer from the aluminum enclosure
p.epsilon = 0.80;
p.sigma = 5.670374419e-8;                    % [W/(m2.K4)]
p.Arad = 2*p.a*p.L;                          % two radiating faces [m2]

% Total aluminum-PCB-environment path, held constant between geometries
Qpcb4Design = 1.4441;                        % four enclosures [W]
Qpcb1Design = Qpcb4Design/4;                 % one enclosure [W]
p.ReqPCB = (25-(-10))/Qpcb1Design;           % [K/W]

% Total internal battery heat generation; zero for the initial study
p.QgenBat = @(t) 0.*t;                       % [W]
```

```
%% 2. SENSOR, THERMAL LIMITS, AND TEST PROFILE

p.tauSensor = 15;                               % thermistor time constant [s]
p.Pmax = 3.0;                                   % resistor power limit [W]
p.TAlMinC = 10;                                 % aluminum requirement [degC]
p.TrefC = 12;                                   % preliminary setpoint [degC]
p.TAl0C = 20; p.Tbat0C = 20;                    % initial temperatures [degC]
p.TextDesignC = -10;                            % linearization environment [degC]
p.tFinal = 310*60;                              % two cycles [s]
p.nPoints = 2200;

%% 3. INPUT VALIDATION AND LINEAR DESIGN MODEL

assert(exist('lqr','file') == 2, ...
    'The LQI design requires Control System Toolbox.');
assert(p.CAl > 0 && p.CBat > 0 && p.RAlBat > 0 && p.ReqPCB > 0, ...
    'Thermal capacitances and resistances must be positive.');
assert(p.TrefC > p.TAlMinC, ...
    'The control setpoint must exceed the minimum aluminum temperature.');
assert(p.Pmax > 0, 'The maximum resistor power must be positive.');

[A,B,E,C,f0] = linearizedModel(p);
p.Alin = A; p.Blin = B; p.Elin = E; p.Clin = C; p.f0lin = f0;
p.nPhys = size(A,1);

fprintf('\nSTATE-SPACE DESIGN MODEL\n');
fprintf('Number of physical states      = %d\n',p.nPhys);
fprintf('Controllability rank           = %d of %d\n',rank(ctrb(A,B)),p.nPhys);
fprintf('Observability rank             = %d of %d\n',rank(obsv(A,C)),p.nPhys);
fprintf('C_Al                           = %.6f J/K\n',p.CAl);
fprintf('C_bat                          = %.6f J/K\n',p.CBat);
fprintf('R_Al-bat                       = %.6f K/W\n',p.RAlBat);
fprintf('R_Al-PCB,eq                    = %.6f K/W\n',p.ReqPCB);
fprintf('\nMATRIX A\n'); disp(A);
fprintf('MATRIX B\n'); disp(B);
fprintf('MATRIX E\n'); disp(E);
fprintf('MATRIX C\n'); disp(C);
fprintf('AFFINE VECTOR f0\n'); disp(f0);

%% 4. LQI CONTROLLER AND DETERMINISTIC OBSERVER

% Integral state: xi_dot = Tref-Tmeas = -C*x for deviation variables.
```

```
Aaug = [A, zeros(p.nPhys,1); -C, 0];
Baug = [B; 0];

% Preliminary weights; tune these during the control study.
Qx = diag([2.0,0.10,0.80]); qIntegral = 2.0e-6;
Qaug = blkdiag(Qx,qIntegral); Ru = 0.50;
Kaug = lqr(Aaug,Baug,Qaug,Ru);
lqi.Kx = Kaug(1:p.nPhys); lqi.Ki = Kaug(end);

% Deterministic observer; measurement noise is omitted in this study.
Qobs = diag([2.0,0.50,1.0]); Robs = 0.10;
lqi.Lobs = lqr(A',C',Qobs,Robs)';

fprintf('\nLQI CONTROLLER\n');
fprintf('Largest real part of controlled poles = %.6e 1/s\n', ...
    max(real(eig(Aaug-Baug*Kaug))));
fprintf('Largest real part of observer poles   = %.6e 1/s\n', ...
    max(real(eig(A-lqi.Lobs*C))));

%% 5. INITIAL CONDITIONS AND SIMULATION

TAl0 = p.TAl0C+273.15; Tbat0 = p.Tbat0C+273.15; Tmeas0 = TAl0;
x0 = [TAl0; Tbat0; Tmeas0];

% Physical states + integral state + estimated state deviations.
xhat0 = zeros(p.nPhys,1); Y0 = [x0; 0; xhat0];
tspan = linspace(0,p.tFinal,p.nPoints);
options = odeset('RelTol',1e-6,'AbsTol',1e-7,'MaxStep',10);

fprintf('\nIntegrating the LQI simulation...\n');
simulationTimer = tic;
[t,Y] = ode45(@(t,Y) rhsLQI(t,Y,p,lqi),tspan,Y0,options);
fprintf('LQI simulation completed in %.2f s.\n',toc(simulationTimer));
R = processResults(t,Y,p,lqi);

%% 6. PERFORMANCE SUMMARY

Controller = "LQI";
HeaterEnergy_Wh = R.energyWh;
MinimumAluminumTemperature_C = min(R.TAlC);
MinimumAluminumMargin_K = MinimumAluminumTemperature_C-p.TAlMinC;
TimeBelowAluminumLimit_min = R.timeAlBelowMin;
MinimumEquivalentBatteryTemperature_C = min(R.TbatC);
```

```
SensorRMSE_K = R.sensorRmse;
PeakHeaterPower_W = max(R.u);
TimeAt3WSaturation_min = R.timeSaturatedMin;

PerformanceSummary = table(Controller,HeaterEnergy_Wh, ...
    MinimumAluminumTemperature_C,MinimumAluminumMargin_K, ...
    TimeBelowAluminumLimit_min,MinimumEquivalentBatteryTemperature_C, ...
    SensorRMSE_K,PeakHeaterPower_W,TimeAt3WSaturation_min);
fprintf('\nLQI PERFORMANCE SUMMARY\n'); disp(PerformanceSummary);

%% 7. SINGLE RESULTS WINDOW

resultsWindow = figure('Color','w','Name', ...
    'Simplified thermal digital twin - LQI control','NumberTitle','off');
tabGroup = uitabgroup(resultsWindow);
createResultsTab(tabGroup,'LQI',R,p);

%% 8. MODEL-USE NOTES

% 1) The primary requirement is TAl >= TAlMinC throughout the test.
% 2) The equivalent battery node represents the thermal load on aluminum.
% 3) Protecting the distributed battery additionally requires an admissible
%    initial field and no unmodeled heat path to a colder boundary.
% 4) Reintroduce the radial battery model for verification.
% 5) Tune TrefC and the LQI Q/R weights during the control study.
% 6) The physical actuator constraint is 0 <= Qctrl <= 3 W.
% 7) If aluminum violates its limit while the heater is saturated, control
%    tuning alone cannot overcome the available-power limitation.

%% LOCAL FUNCTIONS

function dY = rhsLQI(t,Y,p,c)
    n = p.nPhys; x = Y(1:n); xi = Y(n+1); xhat = Y(n+2:n+1+n);
    [u,yDeviation,integralEffect] = lqiCommand(x,xi,xhat,p,c);
    dx = nonlinearThermalDynamics(t,x,u,p);

    dxi = -yDeviation;
    if (u <= 1e-10 && integralEffect < 0) || ...
            (u >= p.Pmax-1e-10 && integralEffect > 0)
        dxi = 0;
    end

    TextK = twoCycleProfileC(t)+273.15;
```

```
    dText = TextK-(p.TextDesignC+273.15);
    innovation = yDeviation-p.Clin*xhat;
    dxhat = p.Alin*xhat+p.Blin*u+p.Elin*dText+p.f0lin+ ...
        c.Lobs*innovation;
    dY = [dx; dxi; dxhat];
end

function dx = nonlinearThermalDynamics(t,x,u,p)
    TAl = x(1); Tbat = x(2); Tmeas = x(3);
    TextK = twoCycleProfileC(t)+273.15;
    Qrad = p.epsilon*p.sigma*p.Arad*(TAl^4-TextK^4);
    Qpcb = (TAl-TextK)/p.ReqPCB;
    QalBat = (TAl-Tbat)/p.RAlBat;
    dTAl = (u-Qrad-Qpcb-QalBat)/p.CAl;
    dTbat = (QalBat+p.QgenBat(t))/p.CBat;
    dTmeas = sensorDynamics(TAl,Tmeas,p);
    dx = [dTAl; dTbat; dTmeas];
end

function [u,yDeviation,integralEffect] = lqiCommand(x,xi,xhat,p,c)
    Tmeas = x(3);
    yDeviation = Tmeas-(p.TrefC+273.15);
    uFree = -c.Kx*xhat-c.Ki*xi;
    u = min(p.Pmax,max(0,uFree));
    integralEffect = -c.Ki*(-yDeviation);
end

function dTmeas = sensorDynamics(TAl,Tmeas,p)
    dTmeas = (TAl-Tmeas)/p.tauSensor;
end

function [A,B,E,C,f0] = linearizedModel(p)
    TrefK = p.TrefC+273.15; Text0K = p.TextDesignC+273.15;
    GAlBat = 1/p.RAlBat;
    GAl = 4*p.epsilon*p.sigma*p.Arad*TrefK^3+1/p.ReqPCB;
    Gext = 4*p.epsilon*p.sigma*p.Arad*Text0K^3+1/p.ReqPCB;
    A = [-(GAl+GAlBat)/p.CAl, GAlBat/p.CAl, 0; ...
          GAlBat/p.CBat, -GAlBat/p.CBat, 0; ...
          1/p.tauSensor, 0, -1/p.tauSensor];
    B = [1/p.CAl;0;0]; E = [Gext/p.CAl;0;0]; C = [0 0 1];
    Qloss0 = p.epsilon*p.sigma*p.Arad*(TrefK^4-Text0K^4) + ...
        (TrefK-Text0K)/p.ReqPCB;
    f0 = zeros(size(B)); f0(1) = -Qloss0/p.CAl;
```

```
end

function R = processResults(t,Y,p,c)
    n = p.nPhys; x = Y(:,1:n);
    TAl = x(:,1); Tbat = x(:,2); Tmeas = x(:,3);
    u = zeros(size(t));
    for i = 1:numel(t)
        xi = Y(i,n+1); xhat = Y(i,n+2:n+1+n)';
        u(i) = lqiCommand(x(i,:)',xi,xhat,p,c);
    end
    sensorError = TAl-Tmeas; TAlC = TAl-273.15;
    R.t = t; R.tmin = t/60; R.TAlC = TAlC;
    R.TbatC = Tbat-273.15; R.TmeasC = Tmeas-273.15;
    R.TextC = twoCycleProfileC(t); R.u = u;
    TextK = R.TextC+273.15;
    R.Qrad = p.epsilon*p.sigma*p.Arad.*(TAl.^4-TextK.^4);
    R.Qpcb = (TAl-TextK)/p.ReqPCB;
    R.QAlBat = (TAl-Tbat)/p.RAlBat;
    R.cumulativeEnergyWh = cumtrapz(t,u)/3600;
    R.energyWh = trapz(t,u)/3600;
    R.sensorRmse = sqrt(trapz(t,sensorError.^2)/(t(end)-t(1)));
    R.timeAlBelowMin = trapz(t,double(TAlC < p.TAlMinC))/60;
    R.timeSaturatedMin = trapz(t,double(u >= p.Pmax-1e-8))/60;
end

function createResultsTab(tabGroup,name,R,p)
    tab = uitab(tabGroup,'Title',name);
    layout = tiledlayout(tab,2,1,'TileSpacing','compact','Padding','compact');
    axT = nexttile(layout,1);
    hExt = plot(axT,R.tmin,R.TextC,':k','LineWidth',1.2); hold(axT,'on');
    hAl = plot(axT,R.tmin,R.TAlC,'LineWidth',1.9);
    hBat = plot(axT,R.tmin,R.TbatC,'LineWidth',1.5);
    hMeas = plot(axT,R.tmin,R.TmeasC,'--','LineWidth',1.4);
    yline(axT,p.TAlMinC,'--r','Aluminum limit');
    yline(axT,p.TrefC,':','Setpoint'); grid(axT,'on'); box(axT,'on');
    xlabel(axT,'Time [min]'); ylabel(axT,'Temperature [degC]');
    title(axT,['Temperatures - ',name]);
    legend(axT,[hExt,hAl,hBat,hMeas], ...
        {'Environment','Aluminum','Equivalent battery','Thermistor'}, ...
        'Location','eastoutside');

    axQ = nexttile(layout,2);
    hAct = plot(axQ,R.tmin,R.u,'LineWidth',1.9); hold(axQ,'on');
```

```
    hRad = plot(axQ,R.tmin,R.Qrad,'LineWidth',1.5);
    hPCB = plot(axQ,R.tmin,R.Qpcb,'LineWidth',1.5);
    hBatFlux = plot(axQ,R.tmin,R.QAlBat,'LineWidth',1.5);
    yline(axQ,0,':k'); yline(axQ,p.Pmax,'--r','Resistor limit');
    grid(axQ,'on'); box(axQ,'on');
    xlabel(axQ,'Time [min]'); ylabel(axQ,'Heat-transfer rate [W]');
    title(axQ,['Actuator and heat-transfer rates - ',name]);
    legend(axQ,[hAct,hRad,hPCB,hBatFlux], ...
        {'Resistor input','Aluminum-environment radiation', ...
        'Aluminum-PCB-environment path','Aluminum-battery'}, ...
        'Location','eastoutside');
end

function T_C = twoCycleProfileC(t)
    timeMin = [0 30 70 100 140 170 210 240 280 310];
    temperatureC = [-10 -10 50 50 -10 -10 50 50 -10 -10];
    T_C = interp1(timeMin*60,temperatureC,t,'linear','extrap');
end
```

REFERENCES

[1] Gilbert, D. (2025, 05, 11). 435 MHz QFH Antenna for Amateur Radio Satellites. https://www.youtube.com/watch?v=AMrPPIv-bcs

[2] SUN, Junzi. The 1090MHz Riddle: A guide to decoding Mode S and ADS-B signals. TUDelft Open, 2021. Available at: https://mode-s.org/1090mhz/. (Accessed: August 8, 2026).

[3] NASA. GSFC-STD-7000B — General Environmental Verification Standard (GEVS) for GSFC Flight Programs and Projects, 2021. https://standards.nasa.gov/standard/GSFC/GSFC-STD-7000

[4] Fagerudd, J. Stress Simulation of the SEAM CubeSat Structure During Launch. Master's Thesis, KTH Royal Institute of Technology, Stockholm, 2014. https://s3vi.ndc.nasa.gov/ssri-kb/static/resources/FULLTEXT01.pdf

[5] NASA. NASA-STD-7001B — Payload Vibroacoustic Test Criteria, 2017. Section 4.4.2: random vibration in three orthogonal axes and ASD control from 20 to 2000 Hz. https://standards.nasa.gov/sites/default/files/standards/NASA/B/0/nasa-std-7001b.pdf